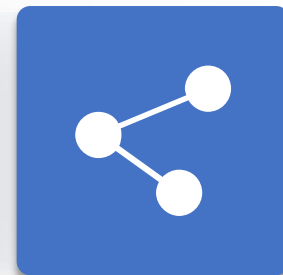


人工智能引论

第6章 神经网络¹

授课教师：李静瑶²





录¹

1 什么是神经网络²

2 神经网络的训练³

3 卷积神经网络⁴

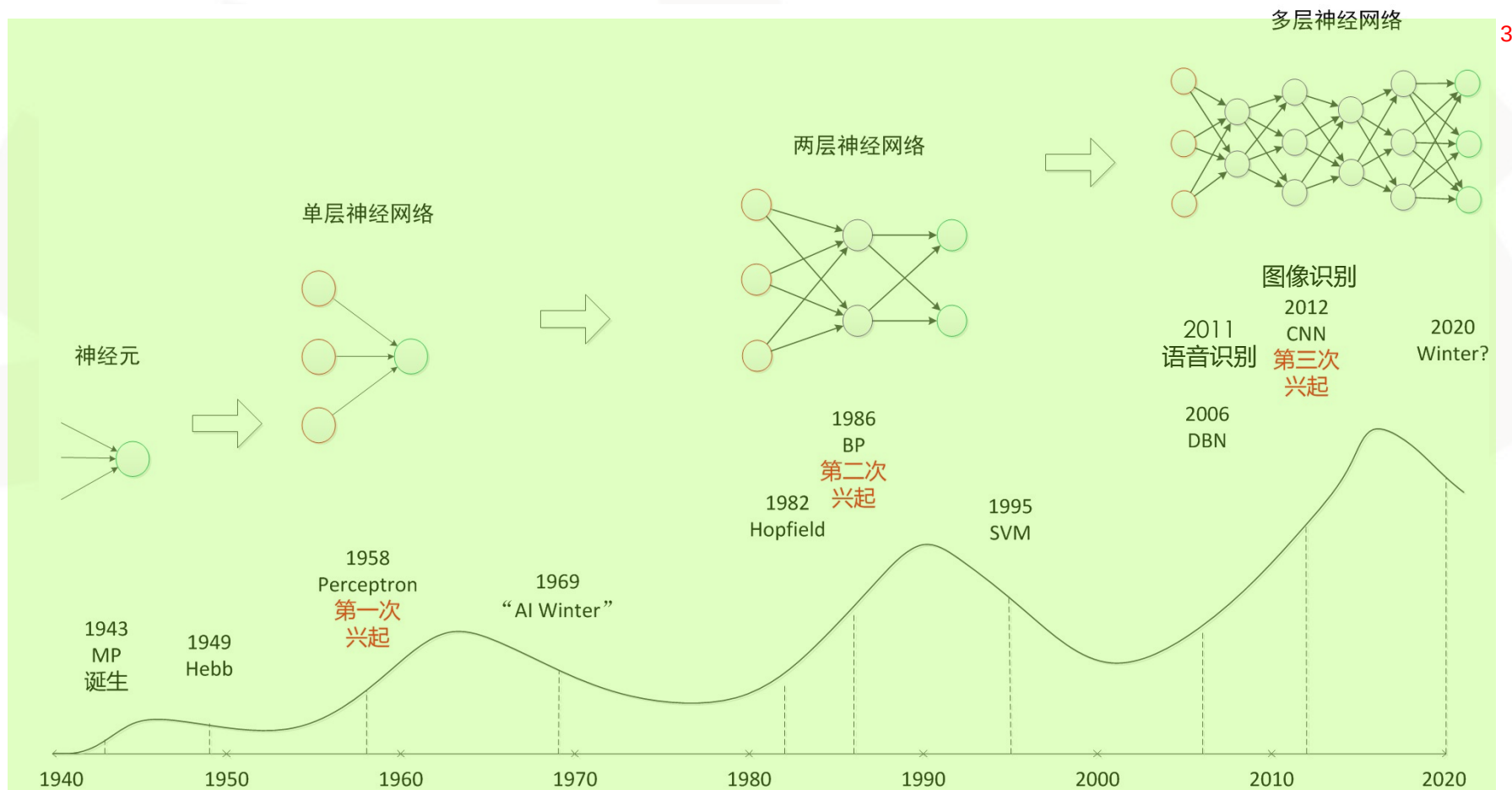
4 梯度消失问题⁵

5 过拟合问题⁶

什么是神经网络¹



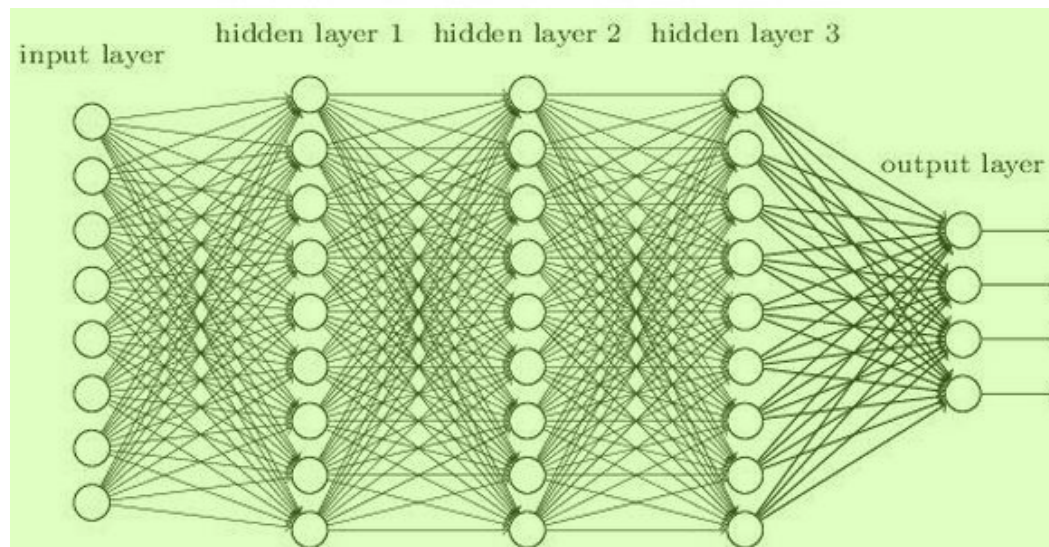
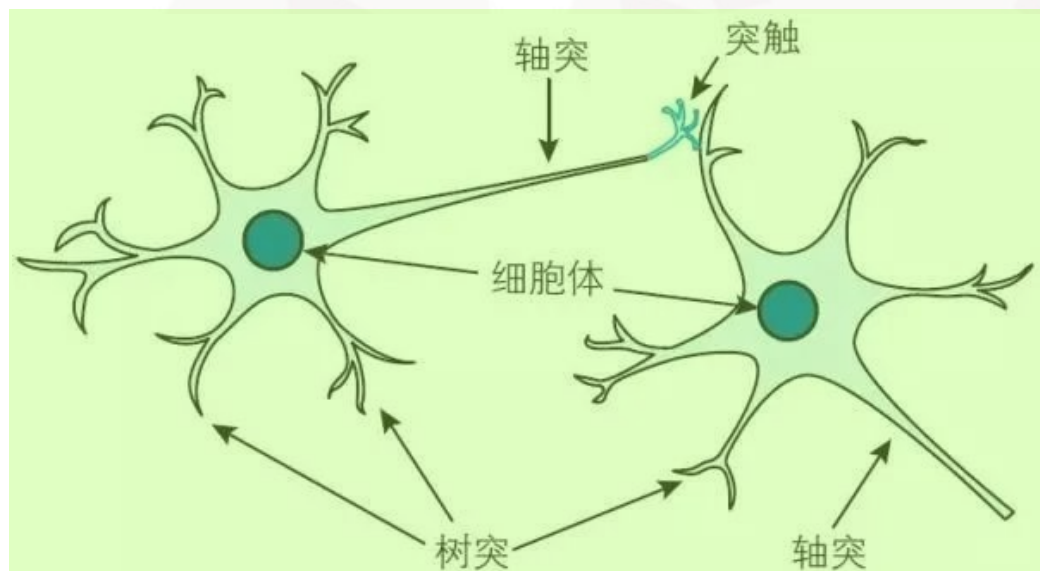
神经网络是如何实现的²



什么是神经网络¹



人工神经网络 (ANN—Artificial Neural Network或者NN)²



什么是神经网络¹



数字识别问题²

³

1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
0	0	0	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	0	0

数字3的图像⁴

⁵

1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1

数字3的模式⁶

什么是神经网络¹



数字识别问题²

³

0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0

数字8的图像⁴

⁵

1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1

数字3的模式⁶

什么是神经网络¹

模式匹配（按位乘 / 点积）²

³

0	1	1	1	0
1	0	0	0	1
0	1	1	1	0
1	0	0	0	1
0	1	1	1	0

•

⁴

1	1	1	1	-1
-1	-1	-1	-1	1
1	1	1	1	-1
-1	-1	-1	-1	1
1	1	1	1	-1

=

⁵

$$\begin{aligned}
 &0 \times 1 + 1 \times 1 + 1 \times 1 + 1 \times 1 + 0 \times (-1) + \\
 &1 \times (-1) + 0 \times (-1) + 0 \times (-1) + 0 \times (-1) + 1 \times 1 + \\
 &0 \times 1 + 1 \times 1 + 1 \times 1 + 1 \times 1 + 0 \times (-1) + \\
 &1 \times (-1) + 0 \times (-1) + 0 \times (-1) + 0 \times (-1) + 1 \times 1 + \\
 &0 \times 1 + 1 \times 1 + 1 \times 1 + 1 \times 1 + 0 \times (-1) \\
 &= 9
 \end{aligned}$$

⁶

$$net = w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n$$

⁷

其中： $w_i (i = 1, 2, \dots, n)$ ：表示数字模式

⁸

$x_i (i = 1, 2, \dots, n)$ ：表示数字图像

什么是神经网络¹



数字3、数字8与模式3的匹配²

³

1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
0	0	0	0	0	0	0	0	0	0	0	0	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	0	0

•

⁵

1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1

= 143

⁴

0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0

•

⁶

1	1	1	1	1	1	1	1	1	1	1	1	1	-1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	-1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	-1

= 115

什么是神经网络¹



存在的问题²

- 笔画多少带来的问题³

- 1的笔画少⁴

- 8的笔画多⁵

- 如何评判匹配的程度?⁶

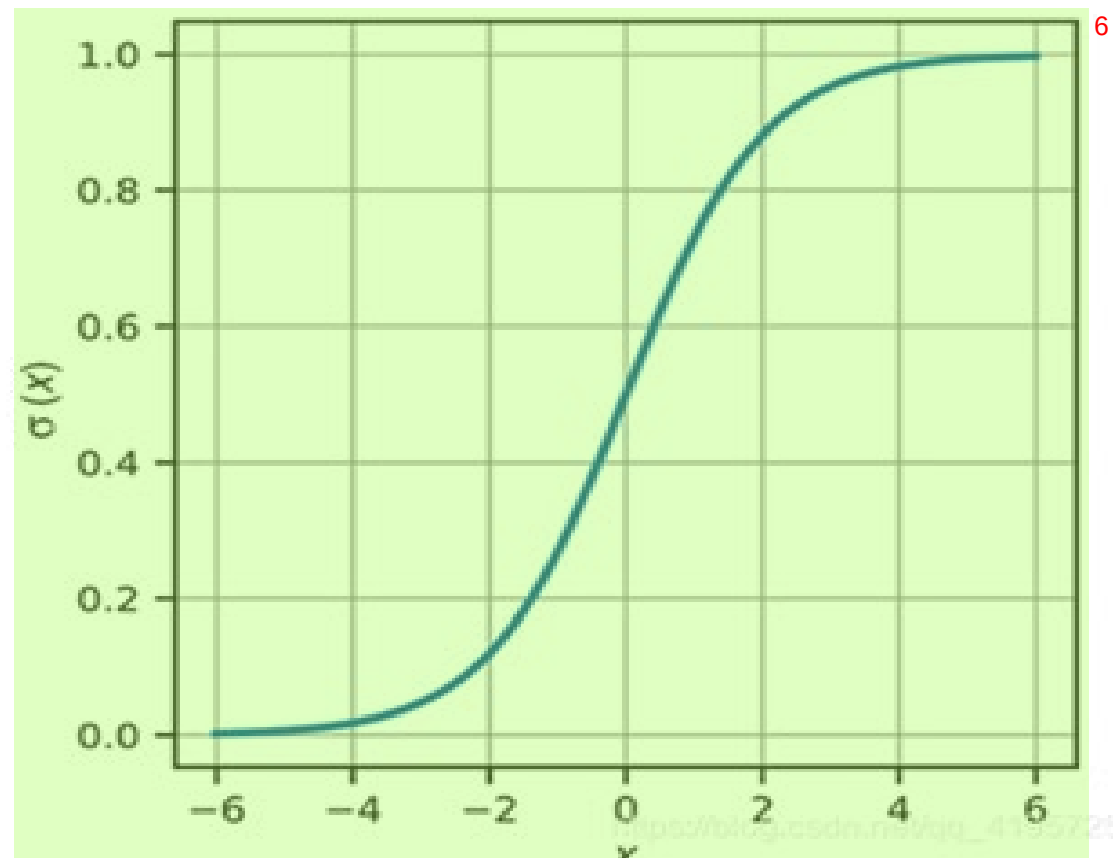
什么是神经网络¹



使用Sigmoid函数²

- 使用Sigmoid函数将匹配结果³变换到0-1之间

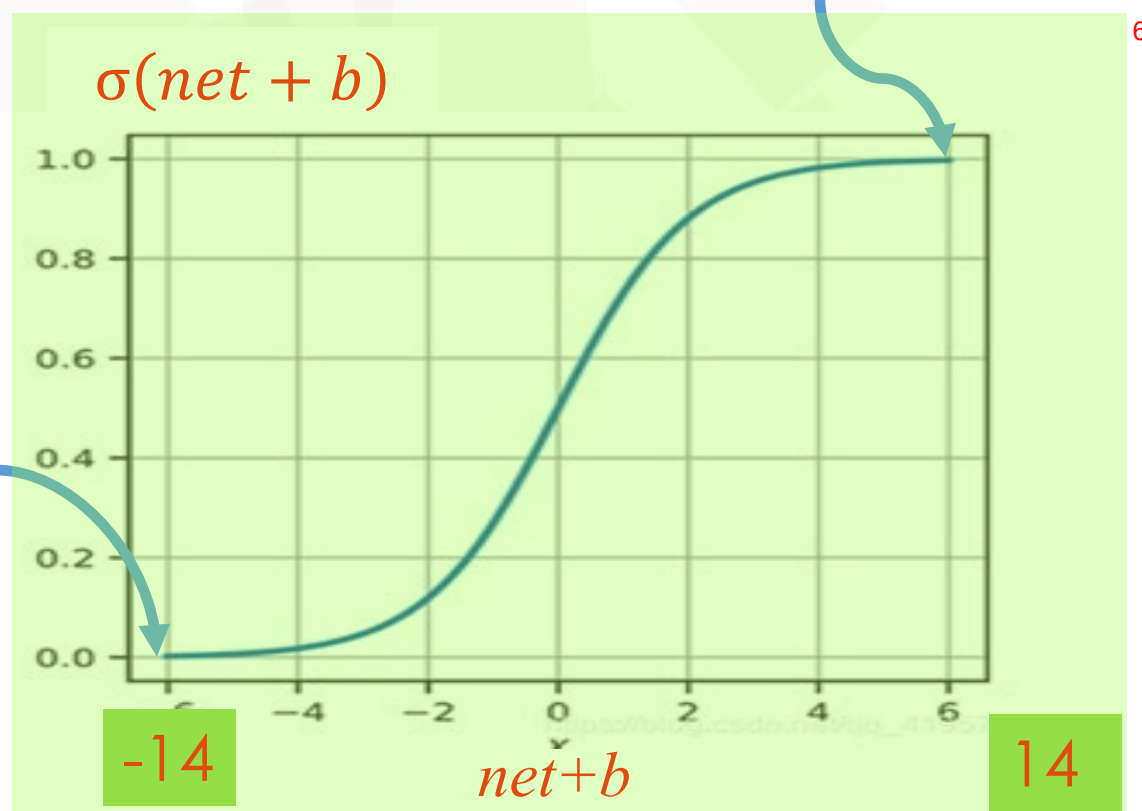
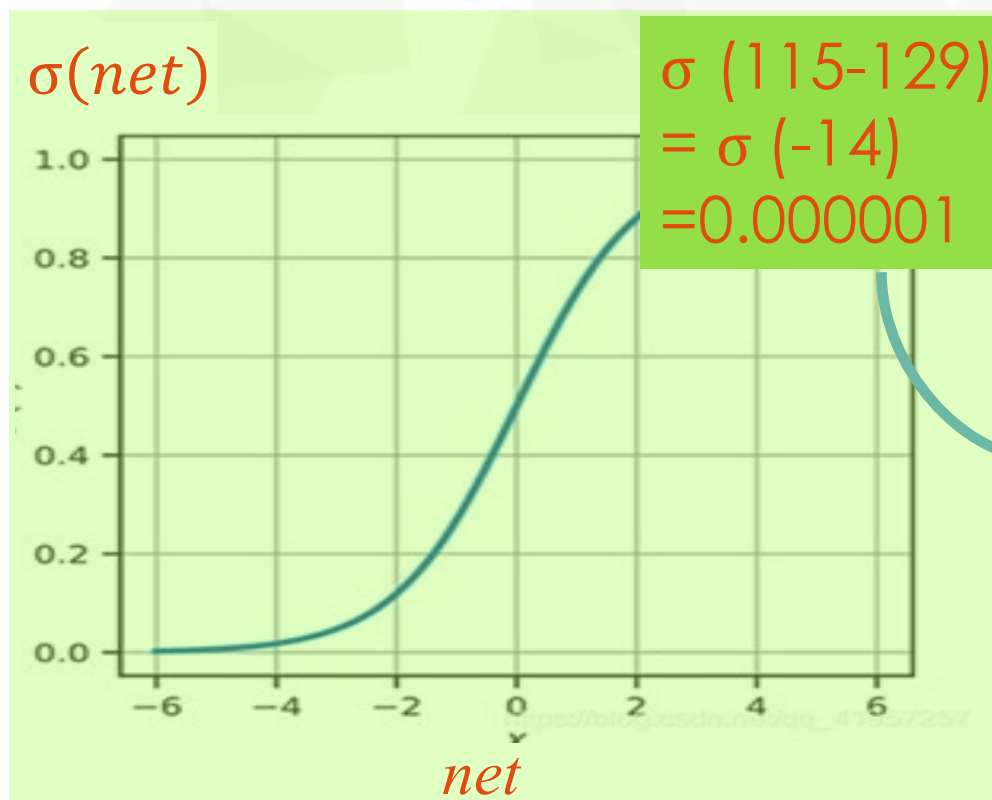
$$\sigma(net) = \frac{1}{1+e^{-net}} \quad ^4$$



什么是神经网络¹

增加偏置项²

- 通过增加偏置项让Sigmoid函数平移³



$$\begin{aligned}\sigma(143-129) &= \sigma(14) \\ &= 0.999999\end{aligned}$$

什么是神经网络¹



增加偏置项²

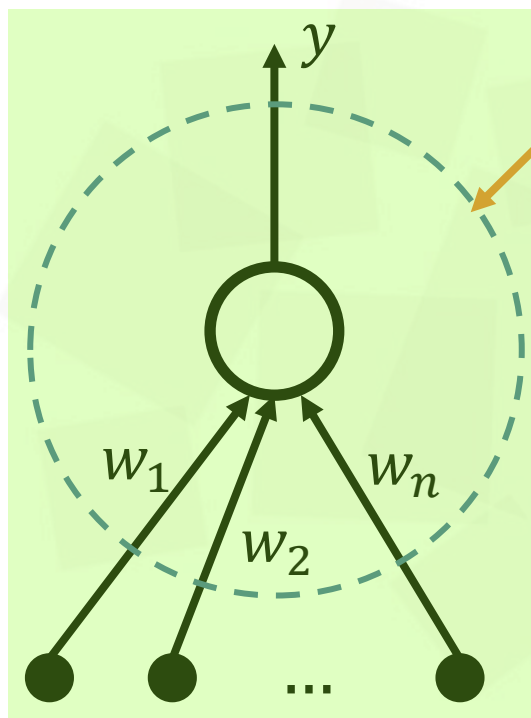
- 模式匹配:³

$$net = w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n + b^4$$

$$y = \text{sigmoid}(net)^5$$

什么是神经网络¹

图示表示²



神经元⁵

模式⁶

$$net = w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n + b$$
⁷

$$y = \text{sigmoid}(net)$$
⁸

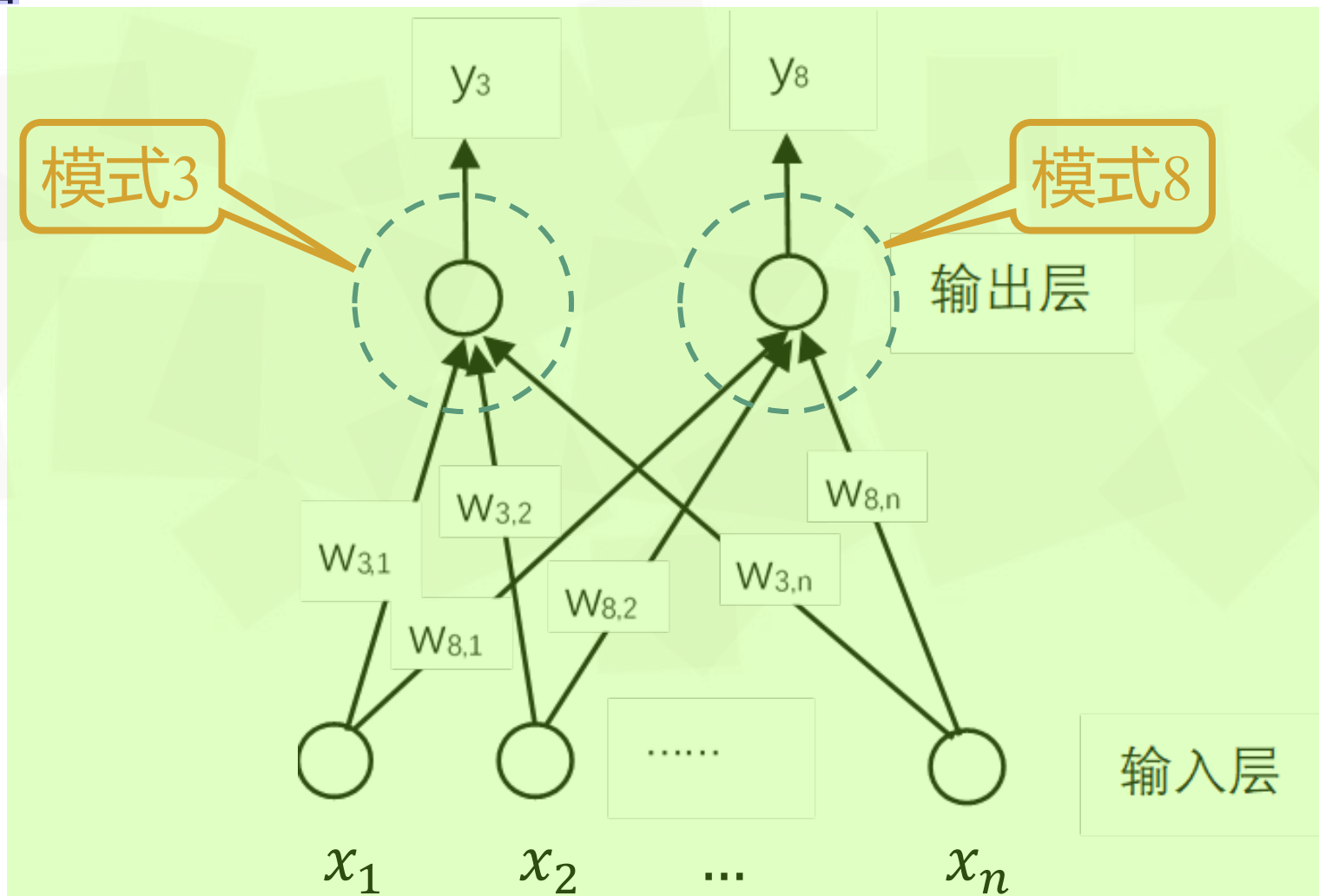
x_1 x_2 ... x_n

⁴

什么是神经网络¹



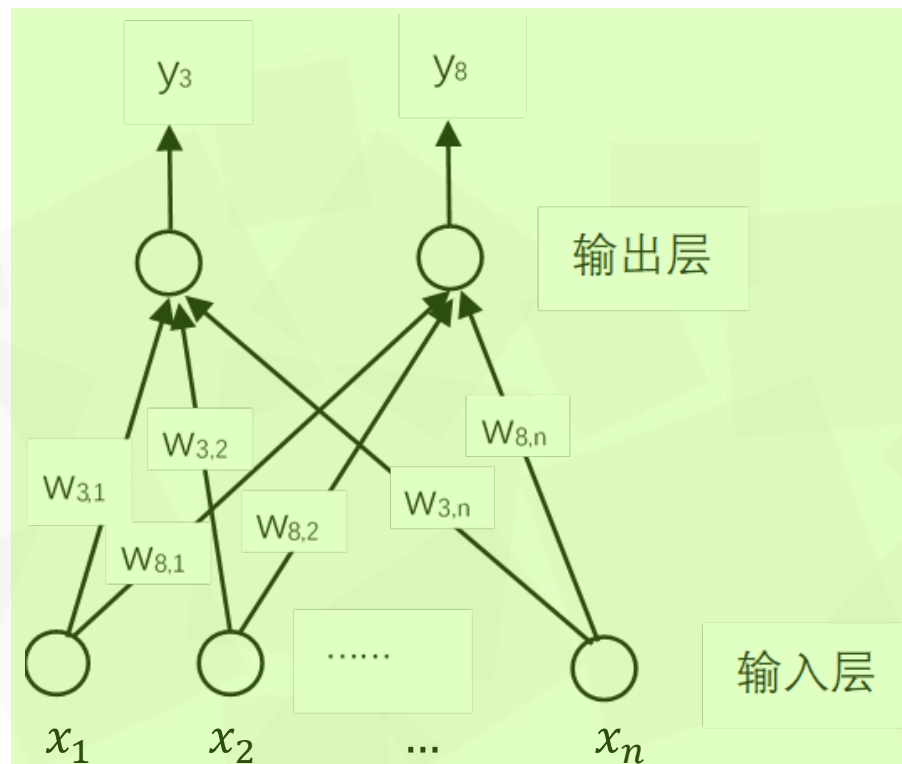
神经网络²



什么是神经网络¹



神经网络²



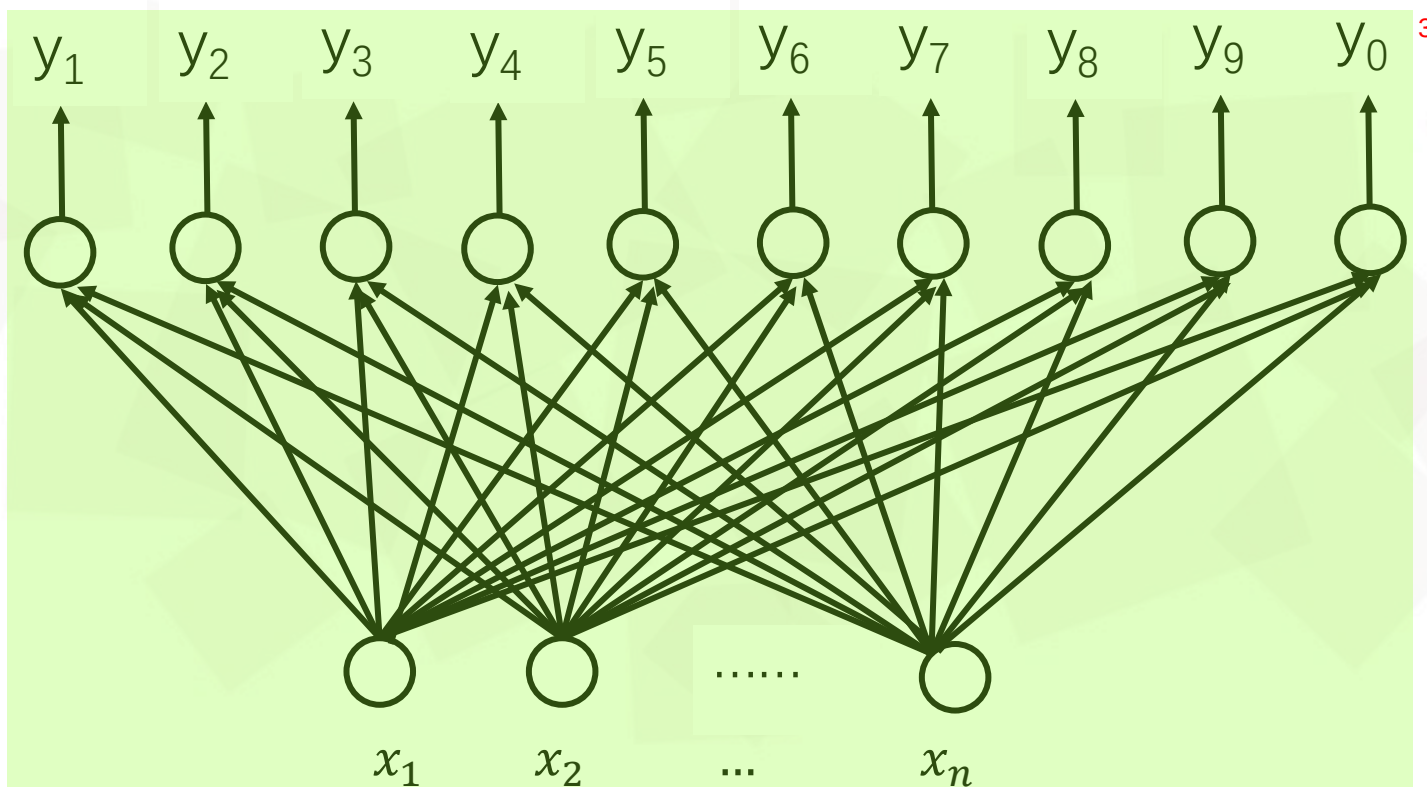
$$y_3 = \text{sigmoid}(w_{3,1} \cdot x_1 + w_{3,2} \cdot x_2 + \cdots + w_{3,n} \cdot x_n + b_3)$$
⁴

$$y_8 = \text{sigmoid}(w_{8,1} \cdot x_1 + w_{8,2} \cdot x_2 + \cdots + w_{8,n} \cdot x_n + b_8)$$
⁵

什么是神经网络¹

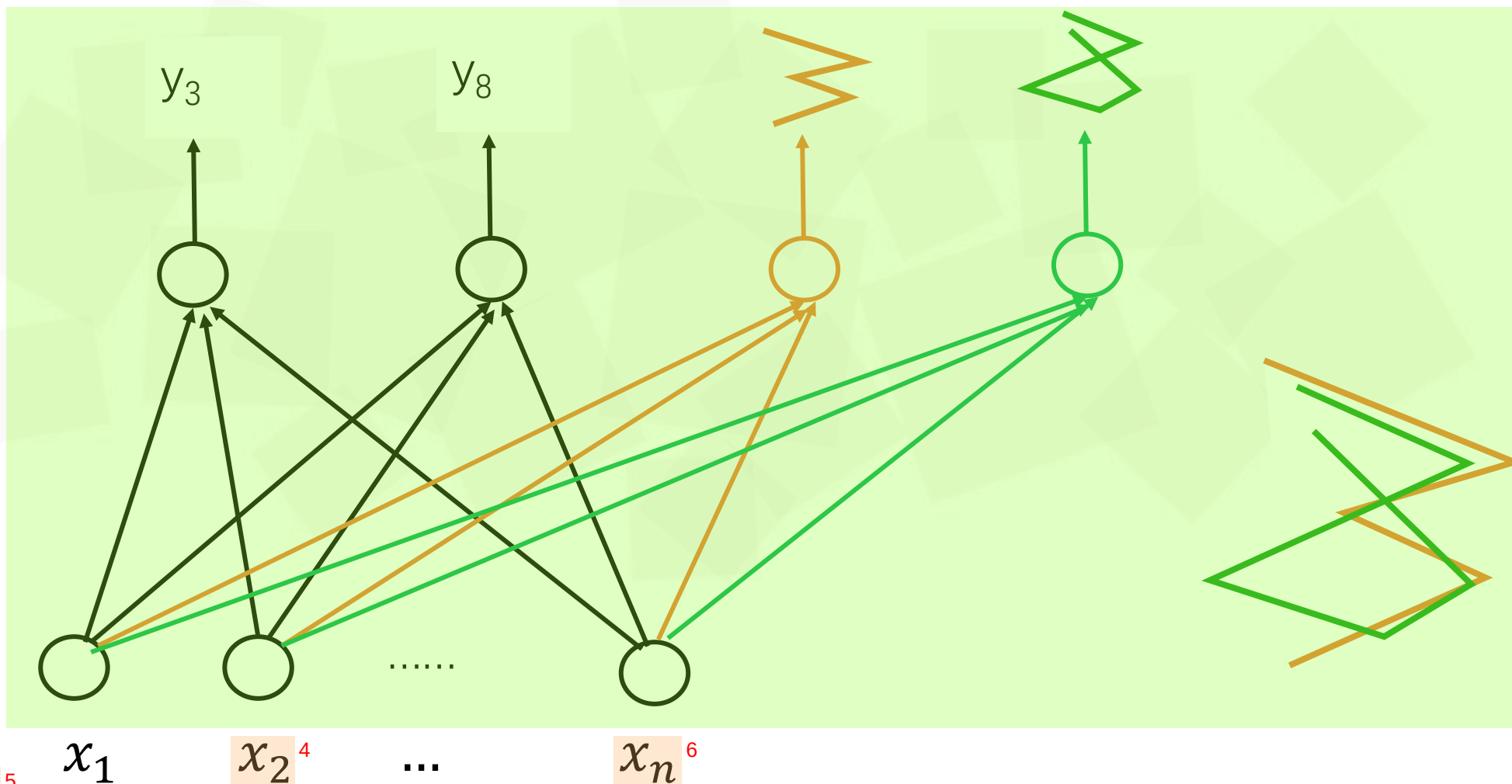


数字识别神经网络²



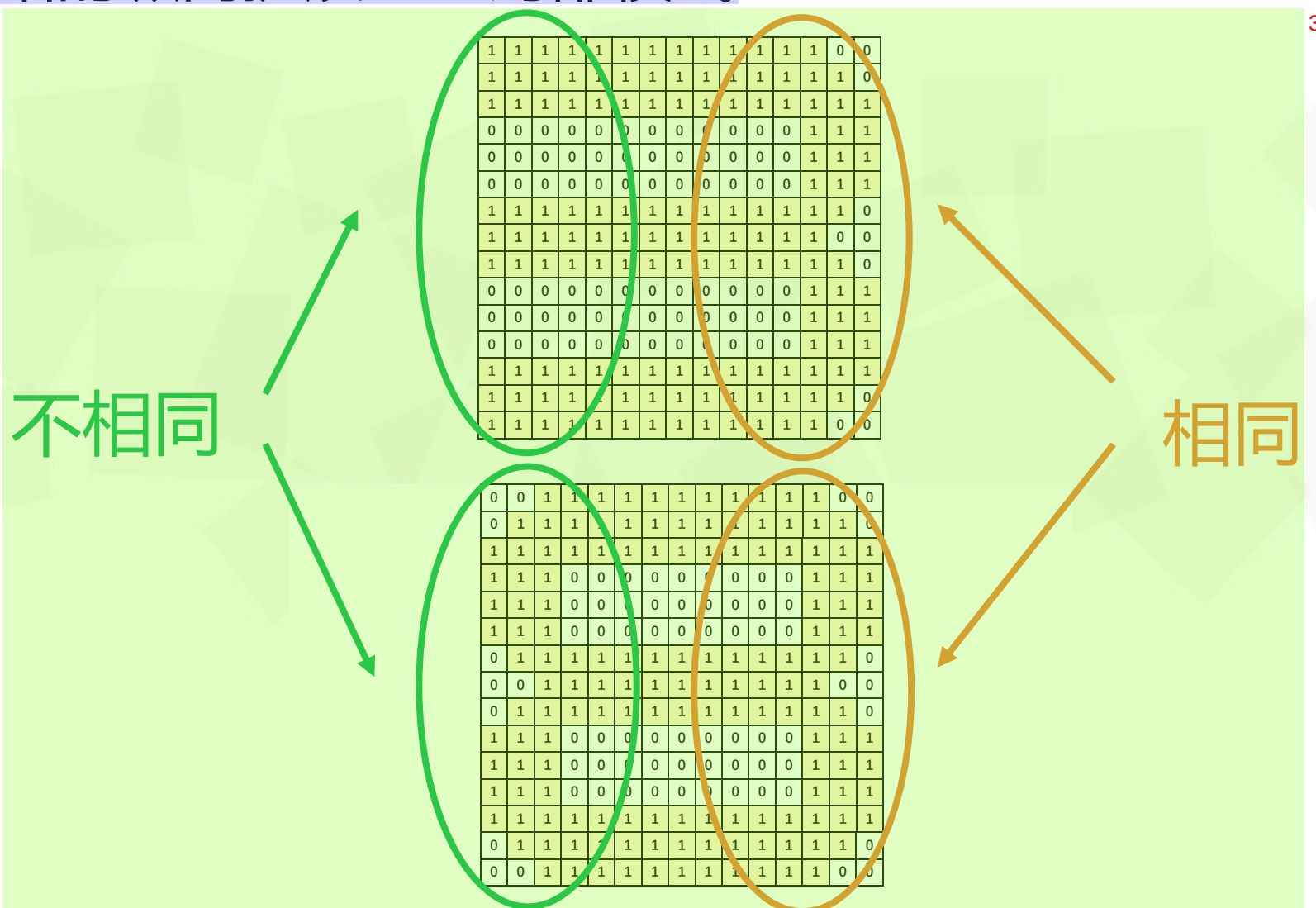
$$y_i = \text{sigmoid}(w_{i,1} \cdot x_1 + w_{i,2} \cdot x_2 + \dots + w_{i,n} \cdot x_n + b_i)^4$$

神经网络的横向扩展——增加模式²



什么是神经网络¹

神经网络的纵向扩展——局部模式²



什么是神经网络¹



让神经网络更深——模式组合²

3
的
右
部

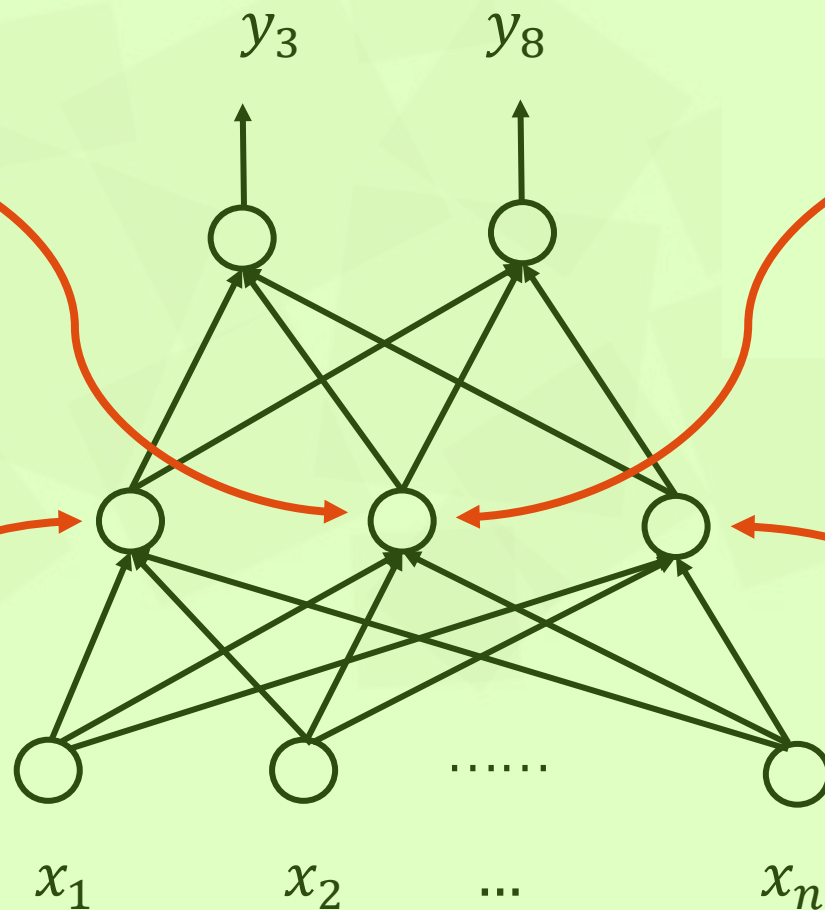
⁴

1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
1	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
0	0	0	0	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0

3
的
左
部

⁷
⁹

1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1



8
的
右
部

⁶
⁸

1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	1	1	1
1	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0
1	1	1	1	1	1	1	0
0	0	0	0	0	1	1	1
0	0	0	0	0	1	1	1
0	0	0	0	0	1	1	1
1	1	1	1	1	1	1	1
1	1	1	1	1	1	1	0
1	1	1	1	1	1	0	0

8
的
左
部

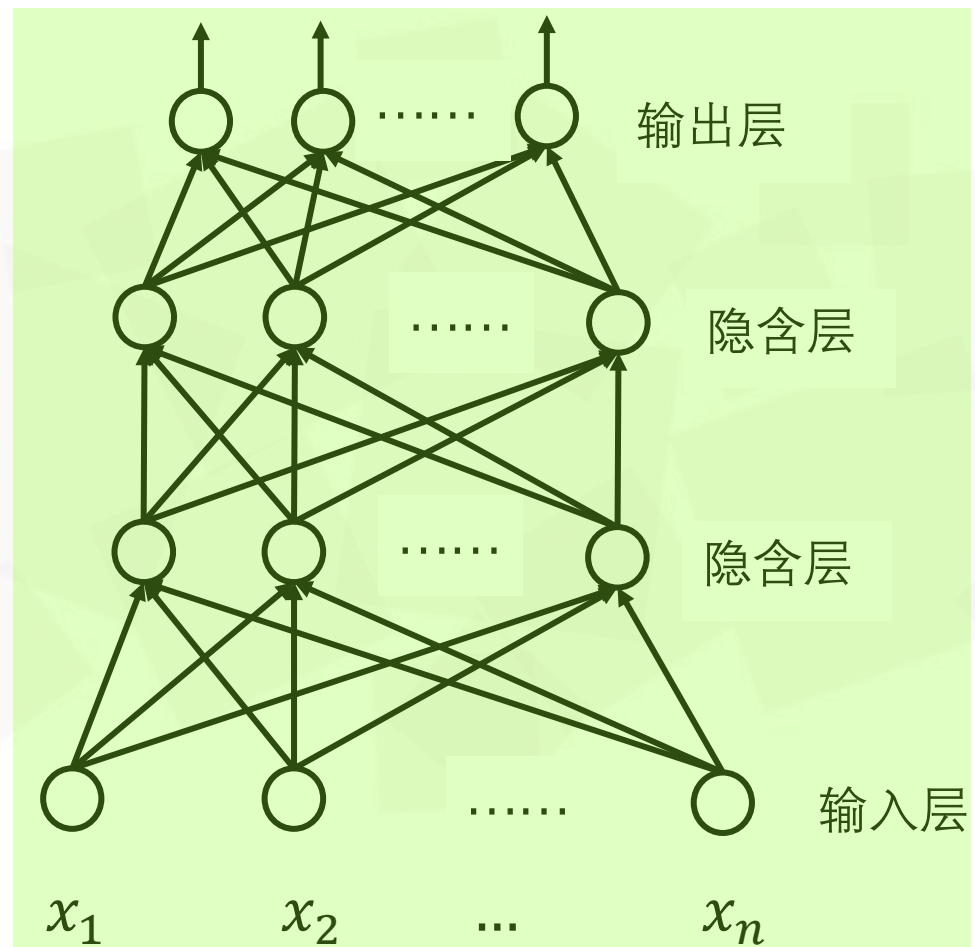
¹¹

0	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1
1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0
0	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1
0	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	0	0	0	0	0
1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1
0	0	1	1	1	1	1	1

什么是神经网络¹



多层神经网络²



什么是神经网络¹



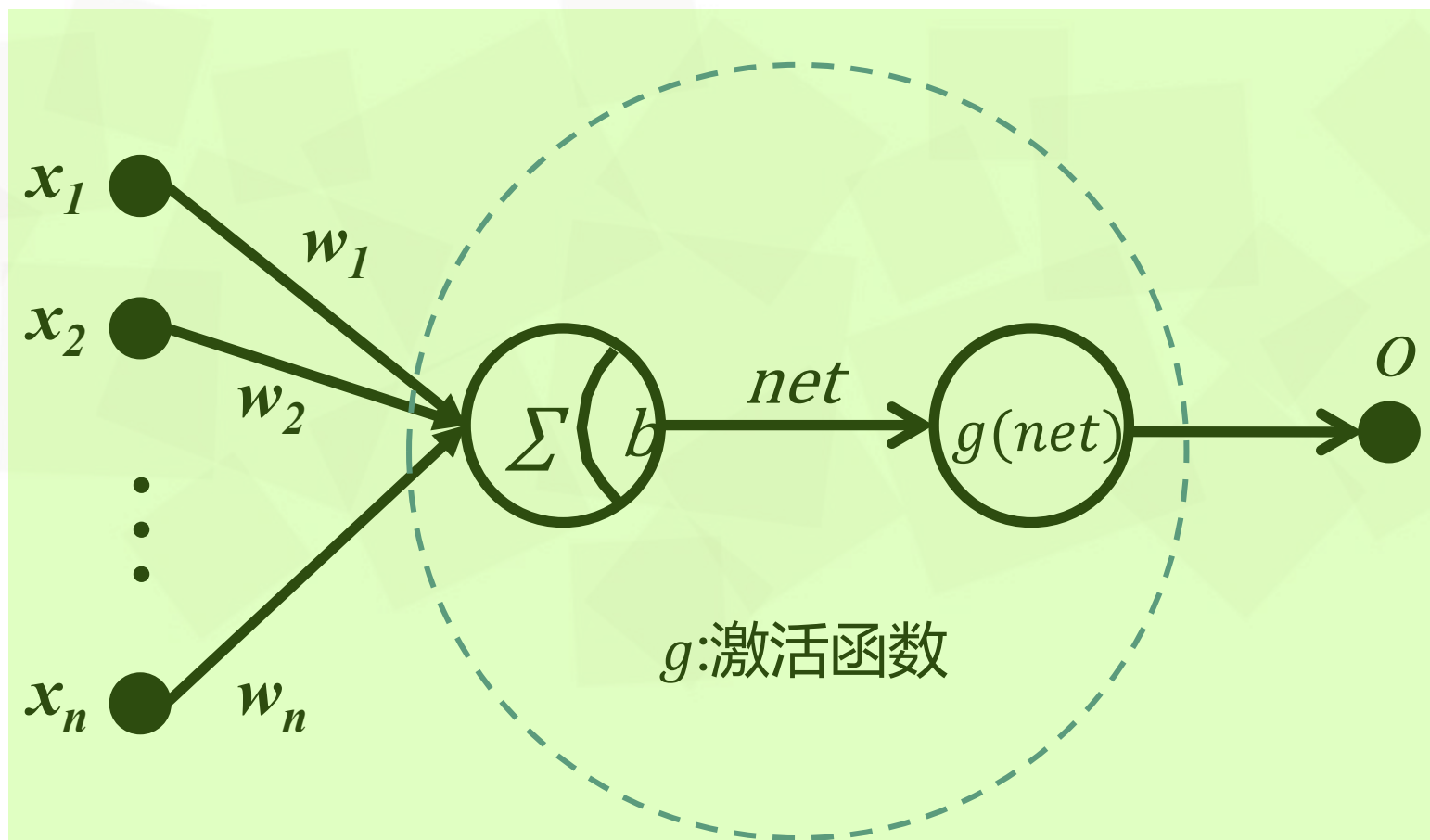
如何获得模式?²

- 模式通过神经元的连接权重表示³
- 通过训练样本，自动学习权重，也就是模式⁴
- 不是人工设计!⁵
- 学习到的模式是一种隐含表达，并不像我们举例的这样清晰⁶

什么是神经网络¹



神经元与神经网络²



什么是神经网络¹

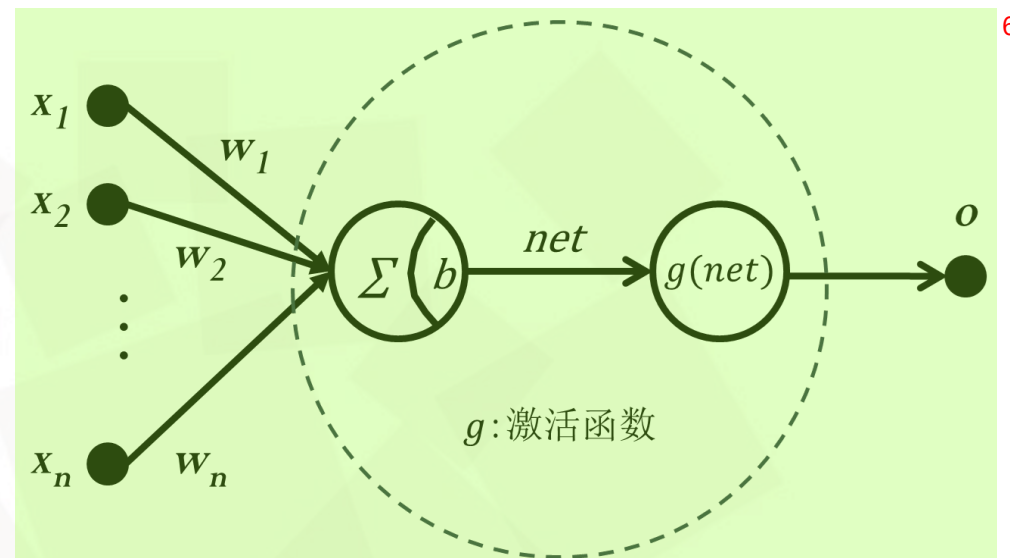
神经元²

$$\begin{aligned}
 net &= w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n + b \\
 &= \sum_{i=1}^n w_i \cdot x_i + b
 \end{aligned}$$

$$o = g(net)$$

$$\text{令: } x_0 = 1, \quad w_0 = b$$

$$\begin{aligned}
 \text{则: } net &= w_0 \cdot x_0 + w_1 \cdot x_1 + w_2 \cdot x_2 + \cdots + w_n \cdot x_n \\
 &= \sum_{i=0}^n w_i \cdot x_i
 \end{aligned}$$



什么是神经网络¹



神经元的向量表示²

- 输入: $x = [x_0, x_1, \dots, x_n]$ ³
- 权重: $w = [w_0, w_1, \dots, w_n]$ ⁴

$$net = w \cdot x$$
⁵

$$o = g(net)$$
⁶

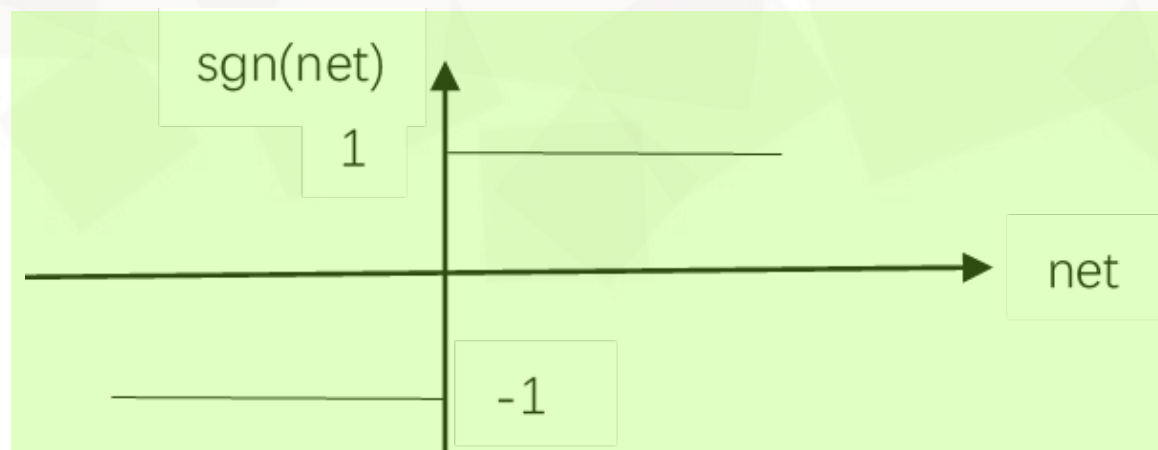
什么是神经网络¹



激活函数²

• 符号函数³

$$g(net) = \text{sgn}(net) = \begin{cases} 1, & \text{当 } net \geq 0 \\ -1, & \text{当 } net < 0 \end{cases} \quad 4$$



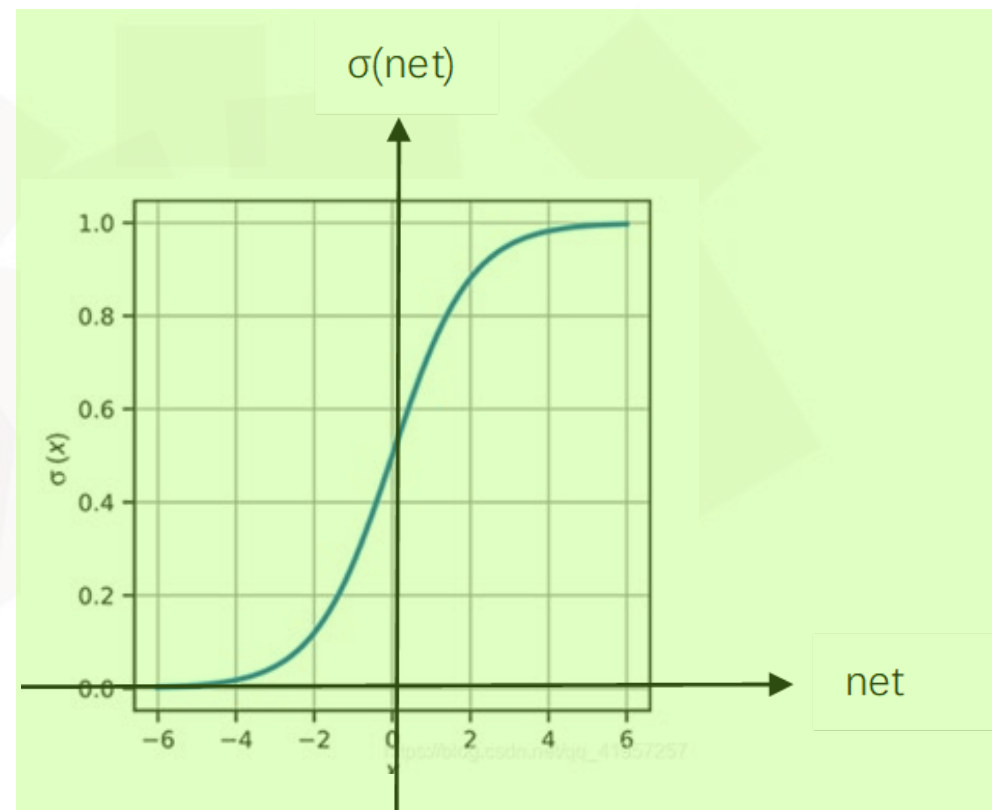
什么是神经网络¹



激活函数²

• Sigmoid函数³

$$g(net) = \sigma(net) = \frac{1}{1 + e^{-net}} \quad 4$$



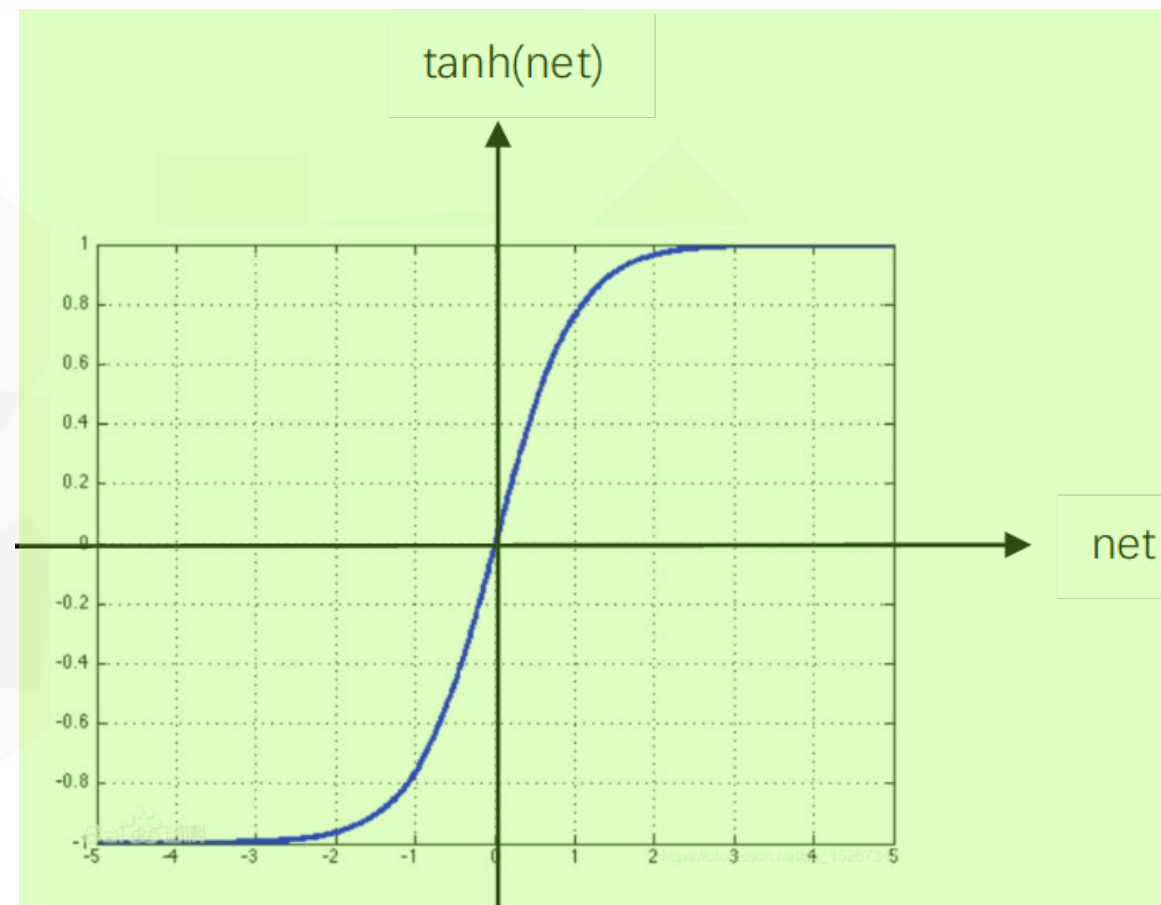
什么是神经网络¹



激活函数²

• 双曲正切函数³

$$g(net) = \tanh(net) = \frac{e^{net} - e^{-net}}{e^{net} + e^{-net}} \quad 4$$



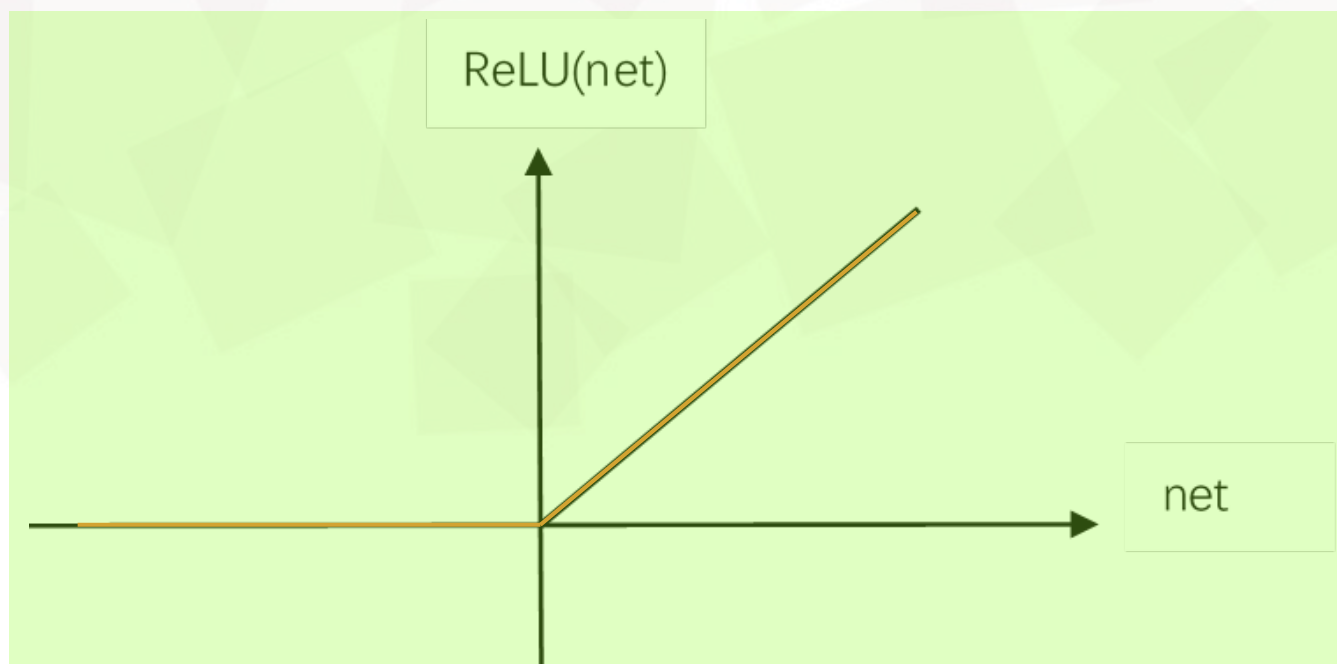
什么是神经网络¹



激活函数²

- 线性整流函数³

$$g(net) = \text{ReLU}(net) = \max(0, net)$$
⁴



什么是神经网络¹



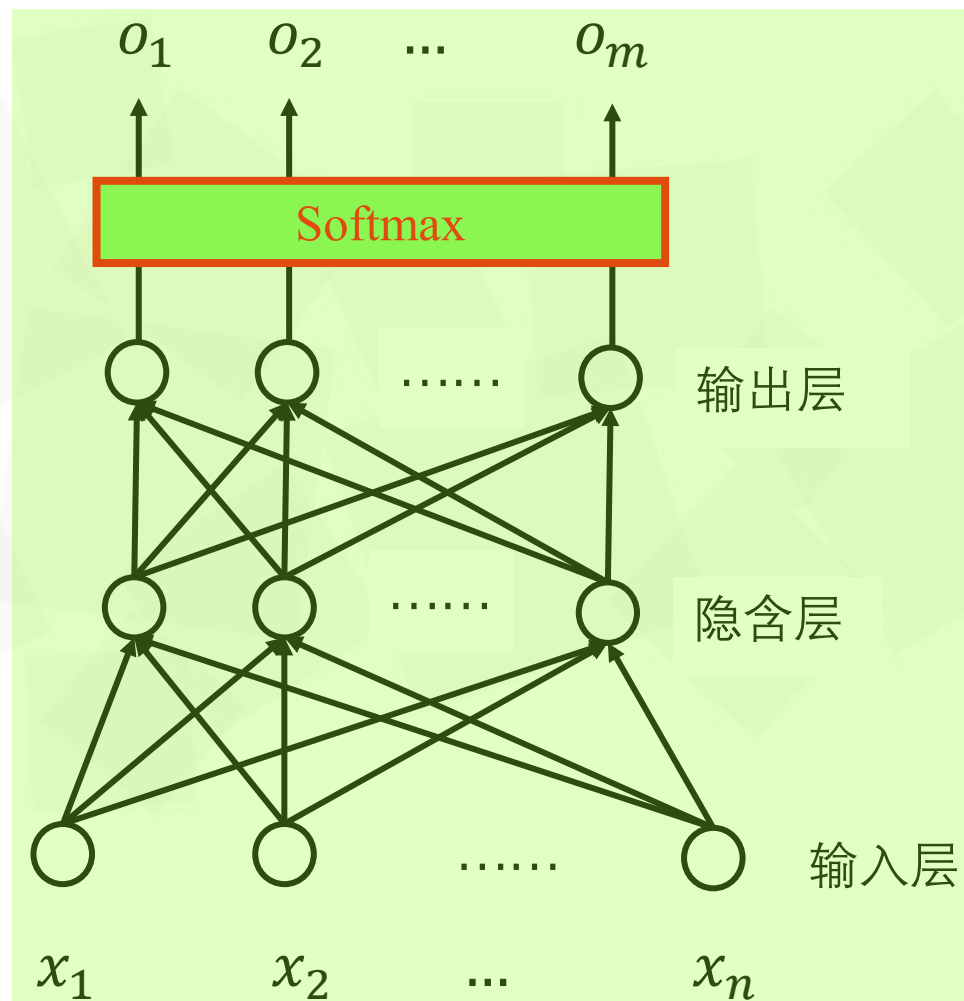
激活函数²

• Softmax函数³

$$o_k = \frac{e^{net_k}}{\sum_{i=1}^m e^{net_i}}$$

$$0 \leq o_k \leq 1$$

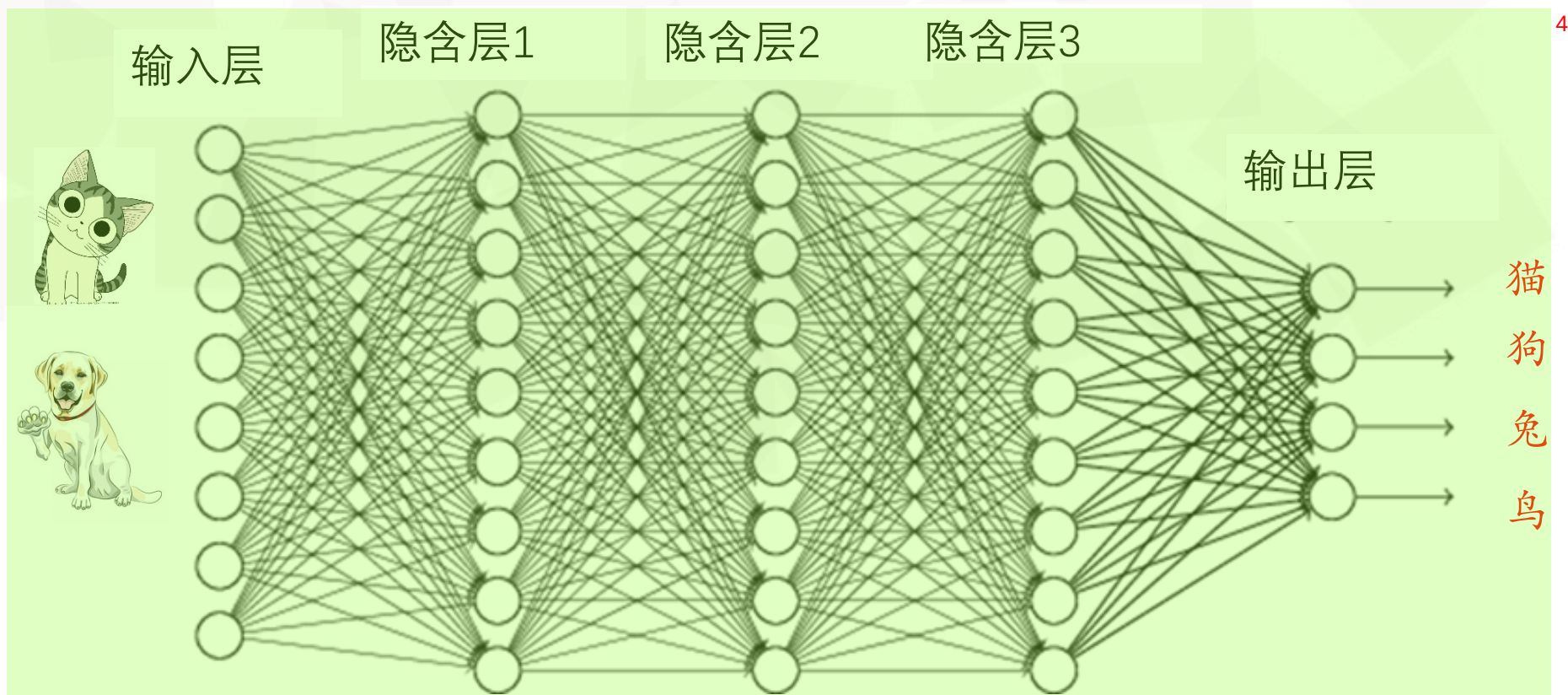
$$\sum_{k=1}^m o_k = 1$$



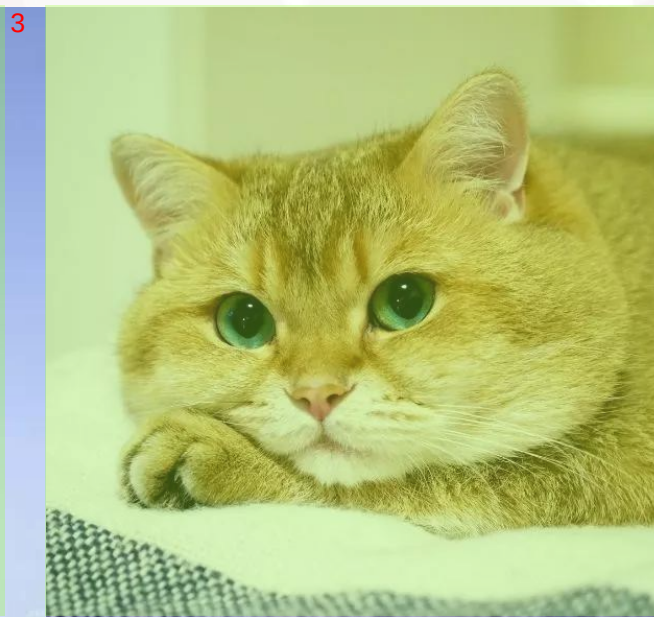
什么是神经网络¹

全连接网络²

- 术语：全连接网络，前馈网络，多层感知机，全连接层，稠密层³



如何认识小动物²



建立数据集²

- 收集各种动物的照片³

- 不同姿势⁴

- 不同角度⁵

- 不同大小⁶

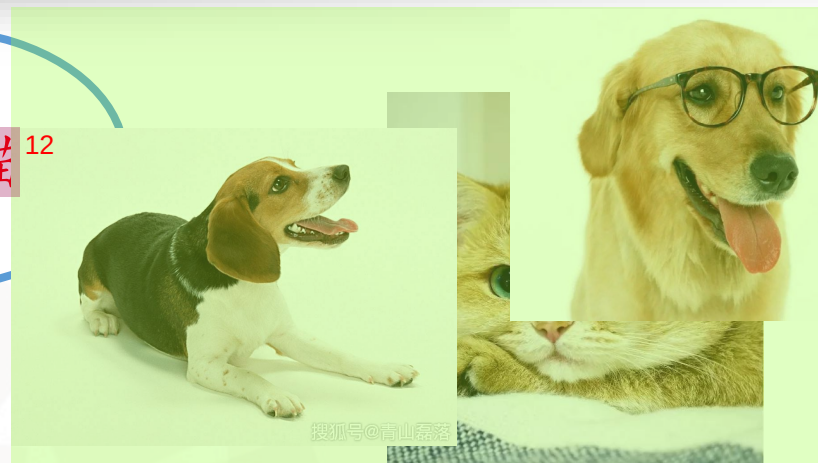
- 数据标注⁷

- 每张照片标注上动物的名称⁸

- 训练集与测试集⁹

- 样本¹⁰

这个是猫¹²



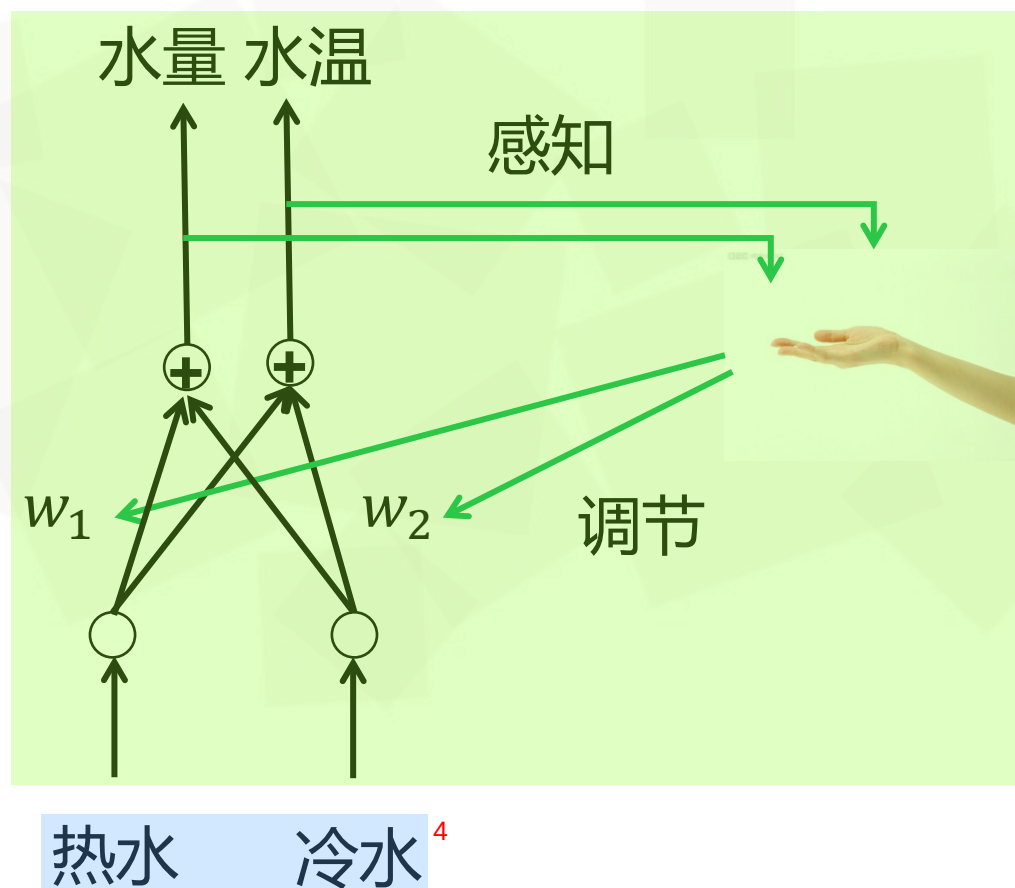
这个是狗!¹⁵



如何训练²



4

淋浴器示意图²

如何评价调节效果 — 损失函数²

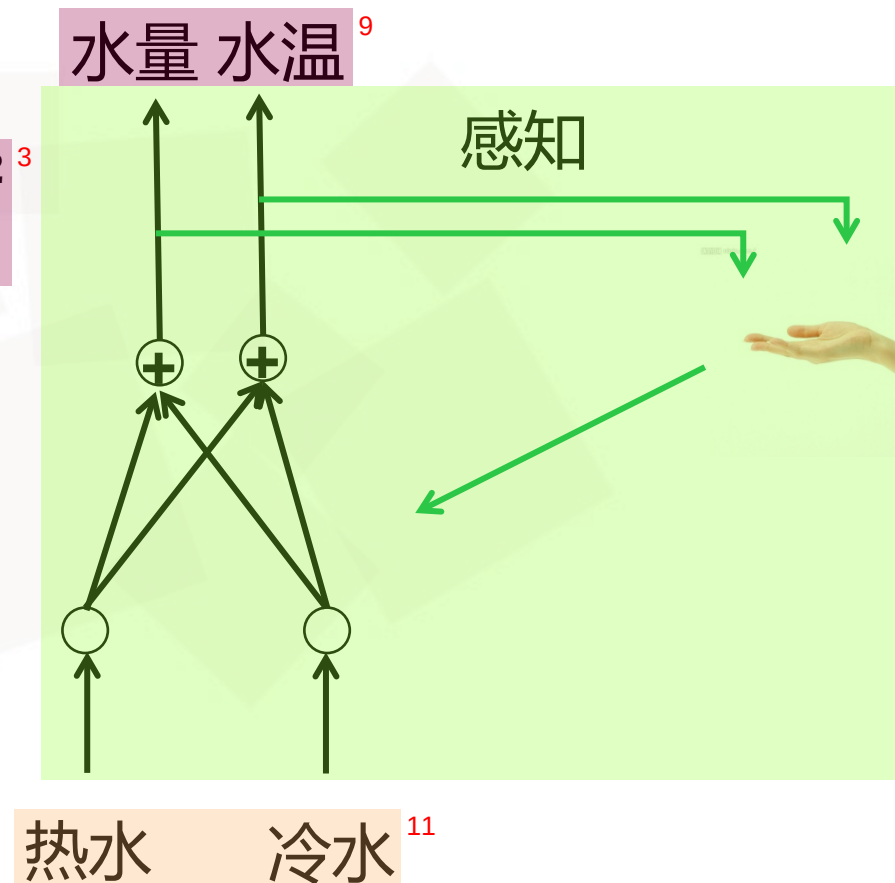
$$E(\text{阀门}) = (t_{\text{水温}} - o_{\text{水温}})^2 + (t_{\text{水量}} - o_{\text{水量}})^2 \quad 3$$

$t_{\text{水温}}$: 希望水温⁴

$t_{\text{水量}}$: 希望水量⁵

$o_{\text{水温}}$: 实际水温⁶

$o_{\text{水量}}$: 实际水量⁷



损失函数 — 误差平方和²

- 对样本d的误差³

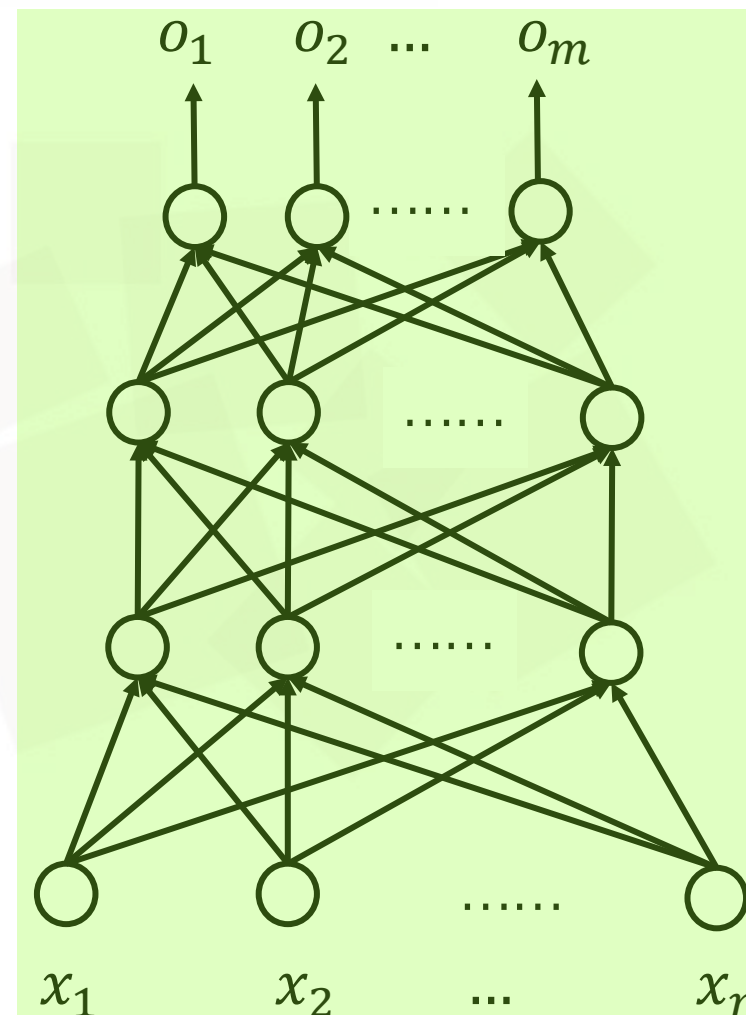
$$E_d(w) = \frac{1}{2} \sum_{k=1}^m (t_{kd} - o_{kd})^2$$

- 对所有样本的误差⁵

$$\begin{aligned} E(w) &= \frac{1}{2} \sum_{d=1}^n E_d(w) \\ &= \frac{1}{2} \sum_{d=1}^n \sum_{k=1}^m (t_{kd} - o_{kd})^2 \end{aligned}$$

- t_{kd} : 对应样本d的第k个希望输出⁷

- o_{kd} : 对应样本d的第k个实际输出⁸

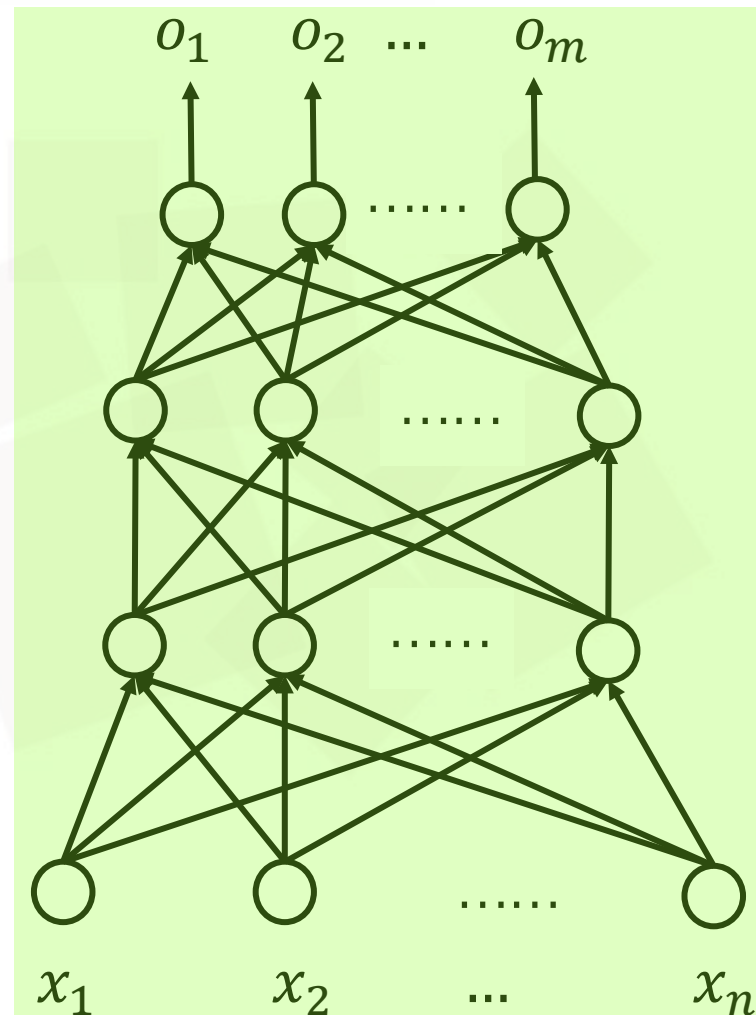


如何训练²

- 损失函数最小化问题³

$$E_d(w) = \frac{1}{2} \sum_{k=1}^m (t_{kd} - o_{kd})^2$$
⁴

$$\begin{aligned} E(w) &= \frac{1}{2} \sum_{d=1}^n E_d(w) \\ &= \frac{1}{2} \sum_{d=1}^n \sum_{k=1}^m (t_{kd} - o_{kd})^2 \end{aligned}$$
⁵



梯度下降法²

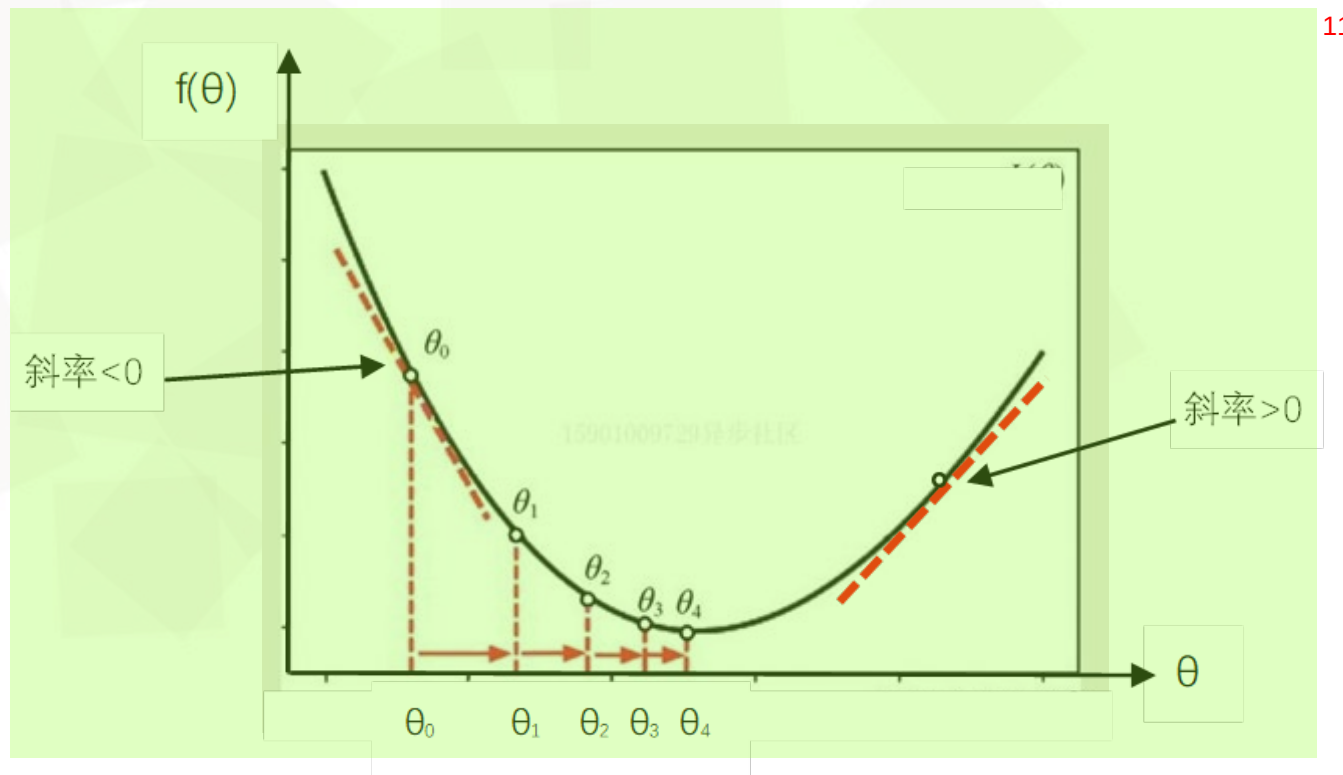
- 求 $f(\theta)$ 的最小值³
- 迭代逼近⁴

$$\theta_{i+1} = \theta_i + \Delta \theta_i$$
⁵

- 两个问题:⁶

- 修改量的大小⁷
- 修改的方向⁸

$$\Delta \theta_i = - \frac{df}{d\theta}$$
⁹

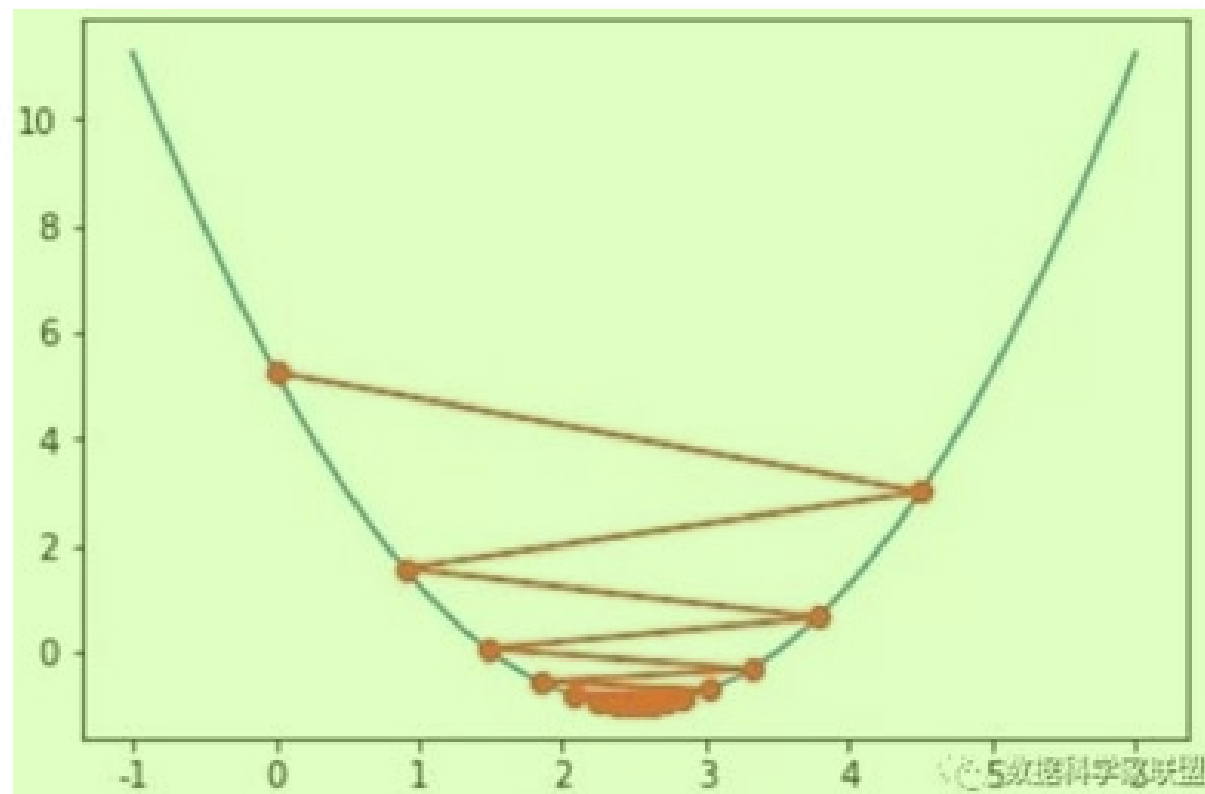


梯度下降法²

- 震荡问题³

- 步长 η ⁴

- $\Delta \theta_i = -\eta \frac{df}{d\theta}$ ⁵



梯度下降法²• 迭代公式:³

$$w_i^{new} = w_i^{old} + \Delta w_i$$

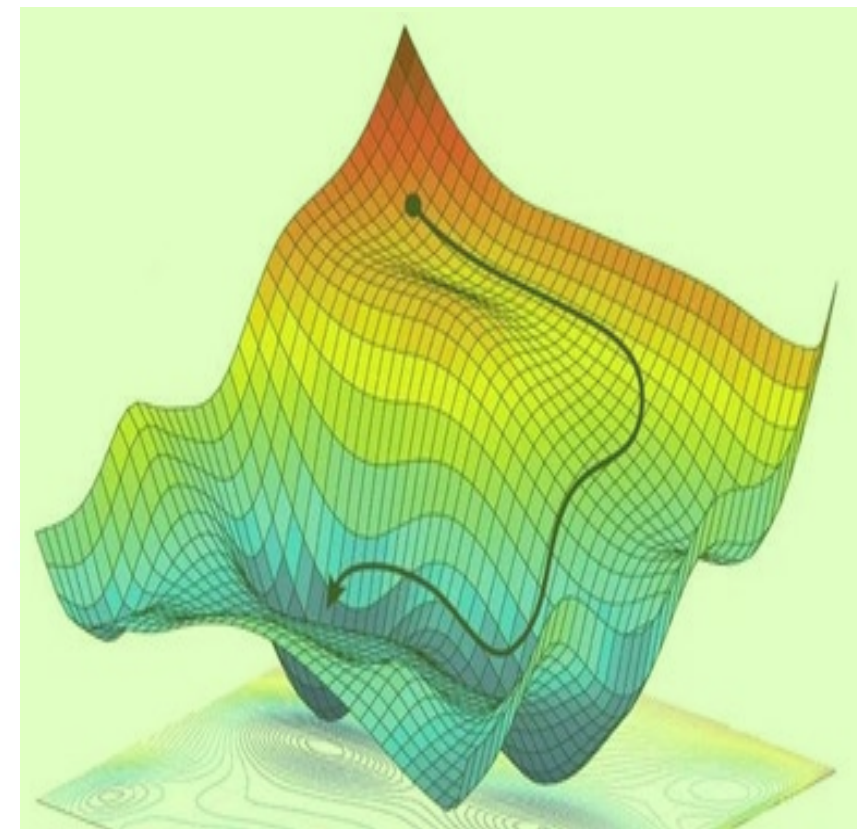
$$\Delta w_i = -\eta \frac{\partial E(w)}{\partial w_i}$$

• 梯度:⁵

$$\nabla_w E(w) = \left[\frac{\partial E(w)}{\partial w_1}, \frac{\partial E(w)}{\partial w_2}, \dots, \frac{\partial E(w)}{\partial w_n} \right]$$

$$w^{new} = w^{old} + \Delta w$$

$$\Delta w = -\eta \nabla_w E(w)$$



梯度下降算法²• 批量梯度下降算法³

- 每次处理全部样本⁴

• 随机梯度下降算法⁵

- 每次处理一个样本⁶

• 小批量梯度下降算法⁷

- 每次处理小批量的样本⁸

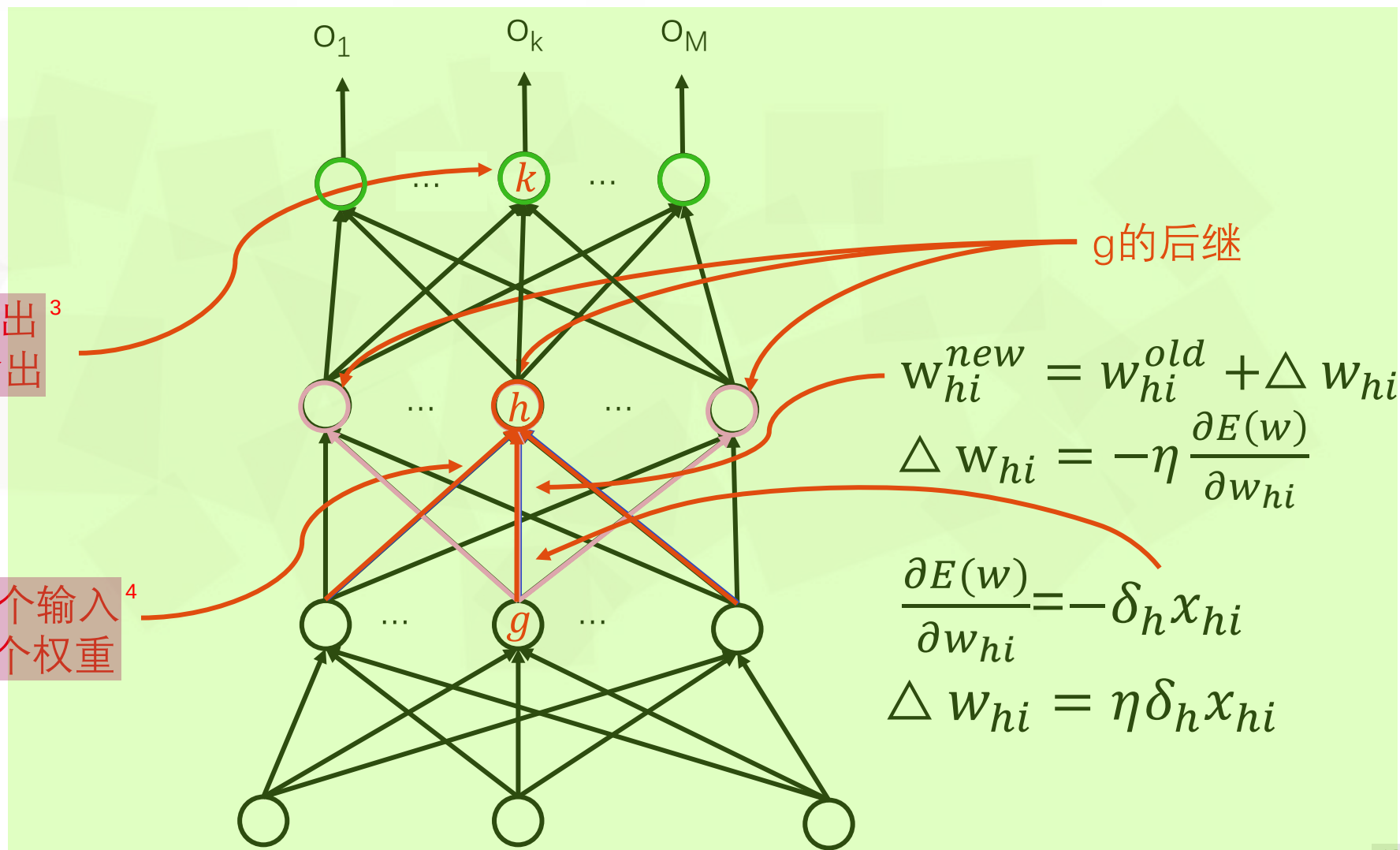
$$E(w) = \frac{1}{2} \sum_{d=1}^n \sum_{k=1}^m (t_{kd} - o_{kd})^2 \quad 9$$

$$E_d(w) = \frac{1}{2} \sum_{k=1}^m (t_{kd} - o_{kd})^2 \quad 10$$

$$E(w) = \frac{1}{2} \sum_{d=k}^{k+B} \sum_{k=1}^m (t_{kd} - o_{kd})^2 \quad 11$$

符号说明²

x_{hi} : h的第i个输入 ⁴
 w_{hi} : h的第i个权重





反向传播算法 (BP: Back Propagation)²

- 又称作误差反向传播算法³
- 给出了一种计算偏导数的方法⁴



3. 计算输出层神经元的 δ_k ²

$$\delta_k = (t_k - o_k) o_k (1 - o_k)$$

2. 得到输出层神经元k的输出¹²

5. 用 δ_k 计算隐含层2的 δ_h ⁴

$$\delta_h = o_h (1 - o_h) \sum_{k \in \text{后继}(h)} \delta_k w_{kh}$$

4. 用 δ_k 值更新权重¹⁴

$$\Delta w_{ki} = \eta \delta_k x_{ki}, w_{ki} = w_{ki} + \Delta w_{ki}$$

7. 用 δ_h 计算隐含层1的 δ_g ⁷

$$\delta_g = o_g (1 - o_g) \sum_{k \in \text{后继}(g)} \delta_k w_{kg}$$

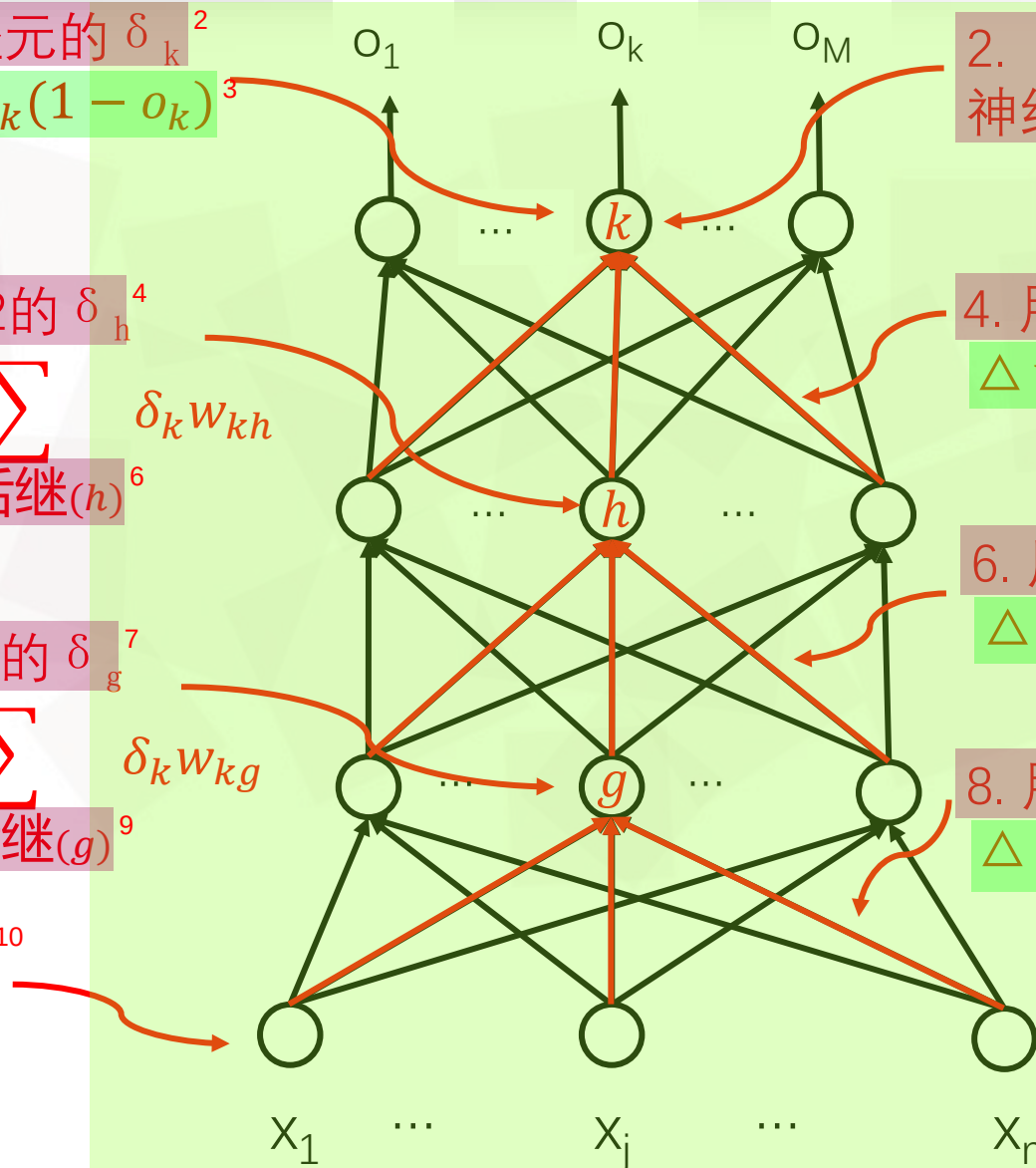
6. 用 δ_h 值更新权重¹⁶

$$\Delta w_{hi} = \eta \delta_h x_{hi}, w_{hi} = w_{hi} + \Delta w_{hi}$$

8. 用 δ_g 值更新权重¹⁸

$$\Delta w_{gi} = \eta \delta_g x_{gi}, w_{gi} = w_{gi} + \Delta w_{gi}$$

1. 输入一个样本
计算所有神经元的输出¹⁰

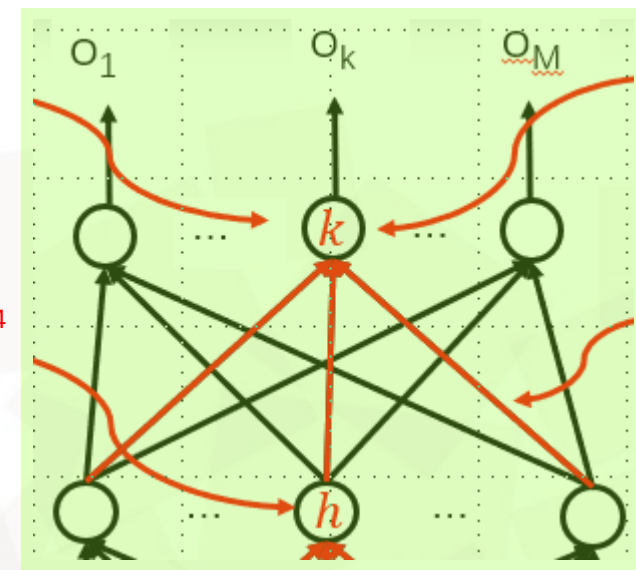




$$o_k = \text{Sigmoid}(\sum_i (w_{ki} \cdot x_{ki}) + b_k)$$
²

$$\text{Loss} = \frac{1}{2} (t_k - o_k)^2$$
³

$$\begin{aligned} \frac{\partial \text{Loss}}{\partial w_{ki}} &= -(t_k - o_k) \cdot \frac{\partial o_k}{\partial w_{ki}} \\ &= -(t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot \frac{\partial (\sum_i (w_{ki} \cdot x_{ki}) + b_k)}{\partial w_{ki}} \\ &= -(t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot x_{ki} = -\delta_k x_{ki} \end{aligned}$$
⁴



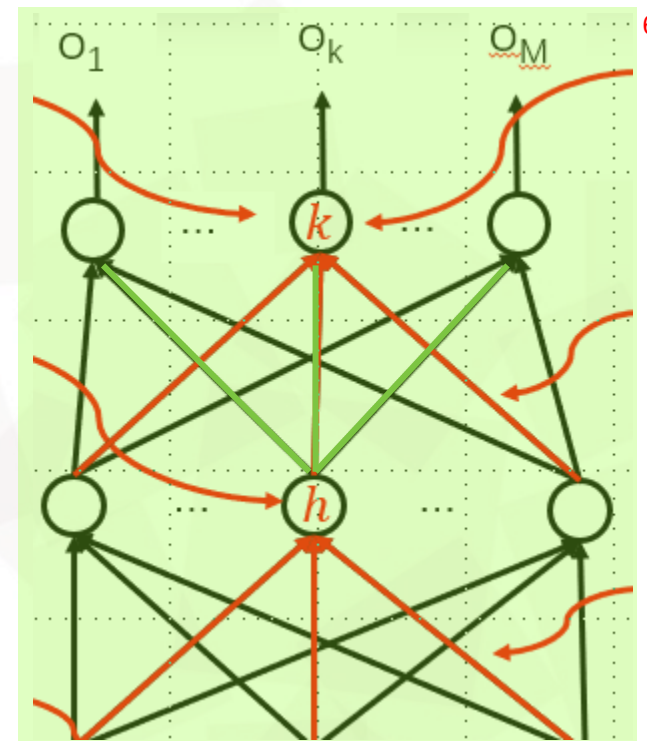


$$o_k = \text{Sigmoid} \left((w_{kh} \cdot o_h) + \sum_{h' \neq h} (w_{kh'} \cdot o_{h'}) + b_k \right) \quad 2$$

$$o_h = \text{Sigmoid} \left(\sum_i (w_{hi} \cdot x_{hi}) + b_h \right) \quad 3$$

$$\text{Loss} = \frac{1}{2} \sum_{k=1}^M (t_k - o_k)^2 \quad 4$$

$$\begin{aligned} \frac{\partial \text{Loss}}{\partial w_{hi}} &= - \sum_{k=1}^M (t_k - o_k) \cdot \frac{\partial o_k}{\partial w_{hi}} \\ &= - \sum_{k=1}^M (t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot \frac{\partial (w_{kh} \cdot o_h)}{\partial w_{hi}} \\ &= - \sum_{k=1}^M (t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot w_{kh} \cdot \frac{\partial (o_h)}{\partial w_{hi}} \end{aligned} \quad 5$$





$$\begin{aligned}
 o_h &= \text{Sigmoid} \left(\sum_i (w_{hi} \cdot x_{hi}) + b_h \right) \\
 \frac{\partial \text{Loss}}{\partial w_{hi}} &= - \sum_{k=1}^M (t_k - o_k) \cdot \frac{\partial o_k}{\partial w_{hi}} \\
 &= - \sum_{k=1}^M (t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot \frac{\partial (w_{kh} \cdot o_h)}{\partial w_{hi}} \\
 &= - \sum_{k=1}^M (t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot w_{kh} \cdot \frac{\partial (o_h)}{\partial w_{hi}} \\
 &= - \sum_{k=1}^M \underbrace{(t_k - o_k) \cdot o_k \cdot (1 - o_k) \cdot w_{kh} \cdot o_h \cdot (1 - o_h)}_{\delta_k \cdot w_{kh} \cdot o_h \cdot (1 - o_h)} \cdot x_{hi} \\
 &= - \sum_{k=1}^M \delta_k \cdot w_{kh} \cdot o_h \cdot (1 - o_h) \cdot x_{hi} = -\delta_h x_{hi}
 \end{aligned}$$



该BP算法的条件²

- 具体条件不同，算法会有差异，总体思路一样³
- 全连接网络⁴
 - 其它形式的网络⁵
- 随机梯度下降算法⁶
 - 批量、小批量梯度下降算法⁷
- 激活函数：Sigmoid函数⁸
 - 其它的激活函数⁹
- 损失函数：误差平方和¹⁰
 - 其它的损失函数¹¹

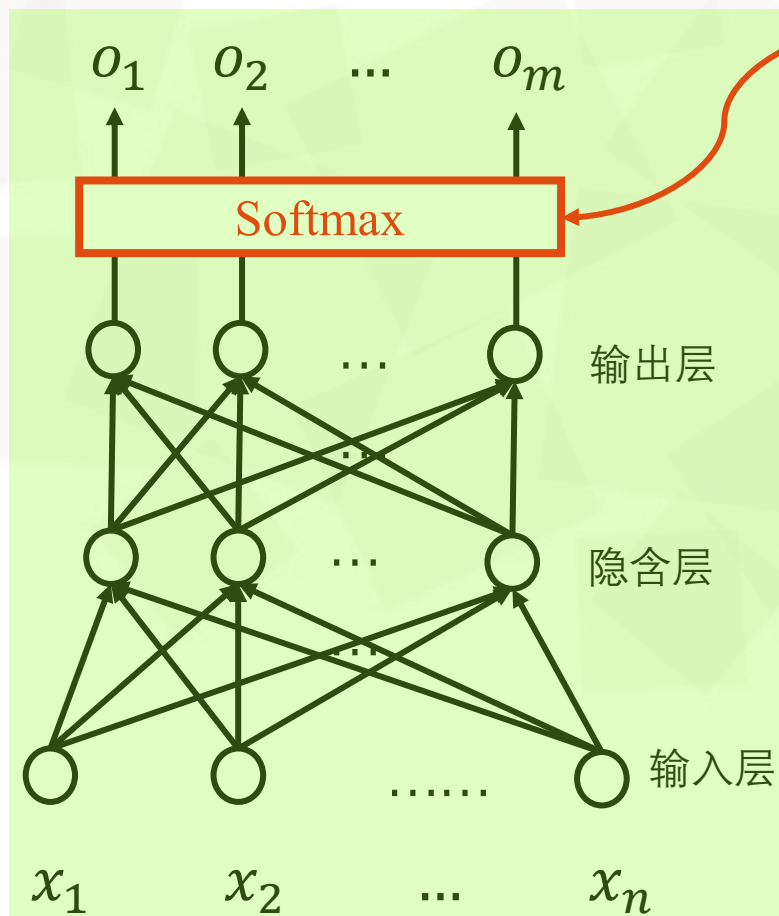
交叉熵损失函数²

$$H_d(w) = - \sum_{k=1}^M t_{kd} \log(o_{kd})$$
³

$$H(w) = \sum_{d=1}^N H_d(w) = - \sum_{d=1}^N \sum_{k=1}^M t_{kd} \log(o_{kd})$$
⁴

t_{kd} : 样本d对应的希望输出值⁵

o_{kd} : 样本d对应的实际输出值, 要求 o_{kd} 是个概率值⁶

交叉熵损失函数²

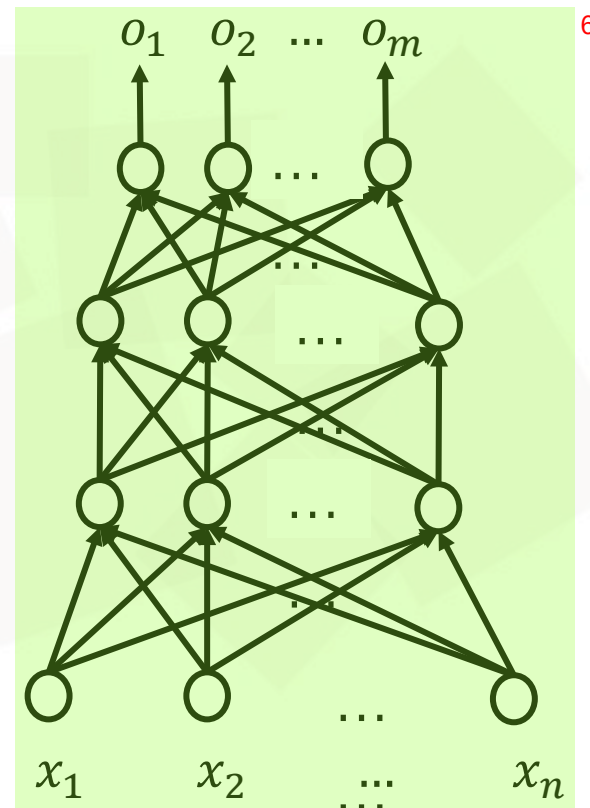
$$o_k = \frac{e^{net_k}}{\sum_{i=1}^m e^{net_i}} \quad 5$$

$$0 \leq o_k \leq 1 \quad 6$$
$$\sum_{k=1}^m o_k = 1$$

交叉熵损失函数²

- 对于分类问题³
- 对于给定输入样本 d ，只有 d 对应的希望输出为1，其他为0⁴

$$H_d(w) = -\log(o_{kd})^5$$





两种损失函数的作用²

- 误差平方和损失函数³
 - 用于输出是具体数值的问题⁴
- 交叉熵损失函数⁵
 - 用于分类问题⁶

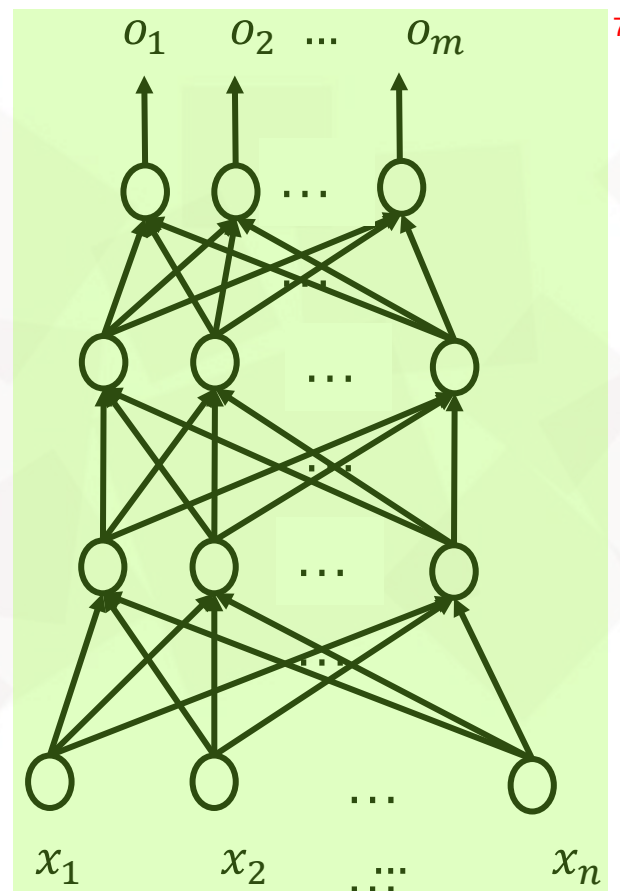


神经网络的训练流程²

- 建立数据集³
- 损失函数⁴
 - 误差平方和、交叉熵⁵
- 梯度下降法⁶
 - 批量梯度下降算法⁷
 - 随机梯度下降算法⁸
 - 小批量梯度下降算法⁹
- BP算法¹⁰
 - 随机梯度下降算法的一种实现¹¹

全连接网络的不足²

- 连接权重过多³
- 影响训练速度⁴
- 影响使用速度⁵



提取局部模式²

0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

边缘信息⁴

输入²

2	1	0	2	3
9	5	4	2	0
2	3	4	5	6
1	2	3	1	0
0	4	4	2	8

权重 (模式)

-1	0	1
-1	0	1
-1	0	1

输出 (匹配结果)

-5	0	1
-1	-2	-5
8	-1	3

$$\begin{aligned} & (-1) \times 2 + 0 \times 1 + 1 \times 0 \\ & + (-1) \times 9 + 0 \times 5 + 1 \times 4 \\ & + (-1) \times 2 + 0 \times 3 + 1 \times 4 = -5 \end{aligned}$$

3

输入²

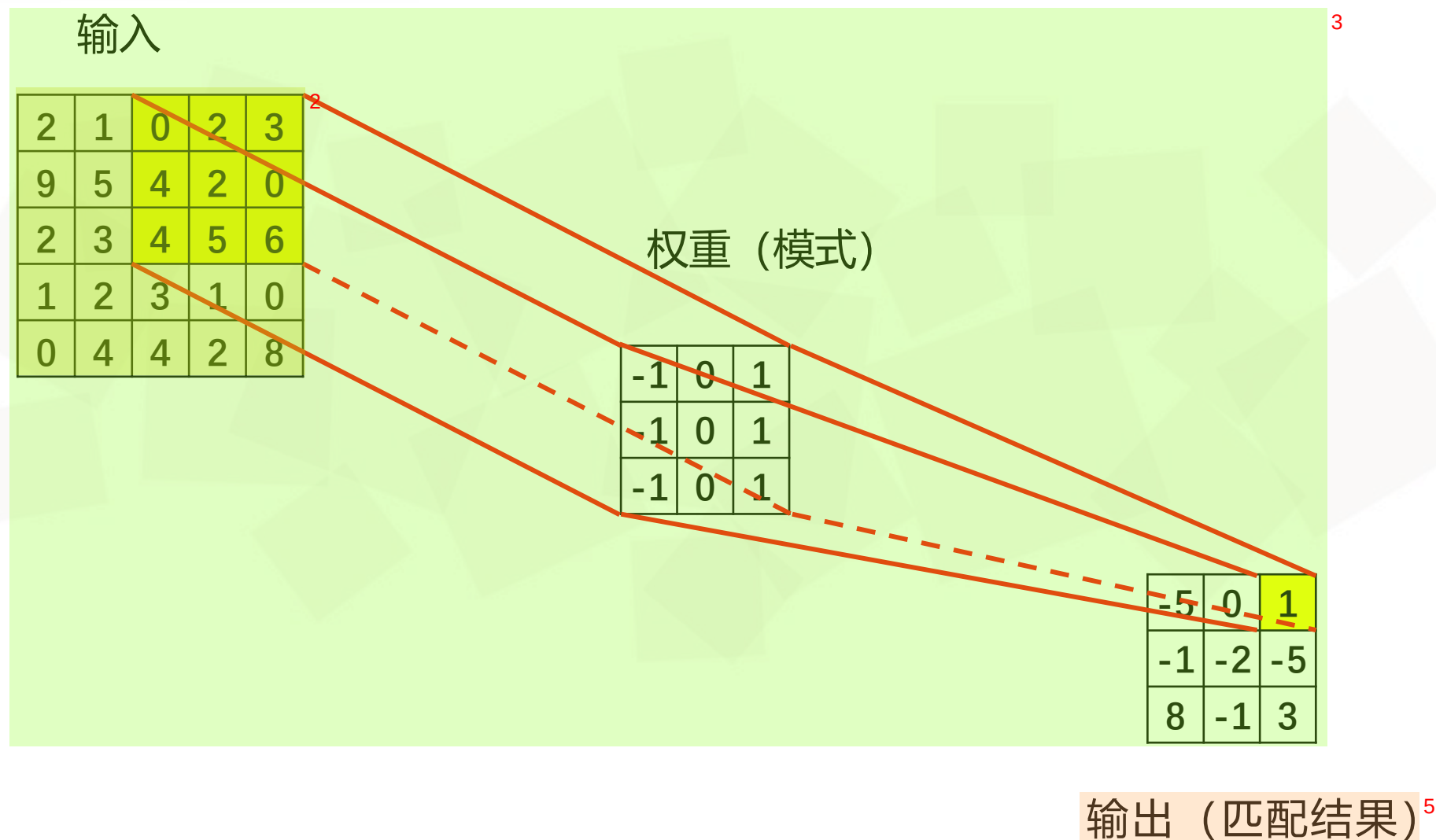
2	1	0	2	3
9	5	4	2	0
2	3	4	5	6
1	2	3	1	0
0	4	4	2	8

权重 (模式)

-1	0	1
-1	0	1
-1	0	1

输出 (匹配结果)

-5	0	1
-1	-2	-5
8	-1	3



输入²

2	1	0	2	3
9	5	4	2	0
2	3	4	5	6
1	2	3	1	0
0	4	4	2	8

权重 (模式)

-1	0	1
-1	0	1
-1	0	1

卷积核
训练获得

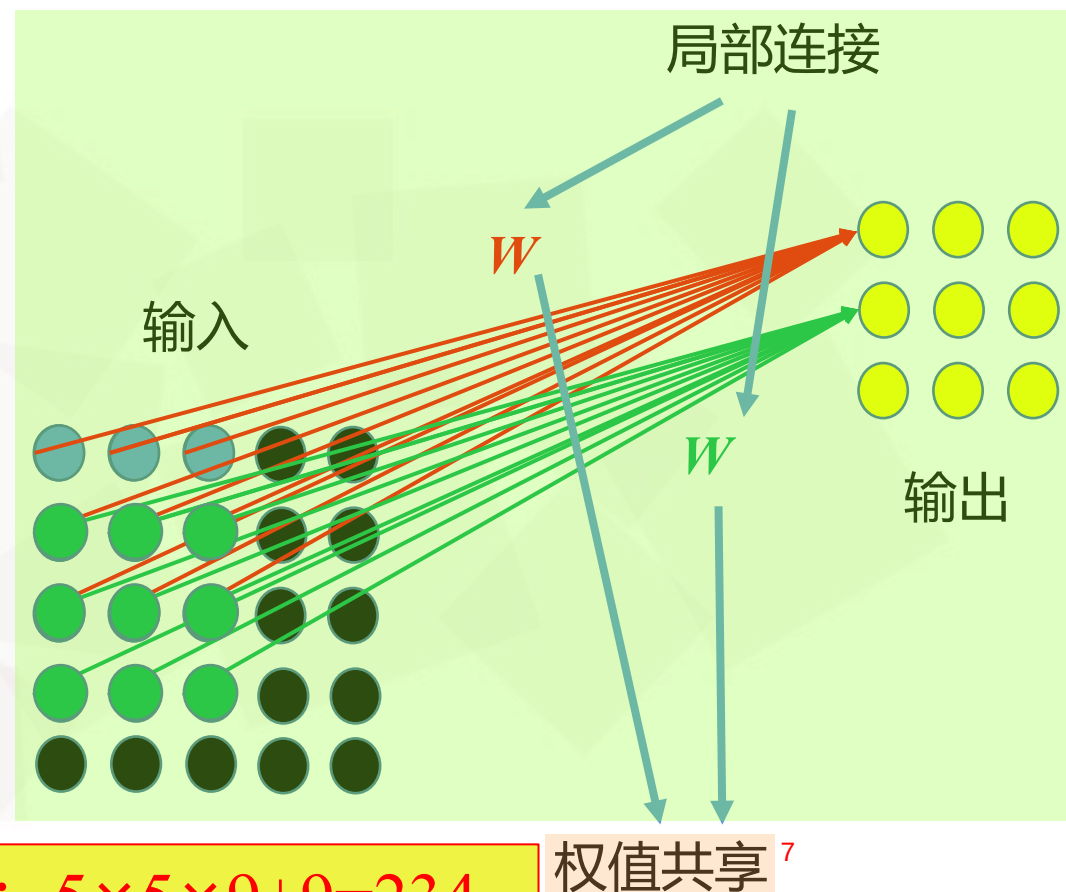
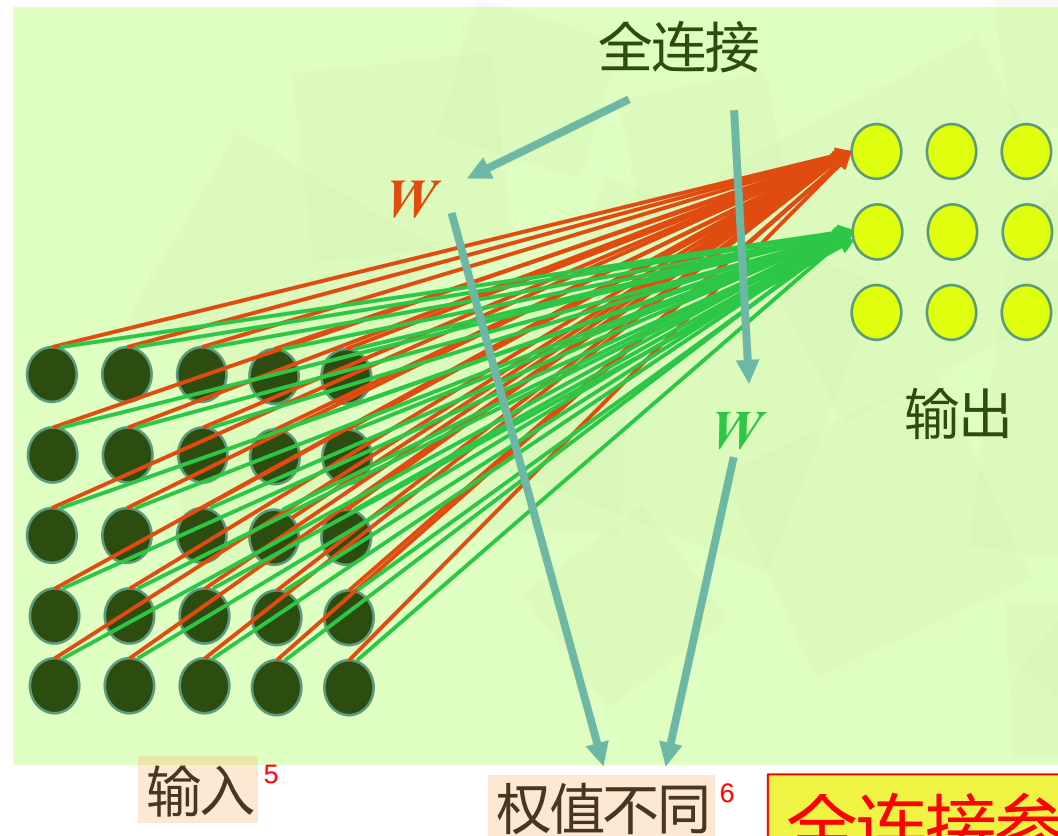
输出 (匹配结果)

5	0	1
-1	-2	-5
8	-1	3

卷积神经网络

-局部连接

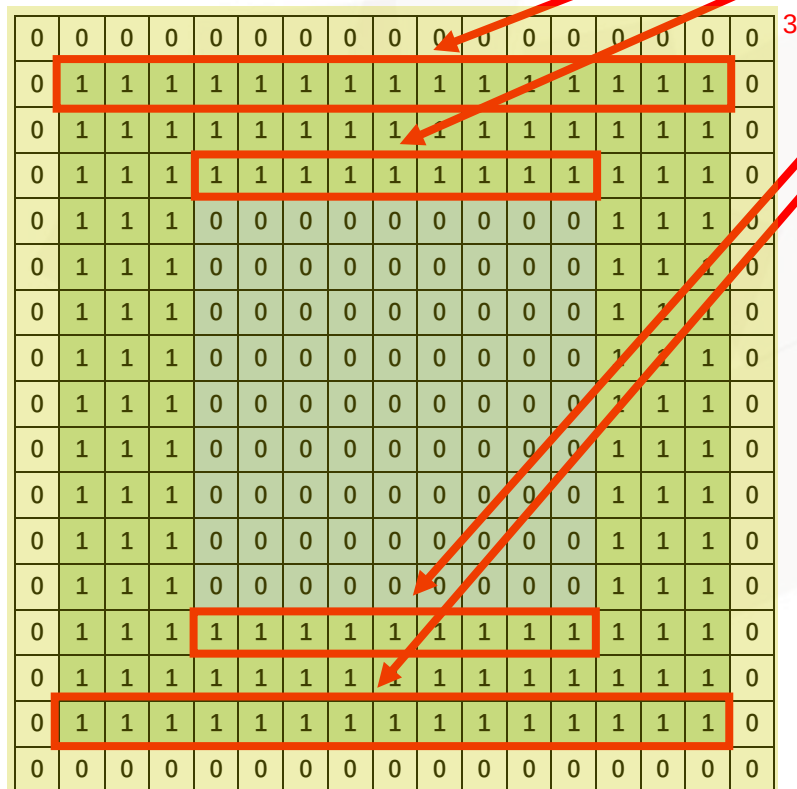
-权值共享

全连接、卷积神经网络对比²

全连接参数: $5 \times 5 \times 9 + 9 = 234$

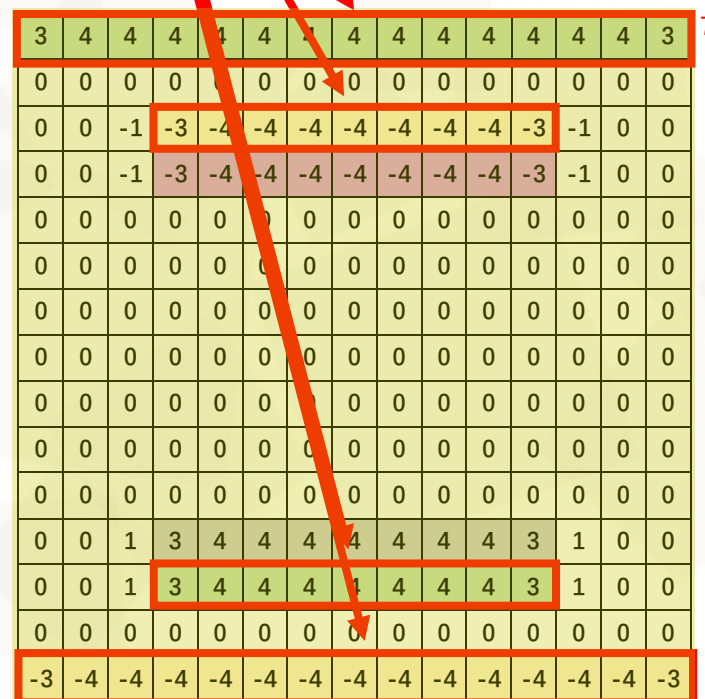
卷积核参数: $3 \times 3 + 1 = 10$

例：边缘提取²



输入 4

正边缘 6



输出⁸

-1	-2	-1
0	0	0
1	2	1

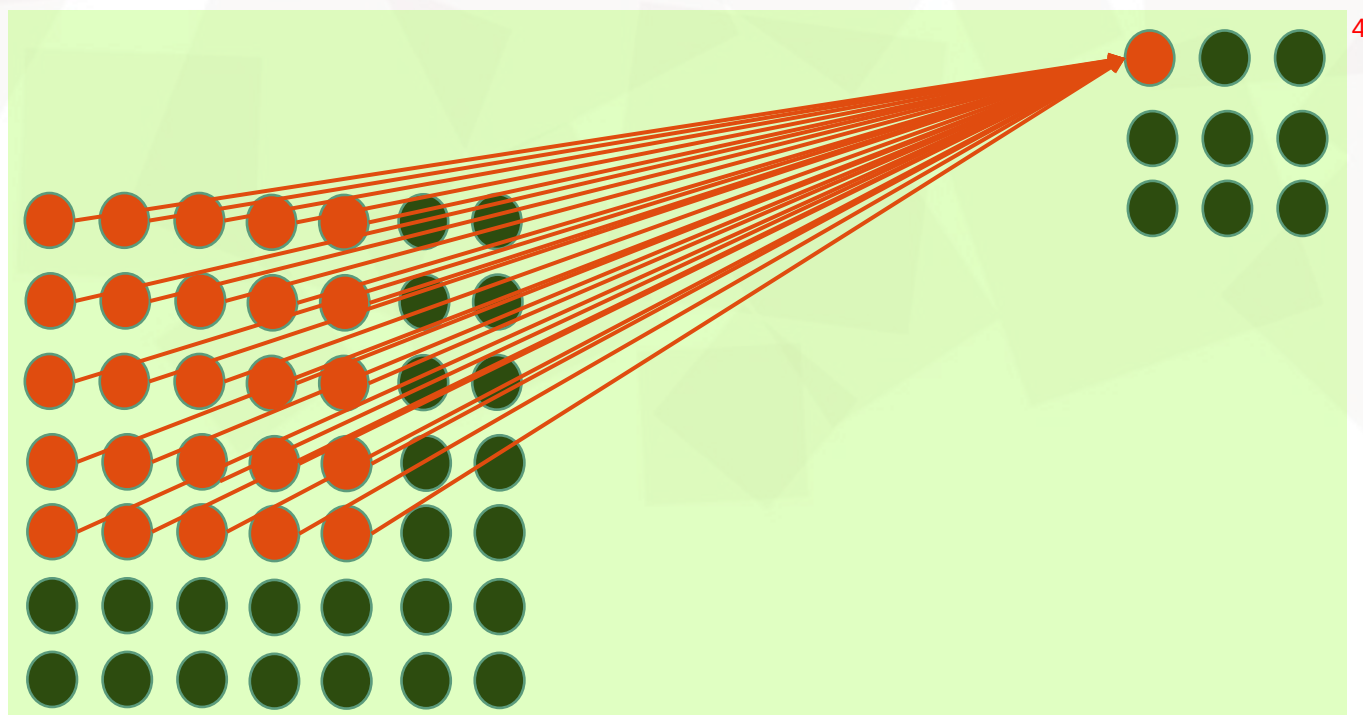
卷积核 10

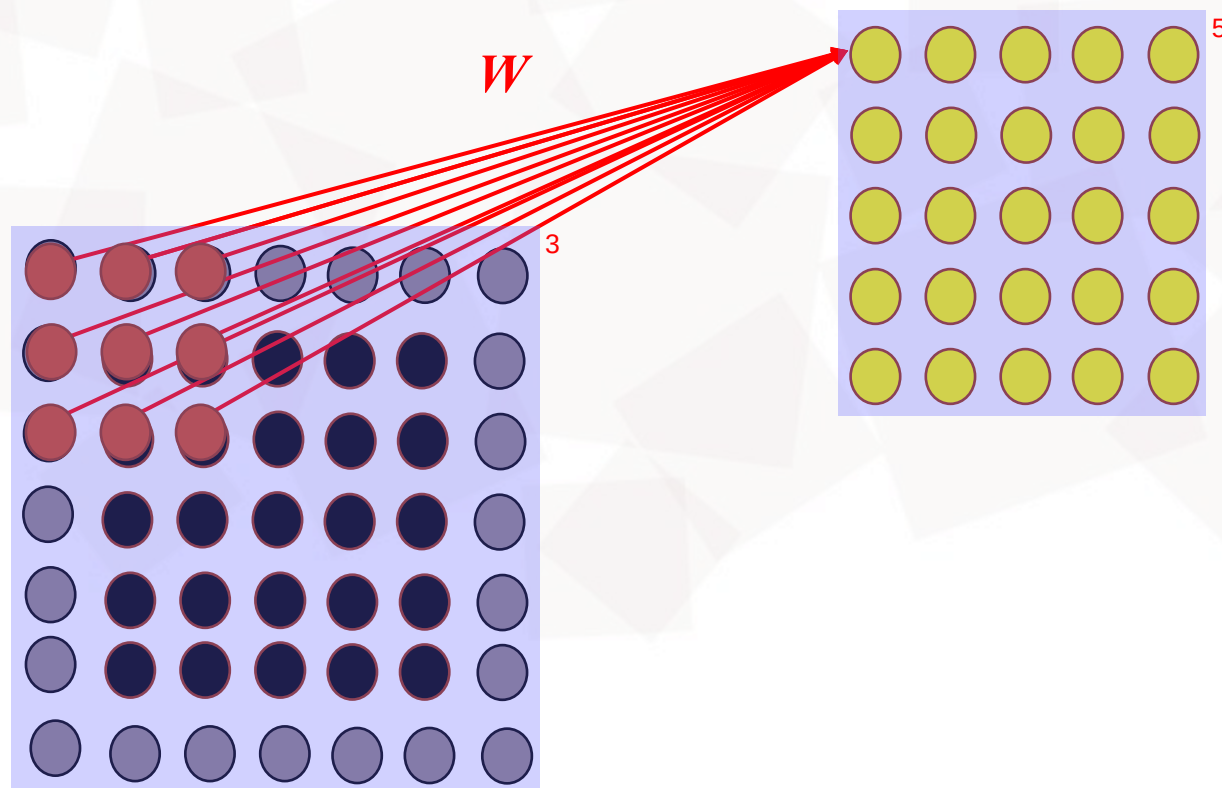
例：边缘提取²

[illegible][illegible][illegible]

卷积核的大小²

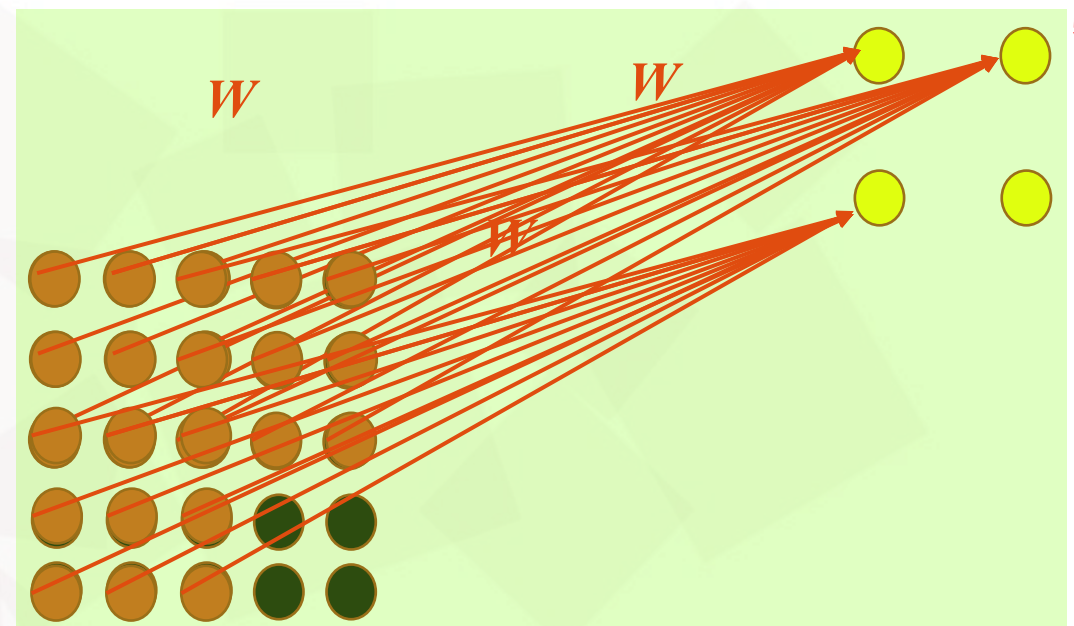
- 对于二维输入，一般是 3×3 、 5×5 、 $7 \times 7 \dots$ ³

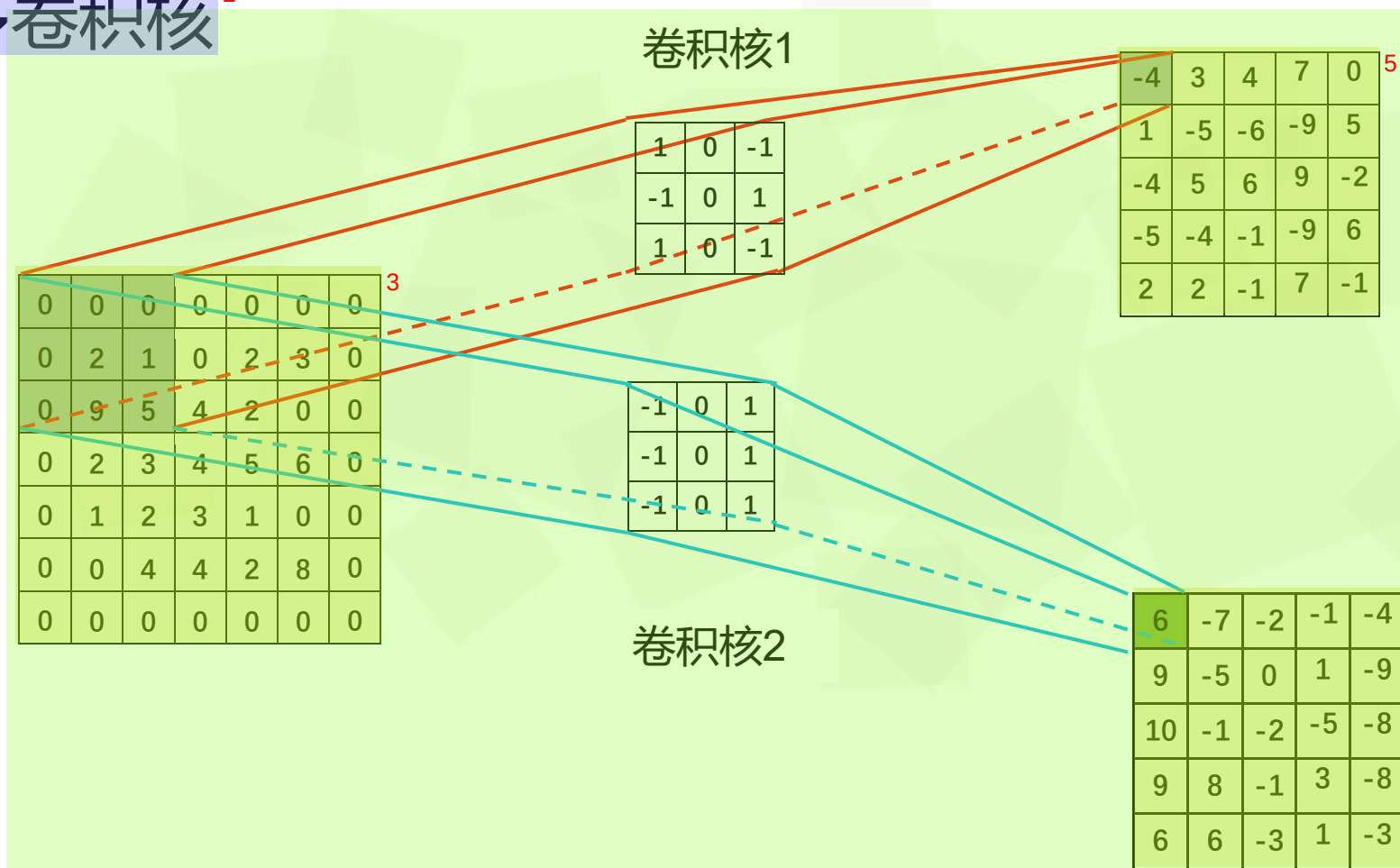


填充²

步长²

- 卷积核每次移动的距离³
- 步长是可以设定的 (以2为例)⁴



多卷积核²输出 (通道1)⁶

$$1*0+0*0+(-1)*0$$

$$+(-1)*0+0*2+1*1$$

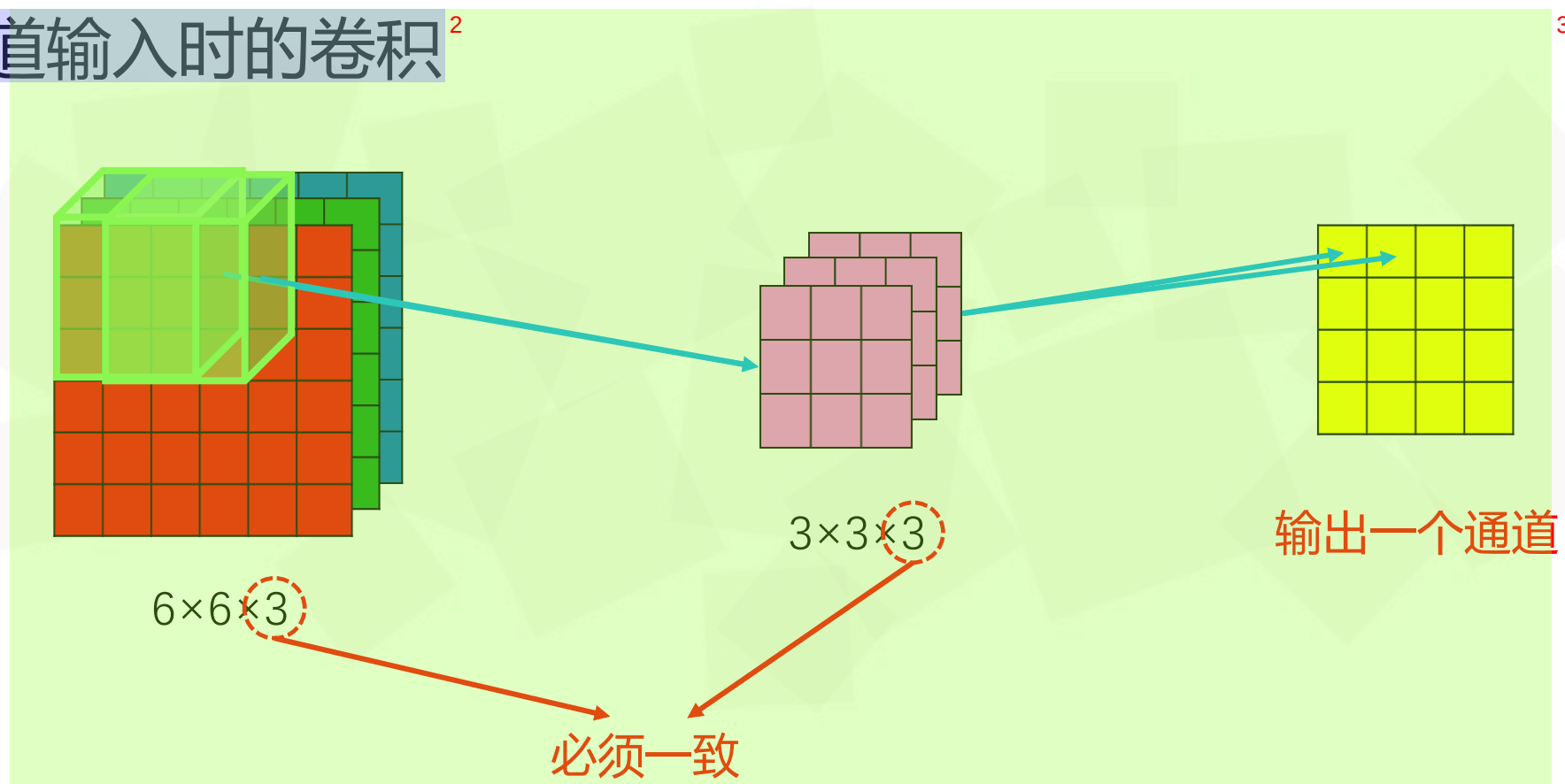
$$+1*0+0*9+(-1)*5 = -4$$

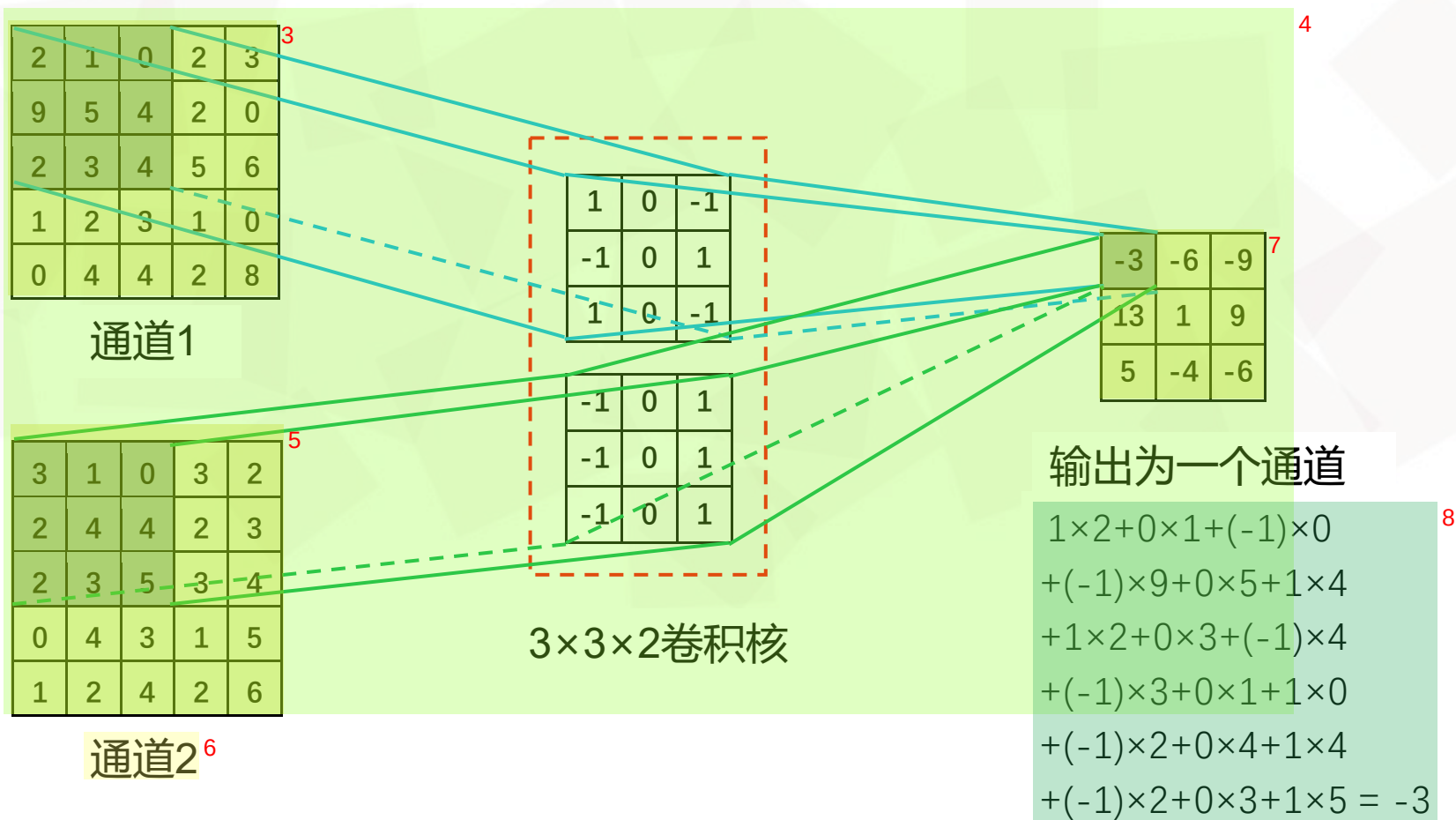
输出 (通道2)¹¹

$$(-1)*0+0*0+1*0$$

$$+(-1)*0+0*2+1*1$$

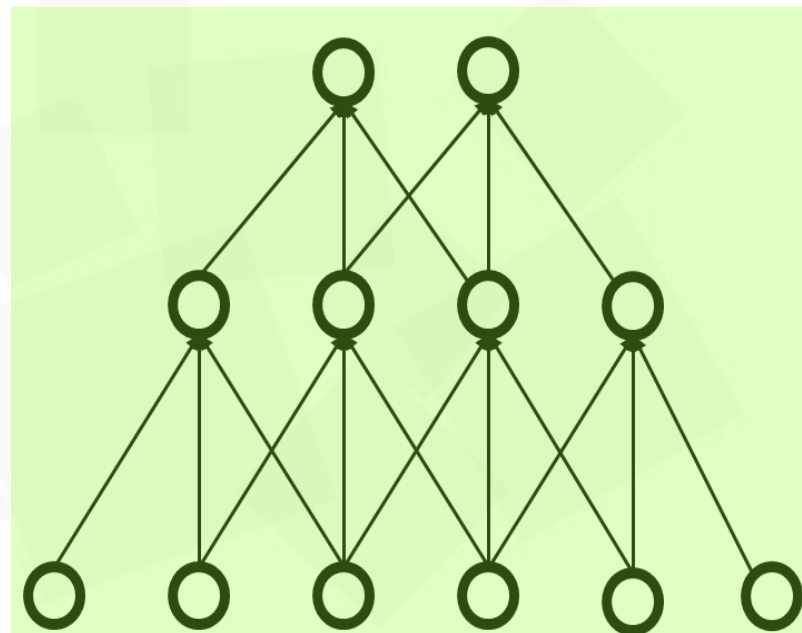
$$+(-1)*0+0*9+1*5 = 6$$

多通道输入时的卷积²

多通道卷积举例²

卷积核的大小²

- 小卷积核：细粒度特征³
- 大卷积核：大粒度特征⁴
- 多层小卷积实现大卷积⁵
 - 两层 3×3 卷积等效一个 5×5 的卷积⁶



池化²

- 一种降维的手段³

⁴

0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	0	0	0	0	0	0	0	0	0	0	1	1
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
0	1	1	1	1	1	1	1	1	1	1	1	1	1	0
0	0	1	1	1	1	1	1	1	1	1	1	1	0	0

⁵

0	1	1	1	0
1	0	0	0	1
0	1	1	1	0
1	0	0	0	1
0	1	1	1	0

池化方法²• 最大池化³

- 取窗口内的最大值⁴
- 窗口大小、步长都可以设定⁵

• 平均池化⁶

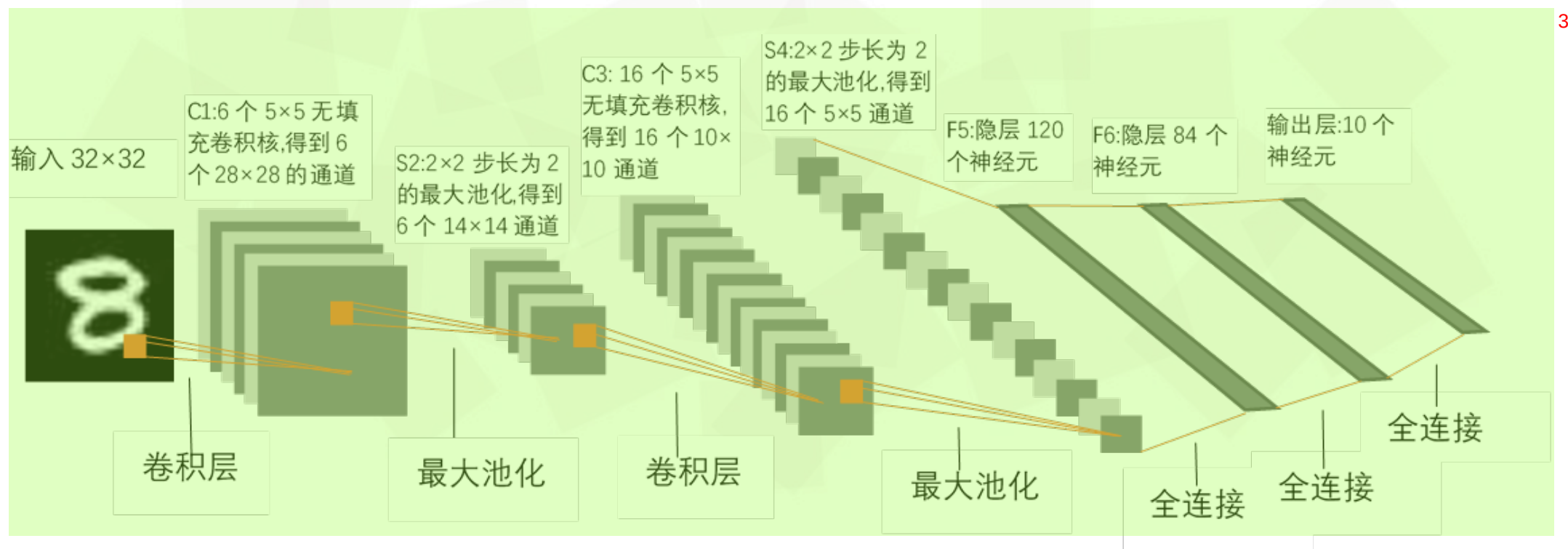
- 取窗口内的平均值⁷

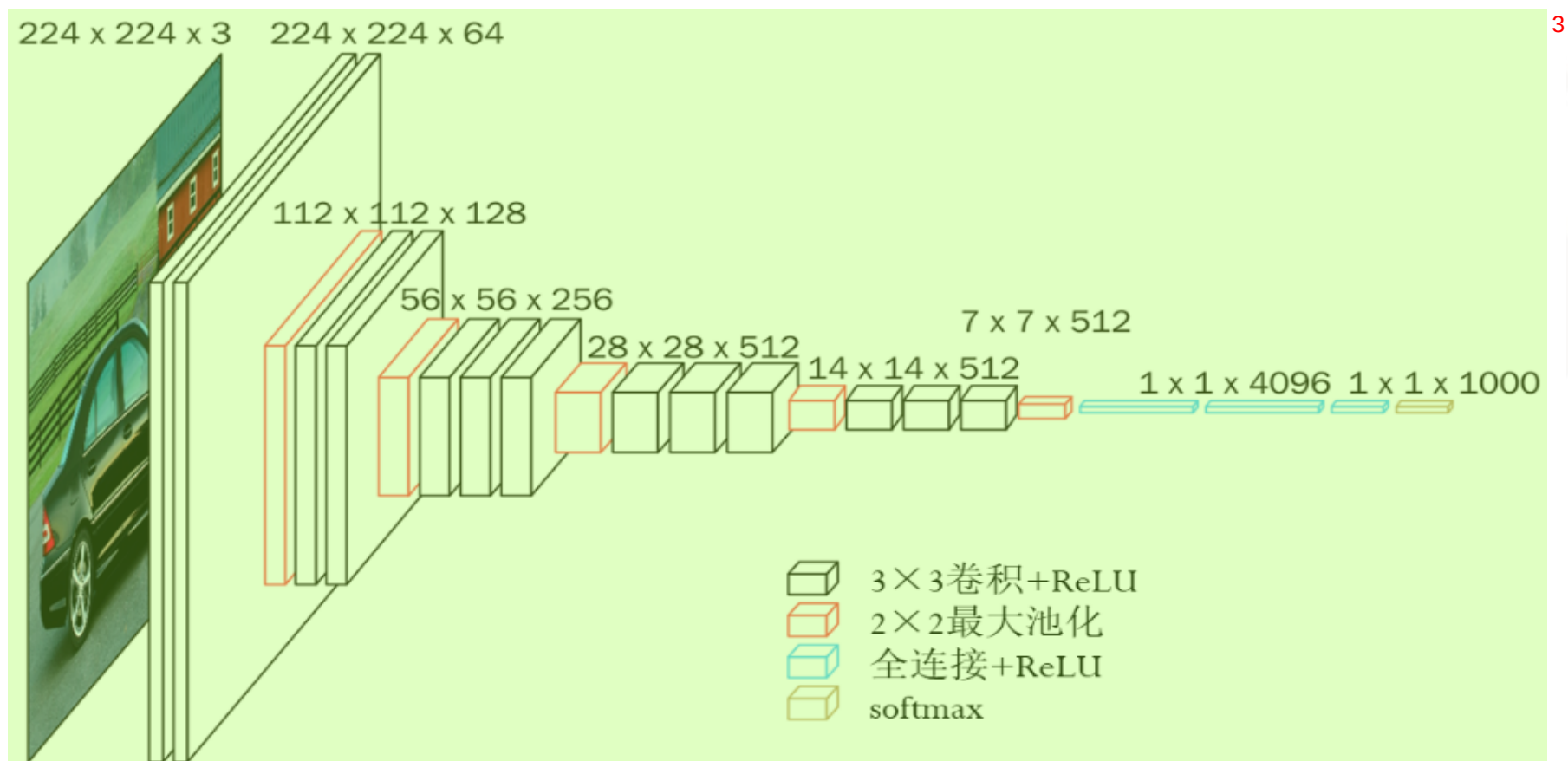
1	1	2	4	5	1
5	6	7	8	0	7
3	2	1	0	4	1
1	2	3	4	3	2
8	6	5	3	3	1
1	0	2	1	2	0

输入为一个通道¹⁰

最大池化¹¹
窗口: 2×2
步长: 2

6	8	7
3	4	4
8	5	3

LeNet神经网络 (Yann LeCun, 1998)²

VGG-16神经网络 (ImageNet)²

梯度消失问题¹



神经网络训练的两大问题：²

- 梯度消失问题³
- 过拟合问题⁴

梯度消失问题¹



什么是梯度消失问题:²

$$\delta_h = \boxed{o_h(1 - o_h)} \sum_{k \in \text{后续}(h)} \delta_k w_{kh}$$

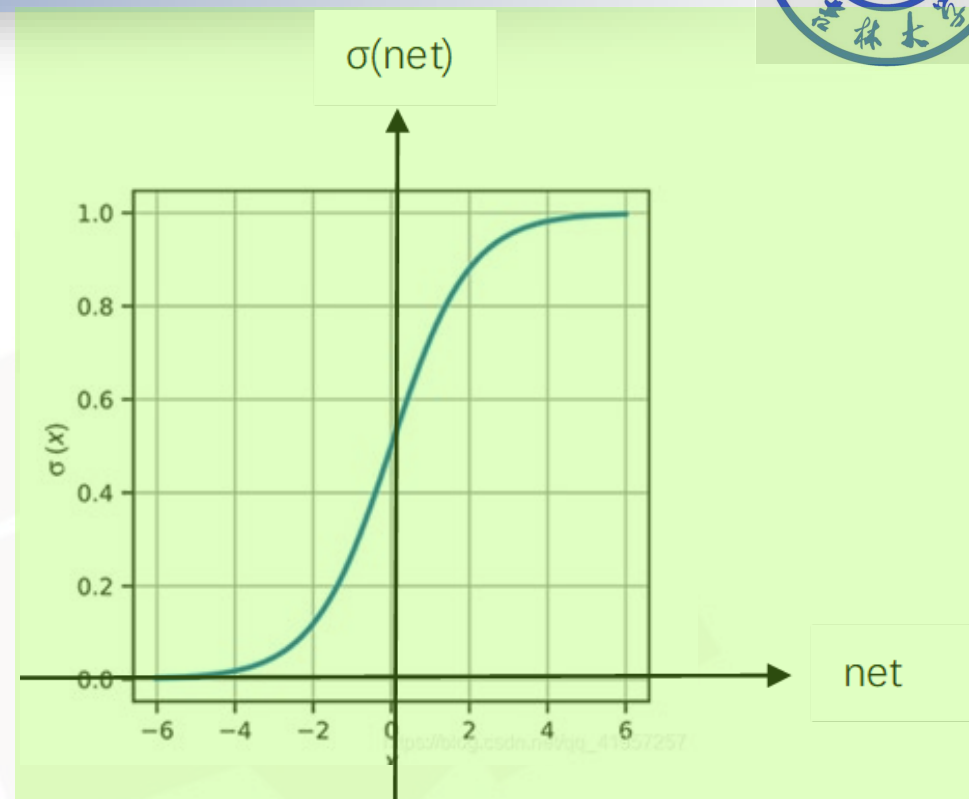
≤ 0.25

- 当神经网络层数增多之后⁴

$$\delta_h \longrightarrow 0$$

- 梯度消失问题⁶

$$\Delta w_{ji} = \eta \delta_j x_{ji}$$

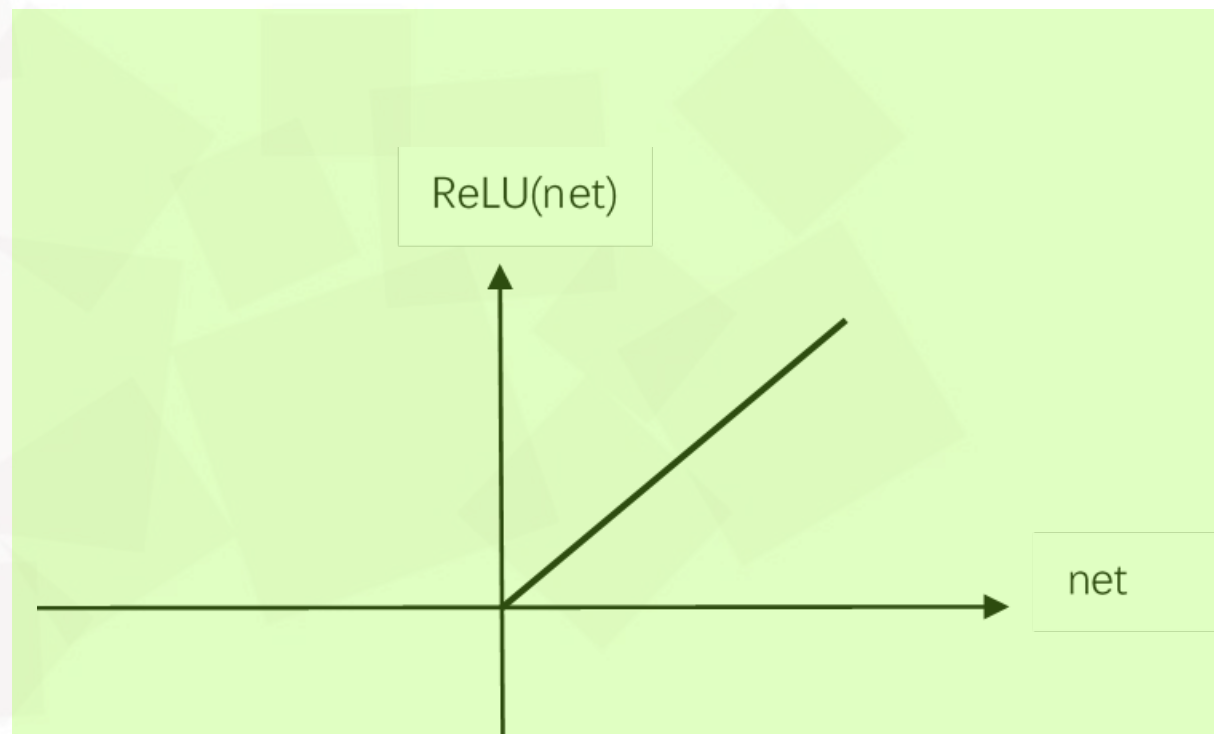


解决思路²

- 使用ReLU激活函数³

Sigmoid
的导数

$$\delta_h = o_h(1 - o_h) \sum_{k \in \text{后续}(h)} \delta_k w_{kh}$$



- ReLU的导数=1⁵



两个解决梯度消失问题的网络结构²

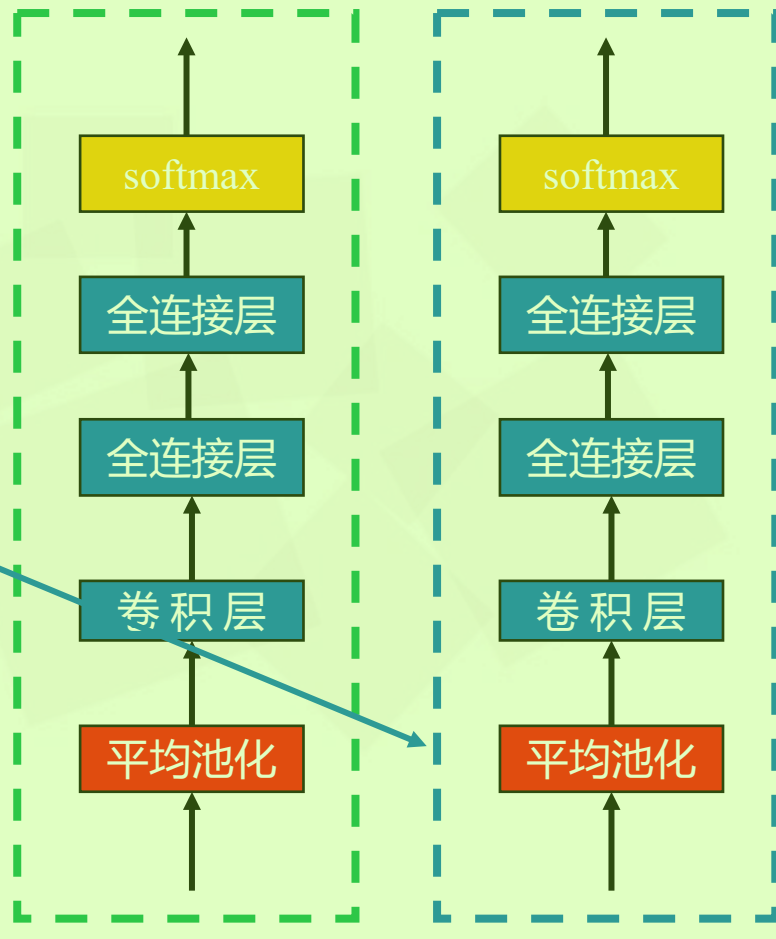
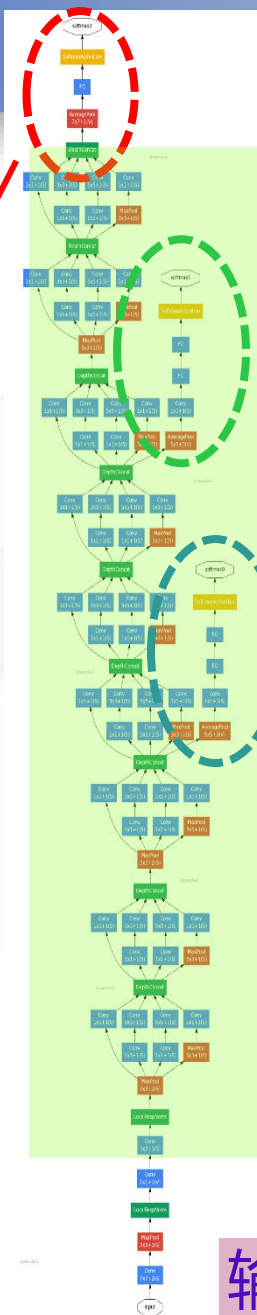
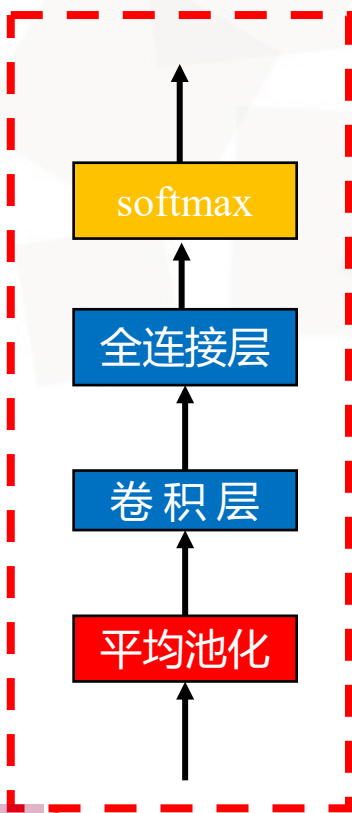
- GoogLeNet³

- ResNet (残差网络)⁴

梯度消失问题¹



GoogLeNet²



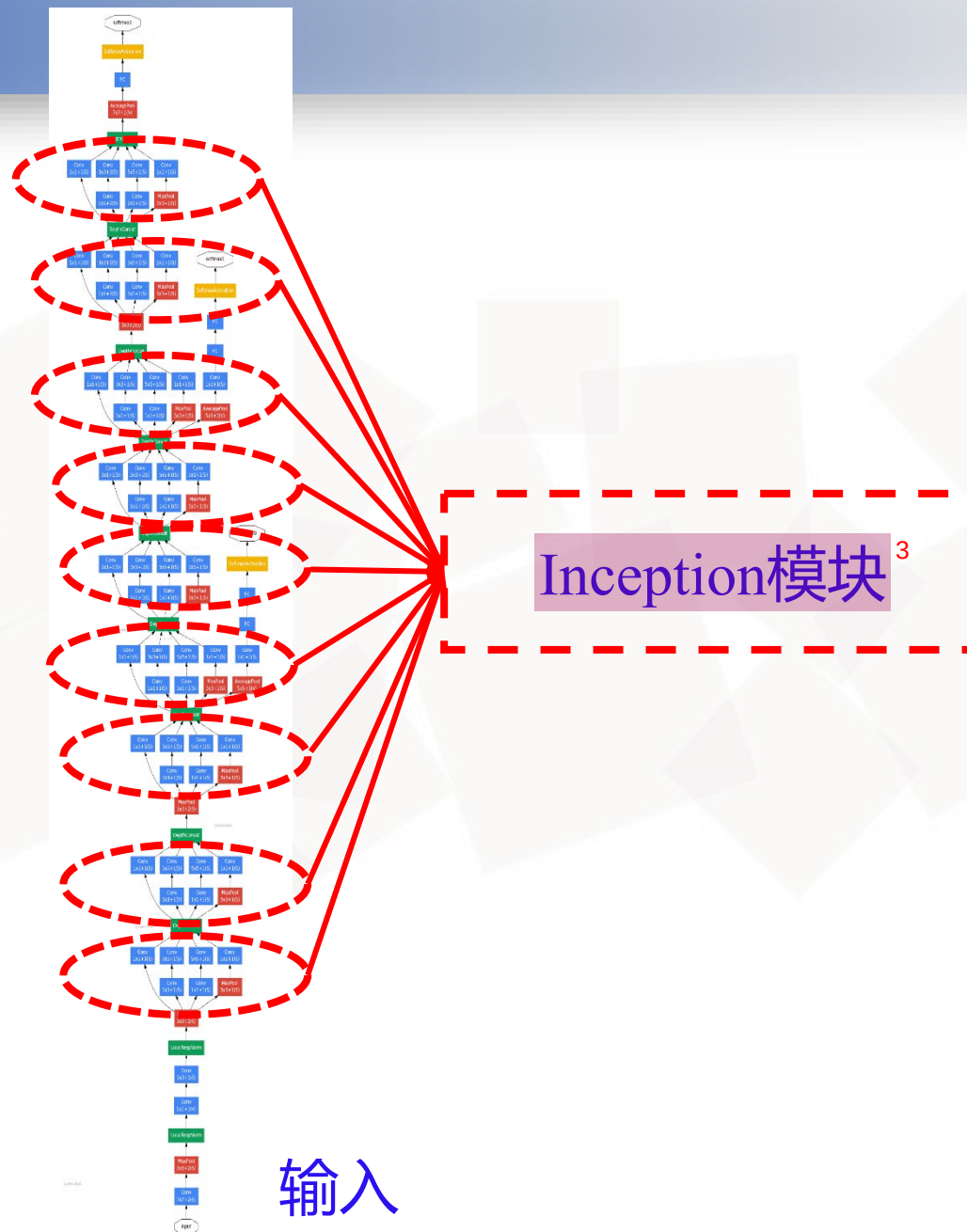
输入⁵

高楼供水系统²



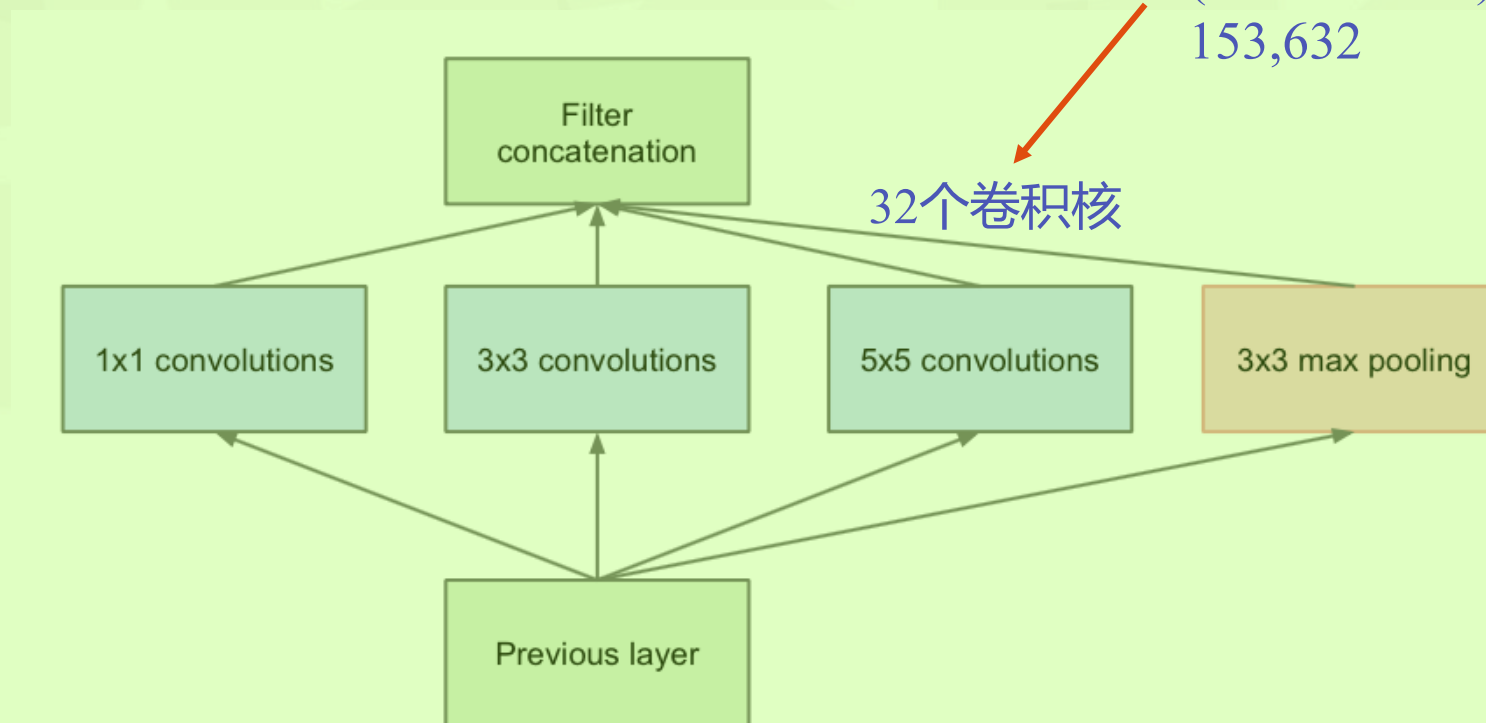
梯度消失问题¹

GoogLeNet²





Inception模块



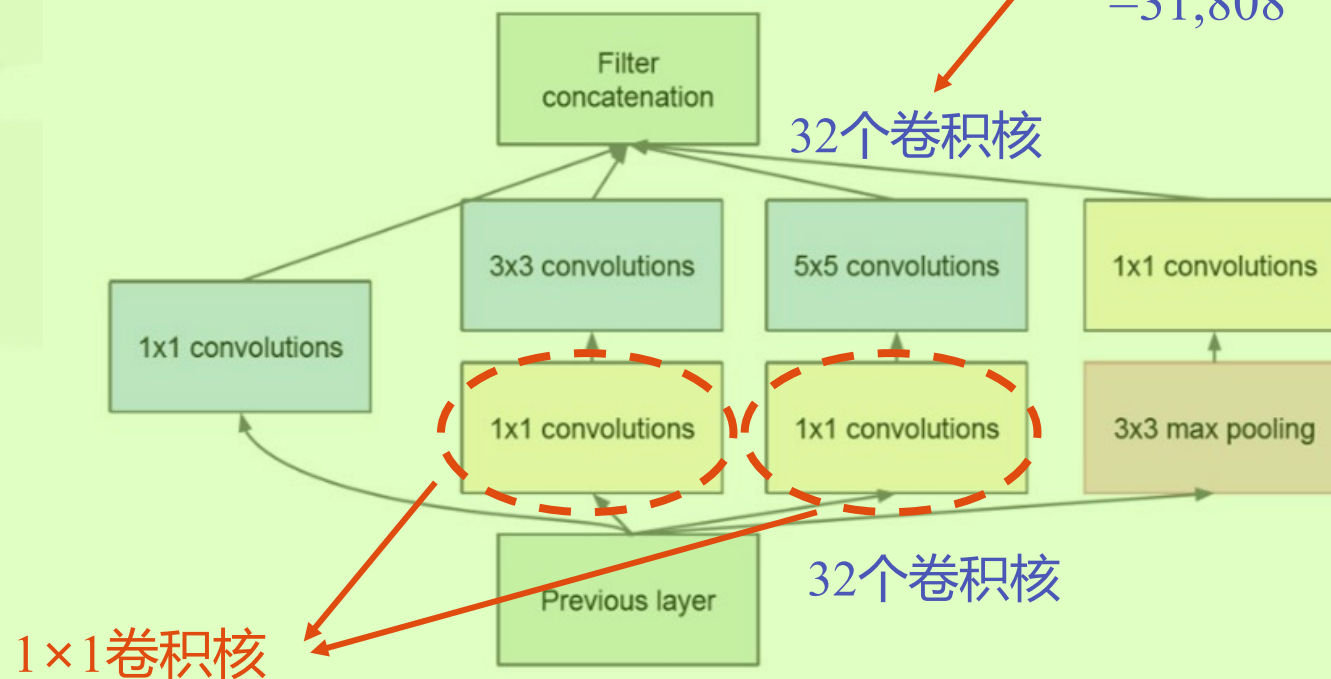
参数个数:
 $(5 \times 5 \times 192 + 1) \times 32 = 153,632$

192个通道³

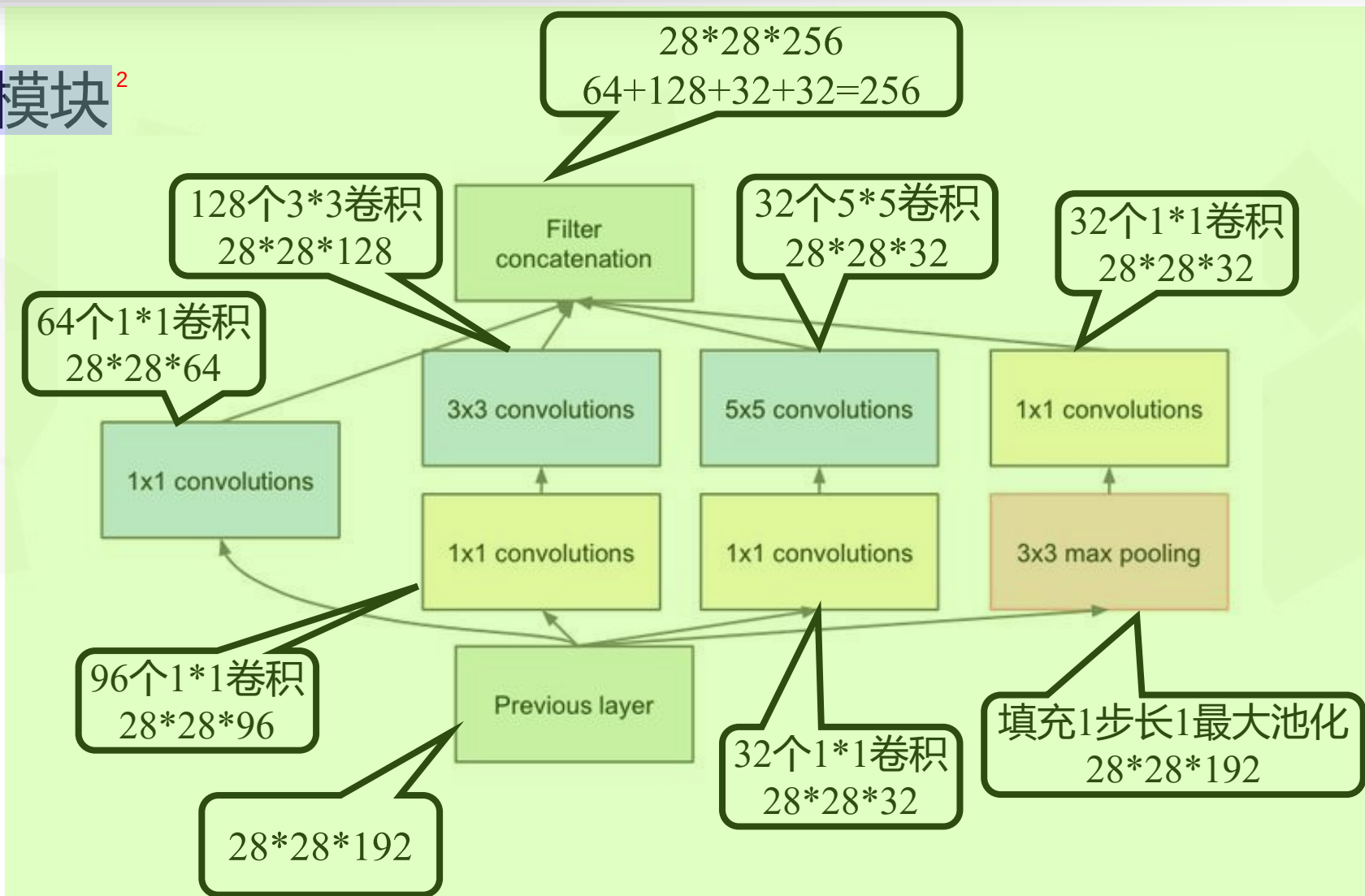


降维的Inception模块

参数个数:
 $(1 \times 1 \times 192 + 1) \times 32 +$
 $(5 \times 5 \times 32 + 1) \times 32$
 $= 31,808$



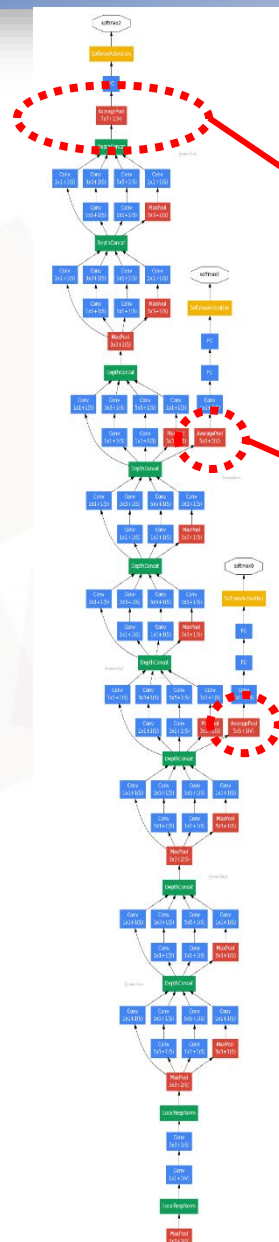
192个通道³

Inception模块²

梯度消失问题¹



GoogLeNet²



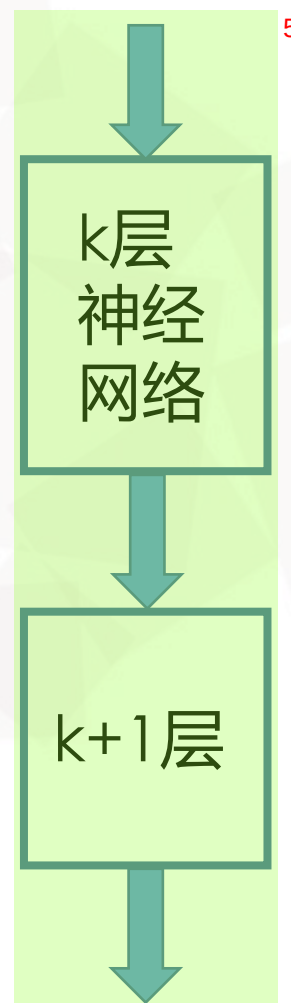
7×7平均池化
步长为1，无填充⁴

5×5平均池化
步长为3，无填充⁵



残差网络 (ResNet)²

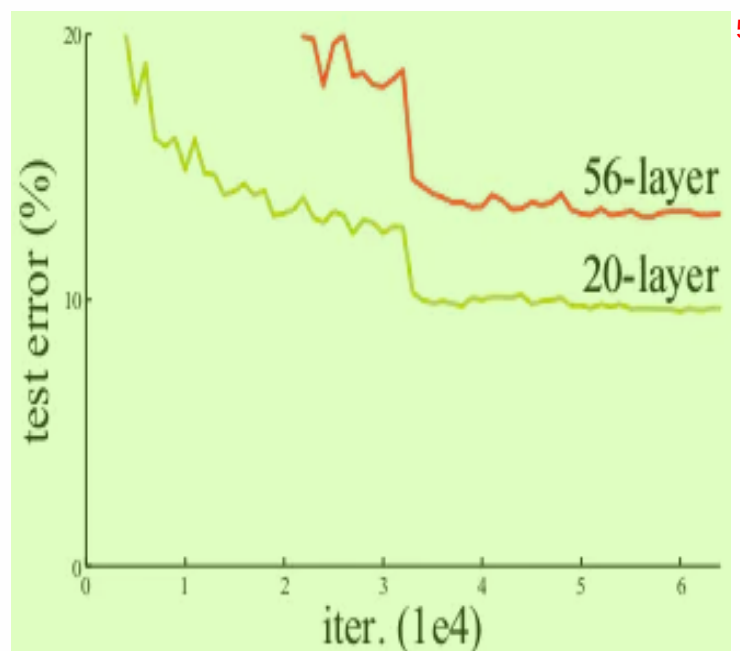
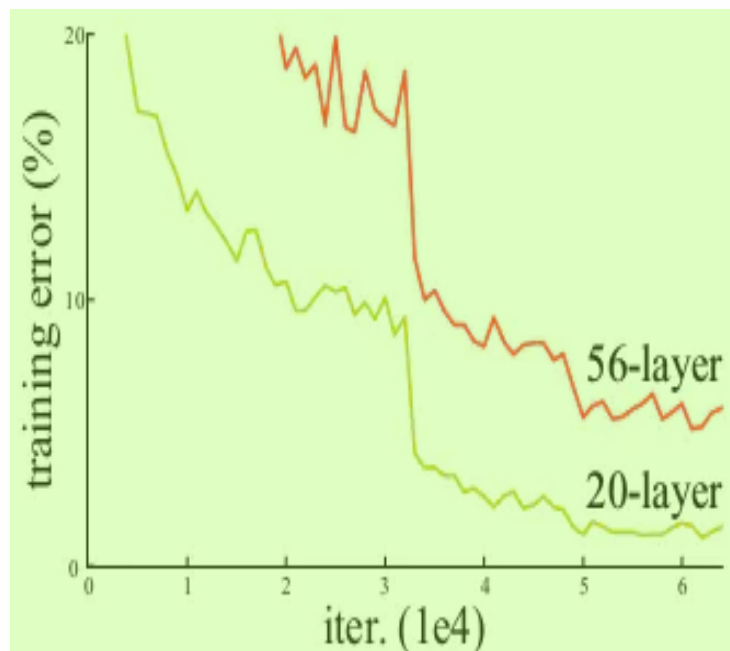
- 神经网络的退化现象³

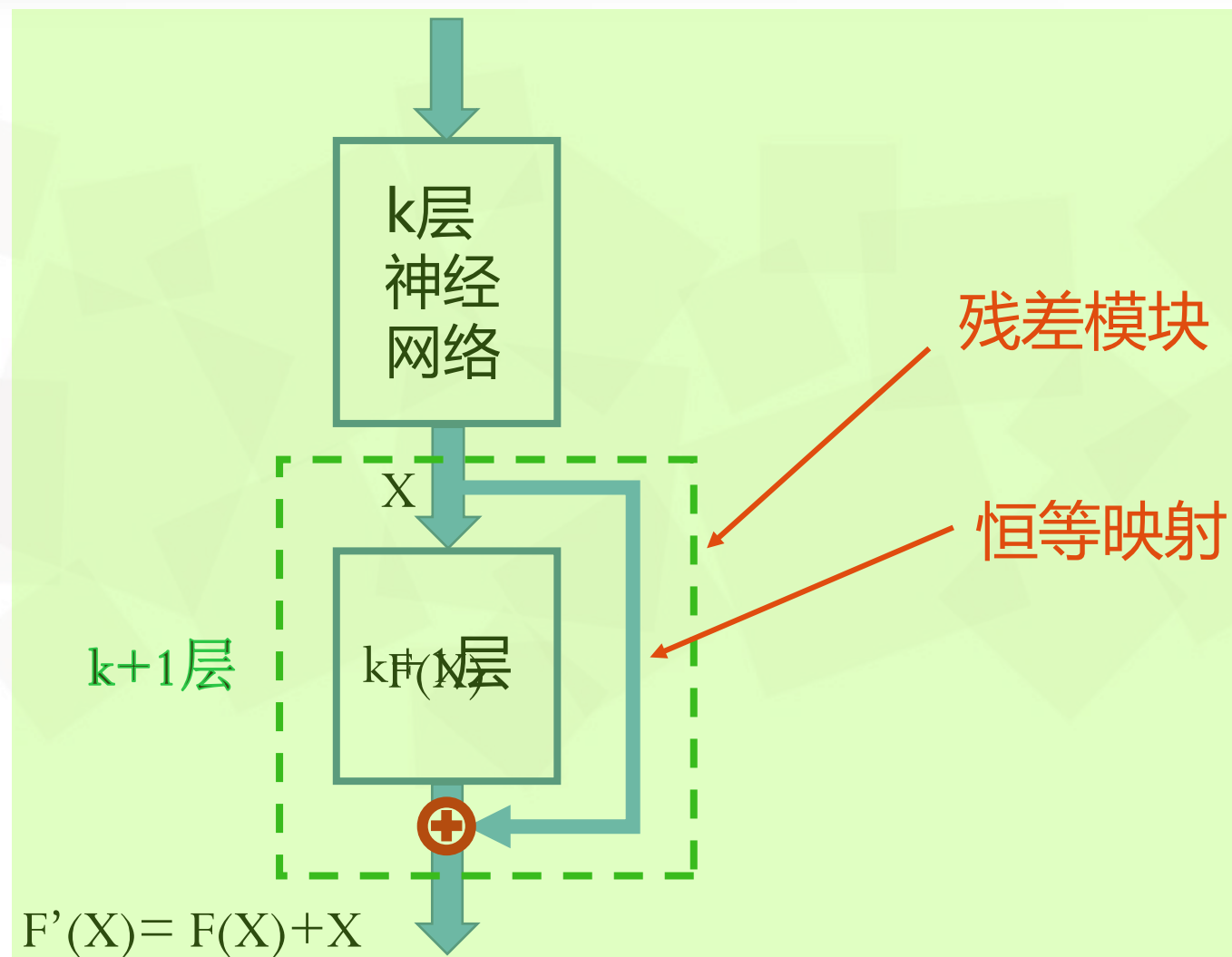


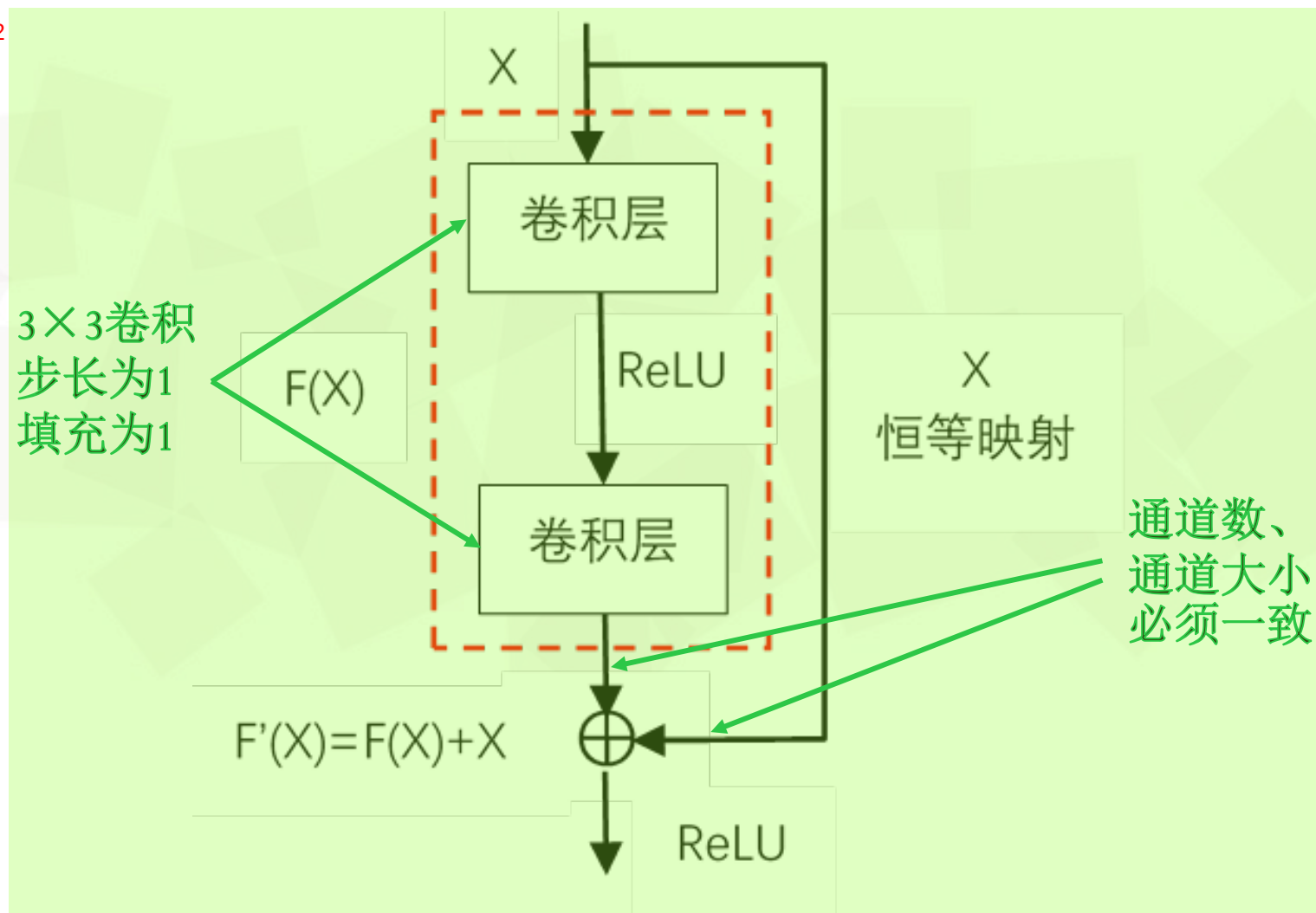


残差网络 (ResNet)²

- 神经网络的退化现象³



解决思路²

残差模块²

梯度消失问题¹

残差网络 (ResNet)²

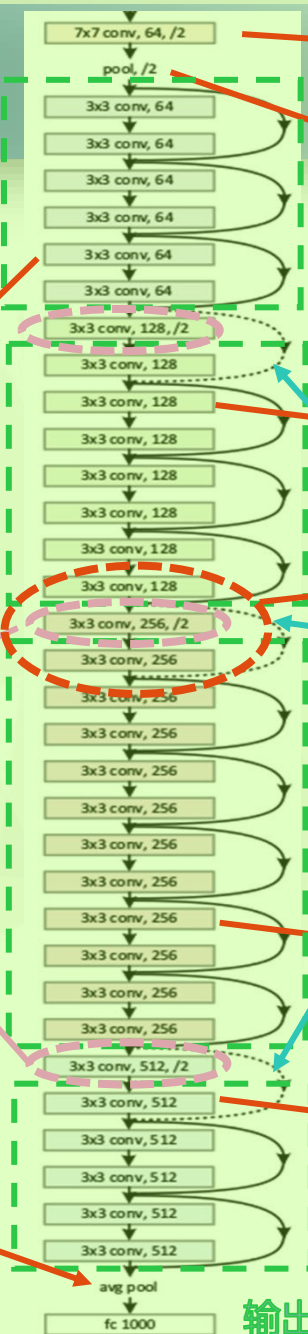
3个残差模块

64个 3×3 卷积核，步长为1，填充为1。

步长为2，长宽各减一半，通道数增加了一倍。恒等映射需要等维处理。

全局平均池化

输入:
 $224 \times 224 \times 3$



输出:
1000

64个 7×7 卷积核，
步长为2，填充为3。

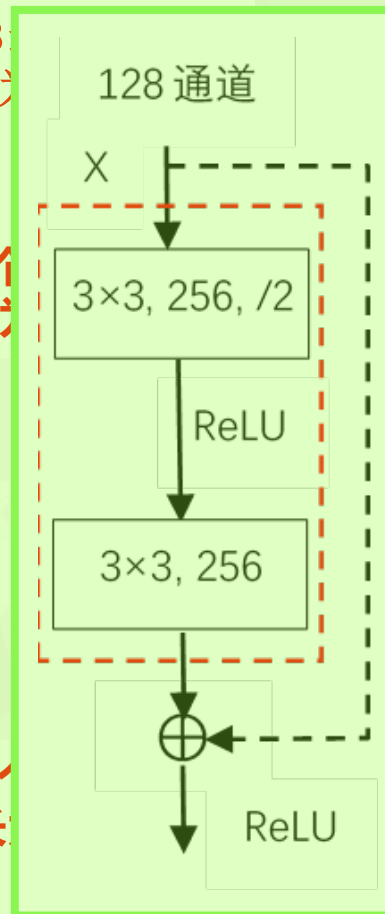
步长为

128个
步长为

虚线

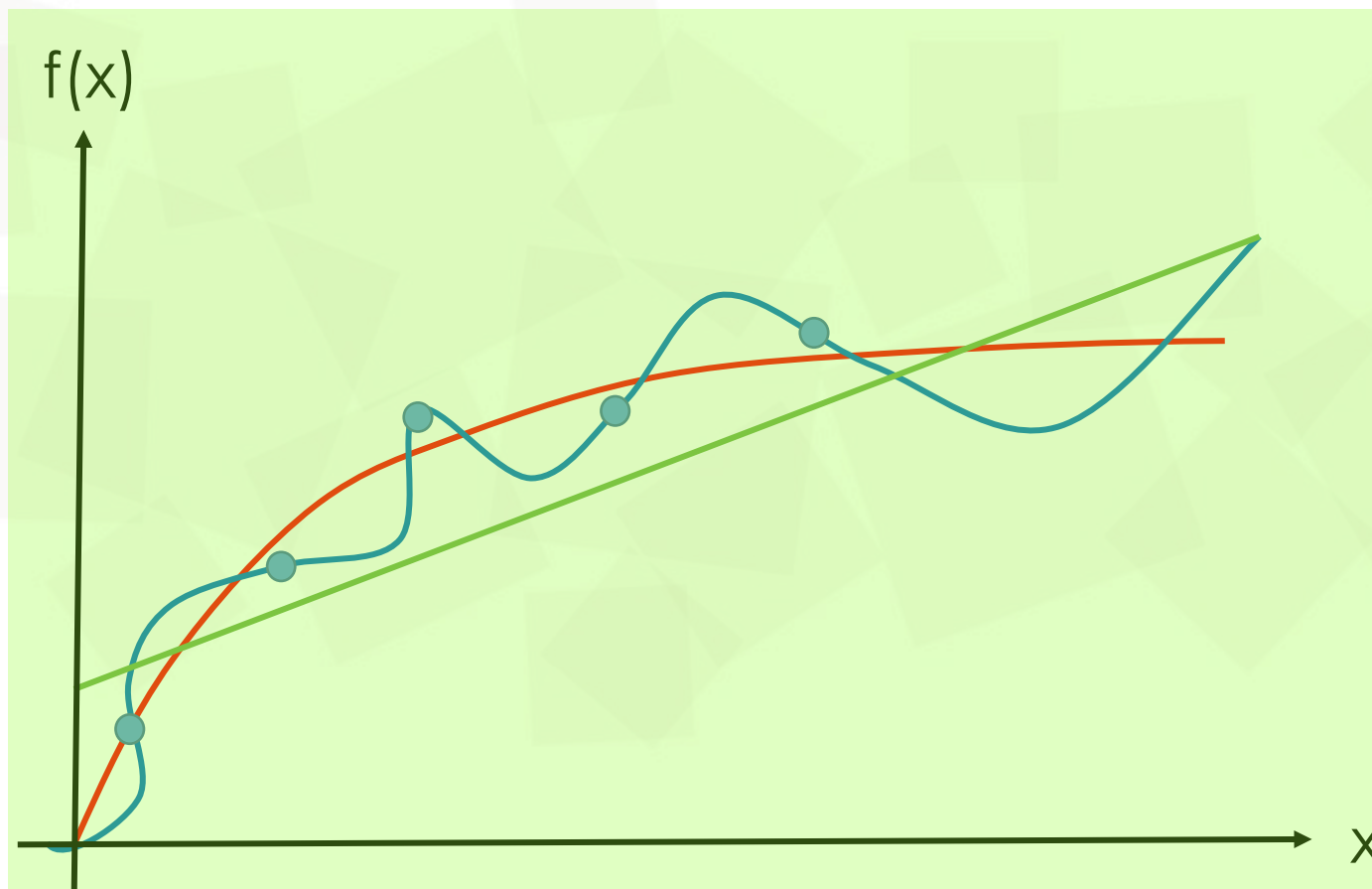
256
步长

512个 3×3 卷积核，步长为1，填充为1。



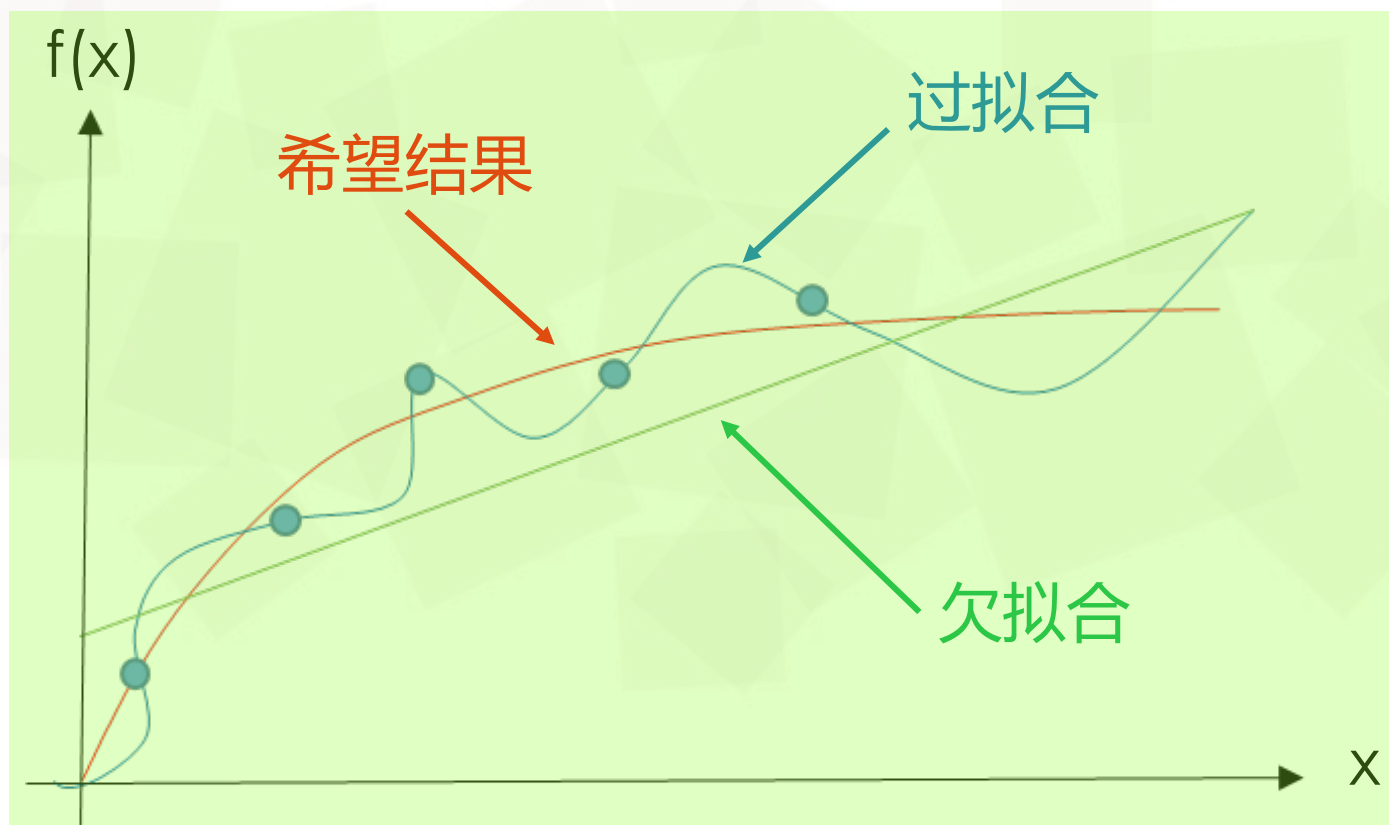


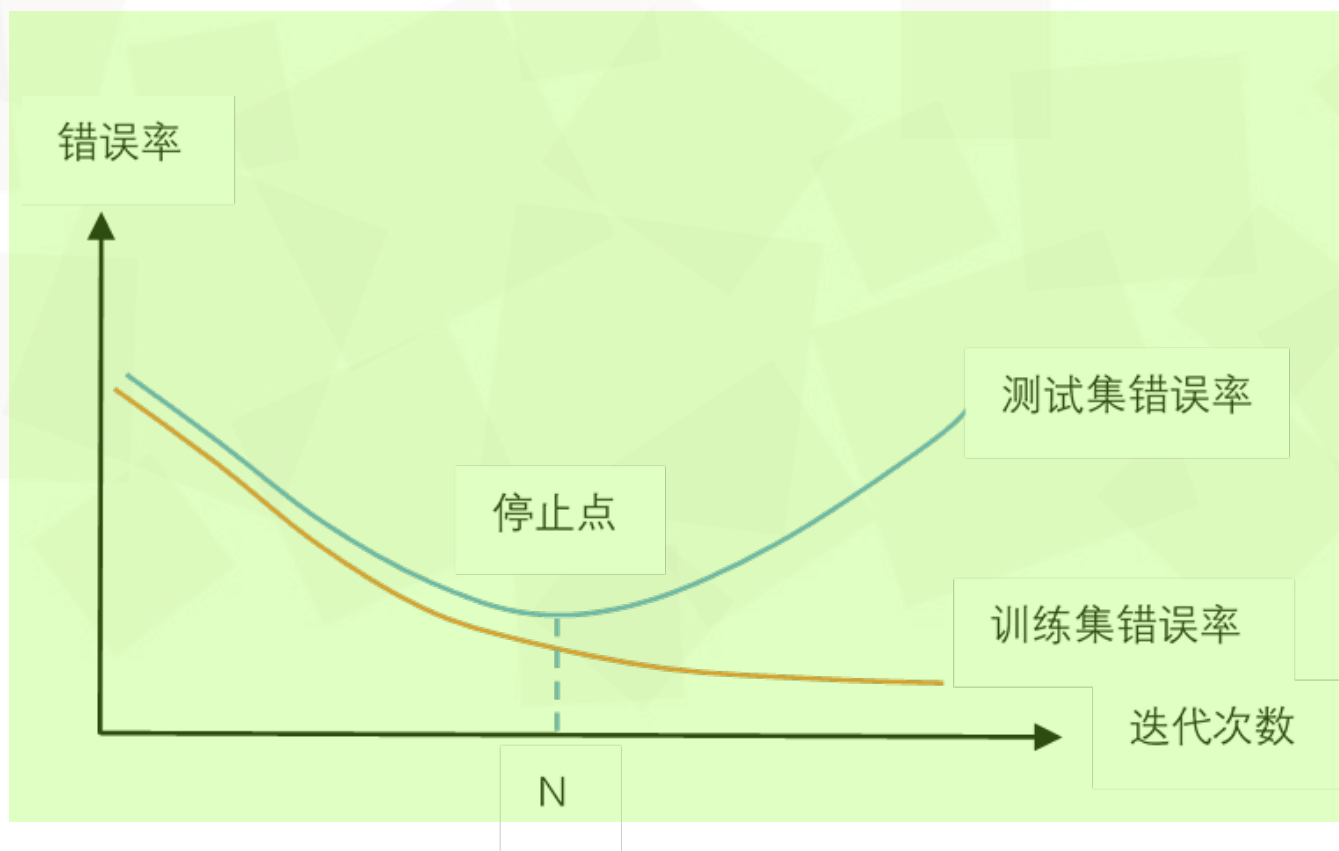
什么是过拟合问题²





什么是过拟合问题²



神经网络的过拟合问题²

减少过拟合的方法：正则化项法²

- 误差平方和损失函数³

$$E_d(w) = \sum_{k=1}^M (t_{kd} - o_{kd})^2$$
⁵

- 加正则化项⁴

$$E_d(w) = \sum_{k=1}^M (t_{kd} - o_{kd})^2 + \|w\|_2^2$$
⁶

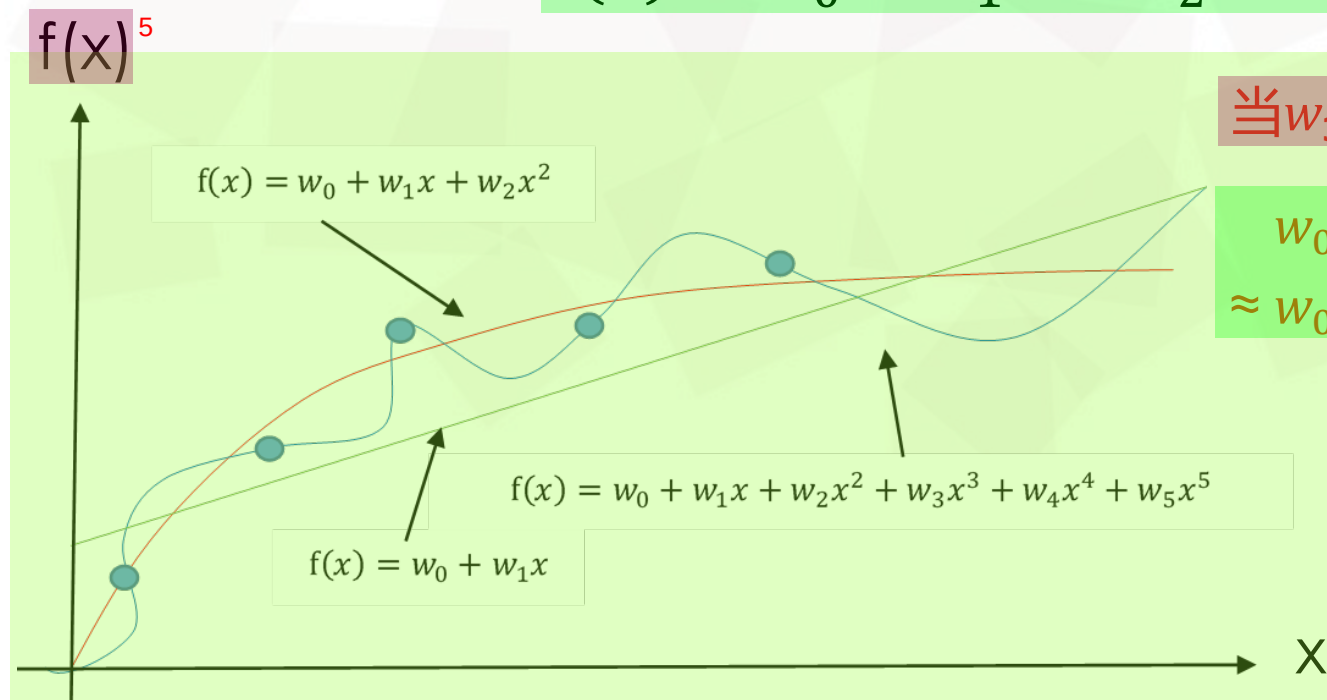
$$\|w\|_2^2 = w_1^2 + w_2^2 + \cdots + w_N^2$$
⁷

- 说明：并不限于误差平方和损失函数和2-范数⁸

正则化项的作用：降低模型复杂性²

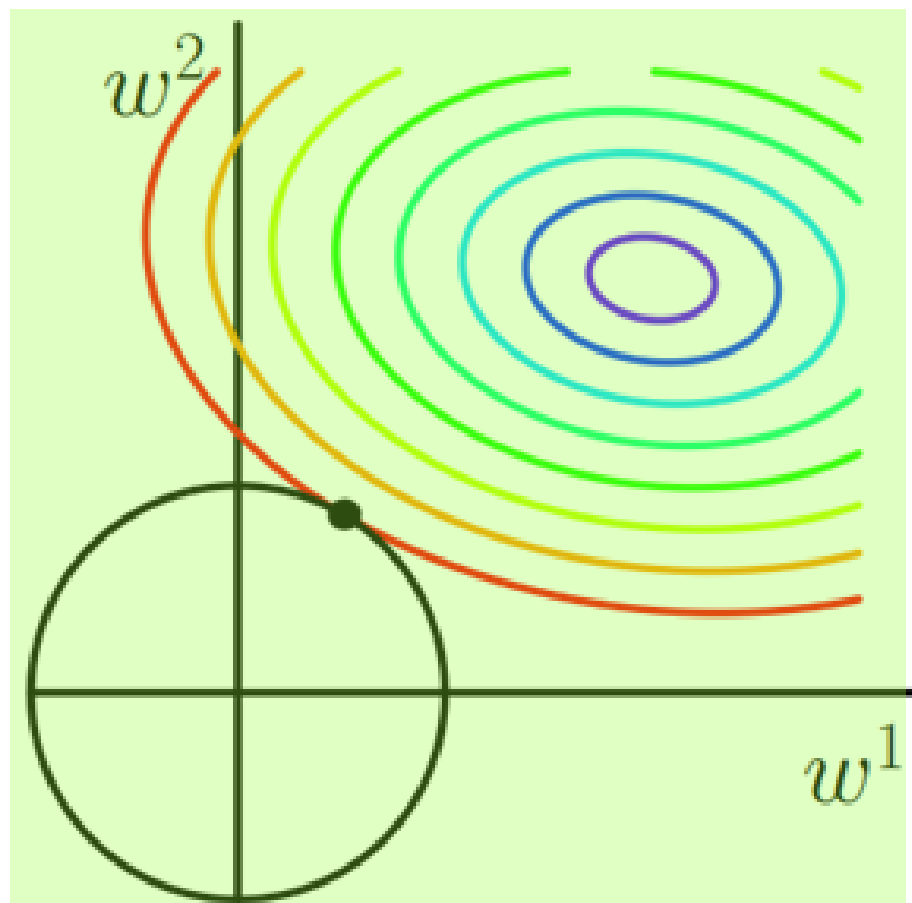
- 曲线的多项式表示：³

$$f(x) = w_0 + w_1x + w_2x^2 + \cdots w_nx^n$$
⁴



当 w_3 、 w_4 、 w_5 比较小时：⁶⁷

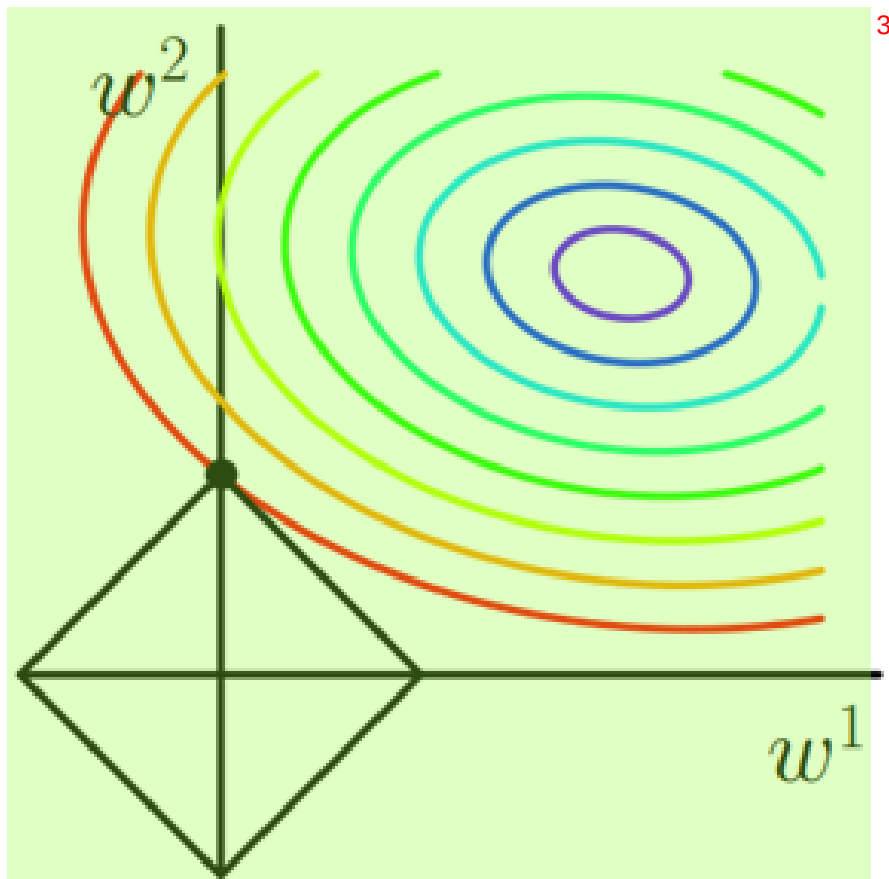
$$w_0 + w_1x + w_2x^2 + w_3x^3 + w_4x^4 + w_5x^5 \approx w_0 + w_1x + w_2x^2$$
⁸

L2 (2-范数) 正则化项²2-范数:⁴

$$\|w\|_2^2 = \sum_i w_i^2$$
⁵

效果:⁶

很多参数值很小，但基本不会为0。抗干扰能力强。⁷

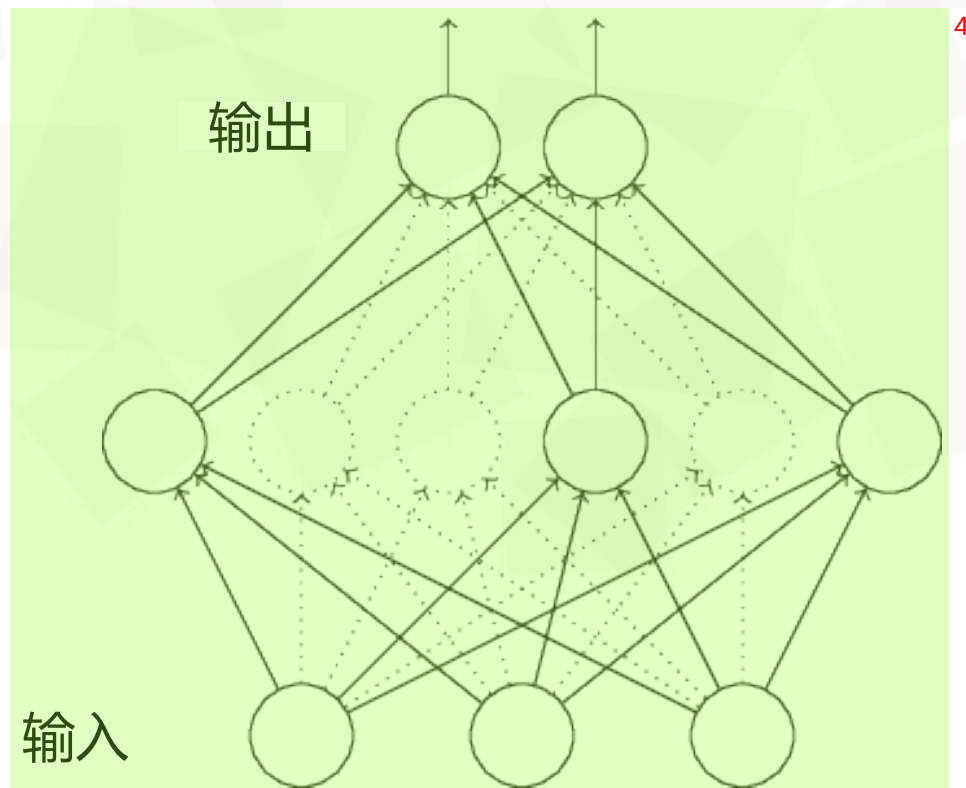
L1 (1-范数) 正则化项²1-范数:⁴

$$\|w\|_1 = \sum_i |w_i|$$
⁶

效果:⁷一些参数为0, 起到特征选择的作用。⁸

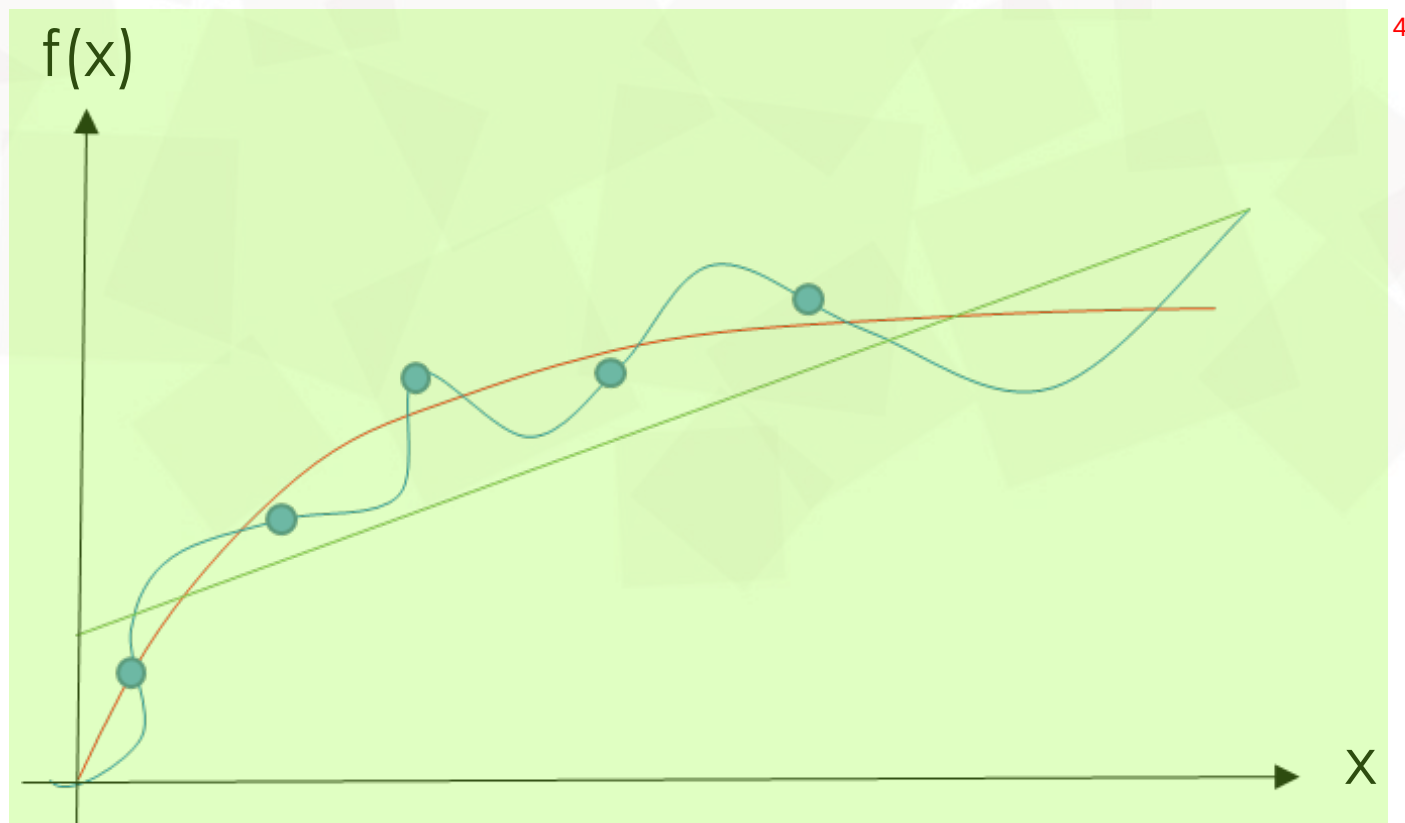
减少过拟合的方法：舍弃法（Dropout）²

- 随机地临时舍弃一些神经元，每次训练一个神经元子集³



减少过拟合的方法：数据增强法²

- 数据越多，过拟合的风险就越小³

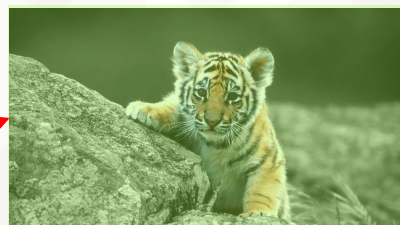




如何获得更多的数据?²

- 数据增强³
 - 通过各种变换增加数据的数量⁴
- 图像数据⁵
 - 缩放、旋转、局部截取、改变颜色等⁶

如何获得更多的数据?²





本课件源自清华大学马少平教授编著的《深入浅出人工智能》配套课件。²